OFFICIAL UNITY GAME JAM 2026 · THEME: LUCKY NUMBER



ZERO-SUM: OMEN

The World & Art Bible

World · Lore · Lands · Characters · Art Direction · Gameplay · Production

Version 3 · TR 1002 · Living document: the Markdown files in `docs/` are the source of truth.

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How to use this bible: when two documents disagree, the order of authority is Art Direction (for visuals) → Character Bible (for characters) → World Bible (for story) → GDD (for mechanics).

PART I

The game in one page

What we are making and why it will feel like nothing else.



In a world where every soul's luck floats above its head as a number, a nameless wanderer with no number at all tells the city its lucky number, and becomes the most wanted person alive.

The fantasy

You are NIL, the world's missing luck in human form. Every number in the Tally spins like a slot reel, and you are the only one who can stop them. Travel lands inspired by Japanese, Chinese, Indian, Greek and American myth, gather a party of eight, and break the seals of the machine that flattened every culture's idea of luck into one zero-sum number.

The play

One verb: tap to stop the number. Three heroes strike, their digits drop into the reel, and patterns pay out as full-screen anime cinematics. Depth comes from team synergy (heroes whose lucky digits add to 10, run in sequence, or are all prime), charms from every culture, and wheels you reshape like a deck.

The look

Painted world, inked people. Hand-painted, sunlit, lived-in lands in the spirit of Studio Ghibli's backgrounds; graphic, neon-edged characters and fights in the spirit of *Cyberpunk: Edgerunners*.

The theme

A world that believes luck is zero-sum learns, from a party drawn from every culture, that luck shared isn't.

PART II

The world

Why numbers matter, the god of luck, and who wants to freeze fate forever.



> *"Every soul a number. Every number a share. The books must balance."* — the Comptroller's catechism, recited at every Dawn Tally

This file is the story source of truth: world, lore, the god of luck, lands, villain, cast, plot and endings. The full scene-by-scene campaign is in `STORY.md`; each land's acts, encounters and bosses are in `ACT_DESIGN.md`. Mechanics live in `GDD.md`; schedule and pipeline in `PRODUCTION_PLAN.md`.

1. The one-paragraph pitch

In **the Tally**, every living soul carries a floating number, its **Omen** (00–99), which is literally its share of the world's luck. Luck is conserved: for one Omen to rise, another must fall. It wasn't always like this. Every people once had its own idea of fate: the Greeks' spun thread, the Indian scribe's ledger of deeds, the Chinese and Japanese underworld books of the dead, the deal struck at an American crossroads at midnight. Then the capital, **Meridian**, built **the Basilica** and bound every one of those ledgers into one machine that gives every soul one number.

NIL wakes in Meridian's slums with no number at all, only a glitching **NaN**. When NIL is made to state a lucky number at a back-alley Naming Rite, the Basilica screams **REMAINDER DETECTED**, and every bounty hunter in the world comes running. To stop the **Final Audit**, which will freeze every fate forever and erase NIL to balance the books, NIL must travel the lands of the Tally, free the ledgers the Basilica stole, gather a party from every culture, and learn the truth: NIL *is* the world's missing luck.

2. Why numbers matter (the rules of the world)

#	LAW	IN THE WORLD	IN THE GAME
1	**Every soul has an Omen** (00–99)	Everyone can see everyone's number floating above their head. It decides loans, jobs, marriages and how dangerous a staircase is for you today.	The Omen over every combatant is its health readout: in an Audit it shows the share of its staked luck (its **Balance**) that is left, 99 down to 00 (GDD §1.6). No HP bars.
2	**The Zero-Sum Law**	Total luck is constant. Fortune is traded, taxed, inherited and stolen. Luck banks lend it at interest; debt collectors drain it.	Fights are transfers of luck. The Zero-Sum pattern (digits summing to 10) "balances the books" and cancels a move.
3	**Resonance (your lucky number)**	At seven, every child performs the **Naming Rite** and states their lucky number. That digit is how the Basilica reads you and how you channel luck.	Each hero has a resonant digit. Their wheel favours it, and landing it charges their ultimate. The player chooses NIL's.
4	**Settlement**	At 00 you are *Settled*. Misfortune finds you within days, and your luck flows back to the pool.	Defeat = Settled. Enemies are "rounded down", not gored.
5	**Audits**	When fates collide, Omens contest. Trained **Auditors** summon a **Fate Wheel** (the old *Rota Fortunae*, Fortune's wheel, now made of digits) and strike with the digits they stop.	The core verb: every number spins, and you stop it. Combat is called an Audit.

Daily life details to show, not tell. Lifts skip floors 4, 9 and 13. Weddings crowd onto the 8th. Gamblers kiss sevens. Beggars sell "luck-ins" (a point of Omen for a meal). Children compare Omens in the schoolyard. Shopkeepers hang a lemon with seven green chillies over the door, or a horseshoe, or a blue evil-eye bead, or a beckoning cat, depending on the street.

Lucky and unlucky numbers across the Tally (the jam theme, built into the setting)

NUMBER	WHERE IT'S LUCKY / UNLUCKY	WHY (REAL-WORLD BELIEF THE LAND IS INSPIRED BY)
0	Śūnyapura (India-inspired)	Zero as a number was formalised in India; *śūnya* means void and also the place that holds everything
2	Kroton Nova (Greece-inspired)	The only even prime; Pythagoreans treated numbers as divine
3	Celtic quarter of Meridian	Triads, the triskele; "third time lucky"
4	Hyakki (Japan-inspired) and the River Empire (China-inspired), unlucky	四 sounds like 死, death, in both Japanese (*shi*) and Chinese (*sì*/*sǐ*)
7	Meridian, Hyakki, the Crossroads	Western lucky seven; Japan's Seven Lucky Gods; "seven come eleven" at the dice table
8	The River Empire, Hyakki	八 *bā* rhymes with 发 *fā* (to prosper); in Japan 八 widens like a fan (*suehirogari*)
9	Hyakki unlucky; River Empire lucky	Japanese *ku* echoes 苦 (suffering); Chinese *jiǔ* echoes 久 (long-lasting)
13	Meridian, the Crossroads, unlucky	Western triskaidekaphobia
17	Kroton Nova, unlucky	In Italian tradition XVII rearranges to VIXI, "I have lived" (i.e. I am dead)

NUMBER	WHERE IT'S LUCKY / UNLUCKY	WHY (REAL-WORLD BELIEF THE LAND IS INSPIRED BY)
99 / 100	Hyakki	Japanese folklore: objects that reach 100 years gain a soul (*tsukumogami*)
108	Śūnyapura	Sacred count of prayer beads

2A. Hap, the god of luck who governs the Tally

> *"Hap deals. People choose."* — the oldest proverb in the Tally, said in every language

Hap is the Tally's one god of luck: not a god *of* a land, but the will inside chance itself. The name is the old word that still hides in *happen*, *happy*, *perhaps* and *mishap*. Hap is invented for this world; it is not any real religion's deity.

What Hap is	Luck with a heart. Playful, generous, unfair, never cruel on purpose. Hap loves surprise more than anything: the long shot, the second chance, the stranger who turns out to be family.
How Hap governs	Hap never commands. It **deals**: every morning, before the Basilica, Hap scattered luck across the world like seed, unevenly, and let people decide what to grow. Every land built its own way of talking to Hap: thread, karma, books of the dead, a contract at midnight. Those ways are the **many ledgers**.
How Hap appears	Never the same face twice. To each land it shows the face that land's stories expect, drawn in that land's art style; in the game it appears as a figure of overlapping faces under a turning wheel of stars, with **two dice for eyes** and every digit 0–9 orbiting its head like a halo. Its voice is the sound of dice in a cupped hand.
Symbol	**The Open Hand**: an open palm with a die resting in it. Shrines to Hap are just that: a hand carved in stone, a die, and somewhere to sit.
Signs of Hap	Coincidences. A coin landing on its edge, a stranger with your birthday, a rainbow at night. People say "Hap's watching" and buy a ticket.
Hap and the land gods	Every land keeps its own revered figures (the Seven Lucky Gods' shrine in Hyakki, the keepers of the old ledgers, Fortuna's wheel in Kroton Nova, Lady Luck at the Crossroads card tables). In this world they are how each culture *talks about* luck; the game never claims they are secretly Hap, and never makes them villains (§10).

What happened to Hap (the truth, revealed across the game): 1. **The Binding** (TR 380–690) took the ledgers: the ways people spoke to Hap. 2. **The Final Binding** (TR 702): the Basilica's heart, the **Even Hand engine**, closed around Hap itself. The Basilica's great amber eye is **Hap's eye, caged**. Since then Hap no longer deals; it is **counted**. Every Dawn Tally divides the caged god into whole-number shares: the Omens. 3. **The Rounding** shaves the fractions off Hap every morning. Three hundred years of shavings pooled into one account, and the account woke up: **NIL is the part of Hap that refused to be counted.** NIL is not Hap, and not a god; NIL is a person made of the god's lost change, and the last free piece of luck in the world. 4. **The Deficit** is Hap starving in the cage. **Noise storms** are its seizures. **The Default** that the theologians fear is not "the Noise returns"; it is **Hap's death**, after which nothing would ever happen by chance again: no surprises, no second chances, no luck at all. 5. **Thale** discovered the cage in TR 999. He believes a caged god is mercy ("No child has died at random since I put chance in a box") and plans the **Final Audit**: settle NIL, count Hap completely, and freeze it forever (**Fixed Fate**). He doesn't know, or won't admit, that a frozen god is a dead one.

Hap in the game: Hap speaks only in dreams. At the end of every act NIL dreams of a dice-starred void where Hap appears with the face of the land just visited and asks one question (the **Hap's Dream** epilogues, Story §Interludes). In play, Hap shows up as **Hap's Favour** (a pity system: luck turns after bad luck), **Hap's Whispers** (coincidence events on the map) and **Open Hand shrines** (rest nodes). See `SYSTEMS.md`.

3. History (the myth told in the intro)

1. **The Noise.** Before numbers, fortune moved like weather: Hap dealt freely, and nobody kept accounts. Floods and fortunes fell where they fell.
2. **The Many Ledgers.** Every people learned to keep accounts with fate in its own way: - In **Kroton Nova**, three sisters, the **Moirai**, spin, measure and cut the thread of every life. - In **Śūnyapura**, the scribe **Chitragupta** records every soul's deeds for the lord of death, and the land's mathematicians gave the world **zero**. - In **the River Empire**, the underworld's Ten Courts keep the **Book of Life and Death** (生死簿), and a Judge holds the brush that writes in it. - In **Hyakki**, the King of Hell, **Enma**, keeps a ledger of the dead, and objects that live a hundred years wake up with souls. - In **the Crossroads Territory**, a stranger waits where two roads meet at midnight with a black book, and anyone can sign a deal for luck.

None of these ledgers agreed. Luck meant karma in one land, thread in another, a contract in a third. The world was messy, plural and alive. 3. **The Basilica.** Meridian, a young trading capital at the centre of the Tally, sent actuaries to every land. Over three centuries they **copied, bought or seized each ledger** and bound them into one brass engine, **the Basilica**, which recomputes one Omen for every soul at sunrise (**the Dawn Tally**). One number for everyone; one law for luck. In TR 702 the engine's heart closed around **Hap itself** (§2A); since then the god of luck is not

worshipped in Meridian, it is *counted*. Meridian became the capital of the world because it holds the merged Ledger. 4. **The Rounding.** Each Dawn Tally divides luck into whole numbers. The fractions, every soul's tiny remainder, are rounded down and dropped. It has happened every day for three hundred years. Nobody noticed. 5. **The Deficit (now).** For forty years the world's total luck has been leaking. **Noise storms** tear open where the books don't balance: patches of pure chance where numbers scramble and glitch-creatures (**Noiselings**) crawl out. Theologians predict **the Default**, the day the Basilica fails. The truth is worse: the Deficit is Hap starving in its cage, and the Default would be the god's death (§2A).

The secret (midpoint reveal, Act III, in Śūnyapura, the land that invented zero): the rounded-down fractions didn't vanish. Three centuries of everyone's lost remainders pooled into a hidden account, and that account became a person. **NIL is the Remainder**, the world's missing luck, embodied. NIL has no Omen because NIL isn't *in* the Ledger. NIL *is what the Ledger owes.*

The final secret (Finale, inside the Basilica): the remainders were shaved from **Hap**. NIL is the last free piece of the god of luck. Settling NIL would complete the count, and Fixed Fate would freeze Hap forever.

4. The villain: Comptroller Lucan Thale

Role	Comptroller of Meridian, master of the Basilica
Look	Tall, ageless; immaculate white-and-brass vestments shaped like a clockwork cathedral; a monocle with a turning gear; his Omen frozen at **50** (the perfect median), rendered in steel, not light
Voice	Calm, kind, exhausted. Narrates the intro myth.
Backstory	Born in Meridian's slums. At twelve, a Noise storm took his family and his street in one minute, by pure chance. He became the greatest actuary in history to make sure chance never decides anything again.
His plan: the Final Audit	1) Capture the Remainder (NIL) and settle it back into the Ledger, closing the Deficit. 2) Freeze every Omen forever ("Fixed Fate"): no luck, no loss, no Noise.
Why he's not wrong	He would save the world from the Default.
Why he's wrong	He flattened every culture's meaning of fortune into one number, and now he would freeze it. His plan also erases a person.
Line	*"Thread, karma, contracts, books of the dead: a hundred superstitions, and every one of them let people die at random. I replaced them with arithmetic."*

How he hunts NIL: the Auditors and the Seals. Each land's bound ledger is locked into the Basilica by a **Seal**, and each Seal is guarded by that land's **Auditor**, a local power who made a deal with Thale. Deliver the Remainder, and your people are protected (frozen high) under Fixed Fate. The upper gate of Meridian opens only with **seven Seal Digits**: one from each of the six Auditors, and **the seventh is NIL's own lucky number**, the one the player chose in the intro.

Why the party fights: 1. NIL is hunted: every Auditor wants the bounty. 2. The gate needs the Seal Digits. 3. Each land suffers under its Auditor's deal, and breaking a Seal returns that land's own idea of luck to its people. 4. Each companion comes from one land and has a personal stake there. 5. If the Final Audit happens, NIL is erased and fate is frozen.

5. The lands: a journey across the world's mythologies

The Tally is a ring of lands around Meridian's plateau. The upper city is sealed, so to go up, you must go around.

ACT	LAND	INSPIRED BY	ART STYLE	LEDGER HELD THERE	LAW OF THE LAND	AUDITOR (BOSS)	SEAL
Prologue	**Meridian Undercity** (the Remainder Wards)	A cosmopolitan modern capital	Anime cyber-noir: rain, neon, Nixie signage	The Basilica itself	*Debt*: everyone owes luck	**Varro Kade, the Ledgerhound**	1
I	**Hyakki (百鬼), the Isles of a Hundred Demons**	Japanese folklore	*Ukiyo-e* woodblock prints + anime: flat colour, bold outlines, Hokusai waves, torii, lantern light	**Enma's Ledger** of the dead	*The Rule of Shi and Ku*: 4 and 9 are cursed; 7 is blessed	**Tamamo, the Nine-Tailed Auditor**	9
II	**The River Empire**: Bāfā Harbor (八发港) above, the Ten Courts (十殿) below	Chinese folk religion and festival culture	Ink-wash (水墨) for the underworld; *gongbi* (工笔) fine line and gold leaf + neon for the harbor	**The Book of Life and Death** and the Judge's Brush	*The Rule of Interest*: luck compounds	**Chancellor Jin Chan (金蟾), the Golden Toad**	8

ACT	LAND	INSPIRED BY	ART STYLE	LEDGER HELD THERE	LAW OF THE LAND	AUDITOR (BOSS)	SEAL
III	**Śūnyapura, the City Where Zero Was Born**	Indian mythology, astronomy and mathematics	Indian miniature painting and Pattachitra outlines + anime; monsoon light; giant stone observatory instruments; geometric stepwells	**Chitragupta's Ledger** of deeds (karma)	*The Rule of Karma*: every action is recorded and returns	**Rahu and Ketu, the Eclipse**	0
IV	**Kroton Nova, the Sieve of the Aegean**	Greek mythology and Pythagorean number mysticism	White marble, Aegean blue, mosaic and red-figure-pottery motifs + anime	**The Thread of the Moirai**	*The Rule of Indivisibility*: only primes pass	**Archon Seventeen** of the Pythagorean Synod	2
V	**The Crossroads Territory**	American folklore: frontier tall tales, the blues crossroads legend, immigrant charms, desert neon	Americana: desert sunsets, Route-66 motels, pulp-poster lettering, art-deco casino neon	**The Black Book of the Crossroads**	*The Rule of the House*: the night loops, and every deal has a price	**The House (Octavian Nine, "Mr. Eighty-Nine")**	7
Finale	**Meridian & the Basilica**	—	Brass clockwork cathedral, monochrome + Nixie amber; every land's art style appears on its wall	All of them	*Fixed Fate*	**Comptroller Lucan Thale**	*yours*

Gods are not the villains. Revered figures (Enma, Chitragupta, the Judge of the Ten Courts, the Moirai) appear as **neutral keepers who test the party**, never as enemies to beat up. The Auditors are mortals, folklore tricksters or legendary monsters who made deals with Thale.

Luck and superstition by region (what each land believes, and how it plays)

LAND	WHAT LUCK MEANS HERE	LUCKY	UNLUCKY	EVERYDAY CUSTOMS	BECOMES IN PLAY
Meridian	A number you own, rent or owe	7, 77, 99	13 (floors skipped), 00	Charms of every culture on one door; kissing sevens; selling "luck-ins"	*Debt*: the Collectors' interest; Charmpunk shops sell every set
Hyakki	A relationship with spirits; old things wake up	7 (Seven Lucky Gods), 8 (*suehirogari*, widening fan)	4 (*shi*, death), 9 (*ku*, suffering)	Daruma eyes, omamori, beckoning cats, throwing salt, no room 4	*Rule of Shi and Ku*: 4s and 9s curse the striker; tsukumogami faces awaken after 100 stops
River Empire	Prosperity that compounds; harmony between the living and the dead	8 (*fā*, prosper), 6 (*liù*, smooth), 9 (*jiǔ*, lasting)	4 (*sì*)	Red envelopes, the money toad, oracle sticks, burning joss paper for ancestors	*Rule of Interest*: Mult compounds; ancestors' gifts at shrine nodes
Śūnyapura	Deeds and their return (karma), auspicious timing	0 (the place that holds everything), 108, 9 (*navagraha*)	Inauspicious hours during the eclipse	Lemon and seven chillies, rangoli, the nine gems, auspicious hours	*Rule of Karma*: parries and Zero-Sums bank Karma; every action returns
Kroton Nova	Purity and proportion; the thread the Moirai spin	Primes, 2 (the even prime), the golden ratio	17 (VIXI), composites (in the Synod's eyes)	Evil-eye beads, spitting three times, Fortuna's wheel	*Rule of Indivisibility*: composite strikes weaken; primes pierce
The Crossroads	A deal: luck is bought, borrowed and owed	7, 11, 21	13, the dead man's hand (aces and eights)	Horseshoes open end up, rabbit's foot, lucky penny, never count money at the table	*Rule of the House*: the night loops; Deals and Wagers
The Basilica	Arithmetic; luck abolished	50 (the median)	Remainders, NaN	Clocks strike 0 at dawn; children perform the Naming Rite at seven	*Fixed Fate*: wheels slow and freeze; the finale takes luck away

Land notes (enemies, set pieces, superstitions)

Meridian Undercity (Prologue) - Enemies: Debt Hounds (cyber-dogs), Collectors (armoured bailiffs), Noiselings (glitch sprites). - Set piece: the Dawn Tally beam sweeping the slums. Its immigrant quarters mix every land's charms on one street. - Superstition: "Don't let a collector read your number twice."

Hyakki (Act I, Japan-inspired) - Enemies: *Kasa-obake* (one-eyed hopping umbrella spirits), *Chōchin-obake* (lantern ghosts), *Kappa* (river imps who lose their strength if they bow), *Oni* enforcers. - Set piece: the **Hyakki Yagyō**, the Night Parade of One Hundred Demons, crossing a red bridge under a full moon; a shrine of the **Seven Lucky Gods**. - Superstitions: no room 4 or 9; daruma dolls get one eye painted when you make a wish and the other when it comes true; the beckoning cat (*maneki-neko*).

The River Empire (Act II, China-inspired) - Bāfā Harbor enemies: Abacus Golems, Debt Junks (armed river boats), Golden Koi spirits (multiply when hit). - Ten Courts: Ox-Head & Horse-Face (牛头马面) are the Courts' wardens and appear as a **trial**, never as villains (they are satisfied only if both are bested in the same round); enemies are Hungry Ghosts (they eat your fen), Jiangshi and Basilica auditors posted below. - Set pieces: the Lantern Festival over the Luck Exchange; crossing Naihe Bridge past Meng Po's pavilion of forgetting. - Superstitions: 8 for prosperity; red envelopes; the three-legged money toad; burning joss paper for the dead.

Śūnyapura (Act III, India-inspired) - Enemies: Vetālas (riddling corpse-spirits from the *Vikram and Betaal* tales, who ask a question before they fight), Brass Yantra Guardians (animated observatory instruments), Monsoon Nāgas (serpent spirits of the stepwells). - Set pieces: a city-sized open-air observatory of stone sundials and astrolabes; a vast geometric stepwell descending into the dark; the Hall of Chitragupta with palm-leaf records to the horizon. - Superstitions: a lemon and seven green chillies at the door (*nimbu-mirchi*) against the evil eye; 108 prayer beads; the nine planetary gems (*navaratna*).

Kroton Nova (Act IV, Greece-inspired) - Enemies: Bronze Automata (Talos-like), Harpies, Composite Knights (the Synod's armoured guard), Sirens who sing numbers. - Set pieces: a white island city built as a giant stone sieve, where citizens whose Omens are composite fall through the holes; the cave where the three Moirai spin. - Superstitions: the blue evil-eye bead (*mati*); spitting three times against bad luck; Fortuna's wheel; unlucky 17 among the Italian-descended merchants.

The Crossroads Territory (Act V, America-inspired) - Enemies: the Headless Horseman of the hollow, Card-Sharp Automata, Tumbleweed Golems, a Ghost Train whose cars are Loop Echoes of your party. - Set pieces: a desert crossroads at midnight under one streetlamp; the casino boomtown of Lucky Strike where the same night repeats; a dust storm made of playing cards. - Superstitions: horseshoes over doors (open end up), rabbit's foot, lucky penny, four-leaf clover, "seven come eleven", never count your money at the table.

The Far Lands (expansion plan, not the main game)

Each a short post-launch act with its own ledger, built with cultural consultants: - **Norse North**: the three Norns at the Well of Urðr; nine worlds; runes. - **Kemet** (Egypt): the Weighing of the Heart against Ma'at's feather; 42 assessors. - **The Long Count** (Maya): the other civilisation that invented zero; a base-20 calendar. - **Anansi's Coast** (West Africa, Akan folklore): the spider who bought all the world's stories from the sky god. - **The House of Wisdom** (Islamic Golden Age): al-Khwārizmī, whose name gave us "algorithm"; the *hamsa*.

6. The cast

NIL, the protagonist (the Remainder)

Digit	The player's choice (0–9)
Look	~19, androgynous; silver-white hair with one Nixie-orange streak; amber eyes; a broken holographic **NaN** where an Omen should be; long asymmetric black actuary's coat with brass ledger clasps; a cracked brass **decoder** fused into the right wrist that leaks orange light and floating digits
Personality	Quiet, observant, dry humour; counts things when anxious; fascinated by other people's numbers because they don't have one
Wants → Needs	Wants a number and a normal life. Needs to accept being what everyone lost, and choose what to give back.
Arc	Prologue: hunted nobody → I: learns the Basilica cheats even the dead → II: learns luck is power and debt → III: **learns they are the Remainder** → IV: rejects the idea that some numbers are worth less → V: refuses to loop or hide → Finale: decides the world's future
Growth (the most important in the game)	Each Seal returns a **memory digit**: NIL's fragments are other people's stolen remainders, so each flashback is someone else's lost moment of luck, in the art style of that land. NIL gains a new verb per act (GDD §6).

The roster (8 companions; NIL + 2 in the party)

HERO	DIGIT	FROM	RECRUITED	CLASS	ONE-LINE
Vesper Sette	7	Meridian (raised in the Crossroads casinos)	Prologue	Gunslinger	Ex-croupier who rigged games for the House, until a rigged hand Settled a child
Maeve Triskel	3	Meridian's Celtic immigrant quarter	Prologue (optional)	Charm-smith	Blacksmith's daughter who forges horseshoes and clovers into weapons
Hina of the Ninety-Nine (九十九 雛)	9	Hyakki	Act I	Tsukumogami trickster	An old paper umbrella that woke up with a soul, one year short of a hundred
Ba Jinling (八金铃)	8	River Empire	Act II	Financier	Loan shark who secretly forgives the debts she buys; the Golden Toad's daughter
Bai Si (白四)	4	River Empire (Ten Courts)	Act II	Executioner	The underworld's headsman, who keeps count so no dead soul is lost

HERO	DIGIT	FROM	RECRUITED	CLASS	ONE-LINE
Shunya (शून्य)	0	Śūnyapura	Act III	Astronomer	Young mathematician at the great observatory who proves what NIL is
Prima of Kroton	2	Kroton Nova	Act IV	Paladin	The Synod's "even prime" novice, the odd one out
Fibi Leone	Fibonacci	The Crossroads	Act V	Dealer	Card prodigy trapped in the looping night, the only one who remembers

Each hero has a 4-scene **bond arc** (Teahouse/Campfire events) that unlocks their Awakening and affects the true ending. Each also carries a **piece of their land's ledger** that matters at the climax.

Vesper Sette (7)

- **Look:** tall, crooked grin, scar across the lip; black hair with a crimson underlayer in a long braid; crimson long coat over a croupier's waistcoat; wide-brim hat with cards in the band; a revolver with a **seven-chamber cylinder** engraved 1–7.
- **Personality:** dry bravado, gambler's charm, guilt hidden behind jokes. Flips a coin before every decision and ignores the result.
- **Backstory:** dealt for the House in Lucky Strike. Ran when a hand she rigged Settled a twelve-year-old.
- **Bond arc:** teaches NIL to read tells → admits the rigged hand → faces the House → deals an honest hand.
- **Ledger piece:** the House's marked deck, proof the Crossroads deals were rigged.

Maeve Triskel (3)

- **Look:** freckled, red hair in three braids; leather apron over work clothes; **horseshoe-knuckle gauntlets**; a four-leaf-clover pin; a triskele (triple spiral) burned into her forge hammer.
- **Personality:** loud, warm, stubborn, superstitious about everything and proud of it.
- **Backstory:** her family fled to Meridian when their land's luck was "rounded" into the Basilica; her grandmother's charms stopped working that day. She makes new ones.
- **Role:** upgrades charms at the forge (crafting tie-in); comic warmth for the party.
- **Ledger piece:** a horseshoe forged from the first Basilica gear, which can hold one Seal open.

Hina of the Ninety-Nine (九十九 雛, 9)

- **Look:** small girl-shaped spirit; a **red-and-white paper umbrella** she rides and fights with (sometimes it opens a single eye and sticks out its tongue, like a *kasa-obake*); short black bob; furisode kimono patterned with numbers 1–99; geta sandals.
- **Personality:** mischievous, ancient and childish at once; obsessed with turning 100.
- **Backstory:** a 99-year-old umbrella left at Enma's shrine. In Japanese folklore objects that reach 100 years gain a soul (*tsukumogami*). Hina woke up early, at 99, because she was soaked in a century of other people's wishes. The Basilica's Fixed Fate would freeze her at 99 forever: never quite real.
- **Bond arc:** NIL helps her decide she's already real.
- **Ledger piece:** a torn page of Enma's ledger stuck in her umbrella ribs.

Ba Jinling (八金铃, 8)

- **Look:** modern jade-and-gold qipao; eight tiny golden bells in her hair ornaments and anklets; a **folding fan whose ribs are abacus rods**; smoky violet eyes; red lips.
- **Personality:** theatrical, sly, mercenary on the surface, enormous heart.
- **Backstory:** Chancellor Jin Chan's daughter. She left the Luck Exchange when she learned how his fortune was built.
- **Ledger piece:** her father's true account book, the proof of the Exchange's fraud.

Bai Si (白四, 4)

- **Look:** very tall, pale; long white hair bound with black funeral ribbon; white mourning robes (white is the mourning colour in Chinese tradition) inked with 四; a huge guandao-style blade hung with four paper talismans; joss paper swirls around him.
- **Personality:** stoic, formal, honour-bound, unexpectedly tender with children.
- **Backstory:** the Ten Courts' executioner, who counts every Settled soul so none is forgotten.
- **Turn:** NIL's decoder shows him three centuries of rounding: his count has always been short. He joins after Judge Cui's trial.
- **Ledger piece:** the true count of the dead.

Shunya (शून्य, 0)

- **Look:** young woman astronomer; dark hair in a long braid with brass astrolabe pins; indigo-and-saffron layered clothing with mirror-work; a **brass astrolabe** she spins like a weapon; palm-leaf manuscripts tied at her hip; ink-stained fingers.
- **Personality:** brilliant, impatient, joyful about ideas, kind to people who are slow.
- **Backstory:** works at the great observatory where her ancestors formalised zero. She calls 0 "the place that holds everything", not "nothing". She has been tracking the Deficit for years and found that it equals a *person's worth* of luck.
- **Role:** delivers the midpoint reveal to NIL with gentleness and maths.
- **Ledger piece:** the proof: the Remainder equation.

Prima of Kroton (2)

- **Look:** short blonde hair, blue eyes; white-and-gold armour with Greek-key borders; a **giant shield shaped like a sieve**, set with blue glass; a halo of floating glyphs 2-3-5-7.
- **Personality:** earnest, loud, honest to a fault; comic relief who becomes the moral backbone.
- **Backstory:** the Synod calls 2 "the even prime", the odd one out. She has always felt like a mistake in a holy system.
- **Bond arc:** discovers the Sieve is a culling machine; must face her mentor, Archon Seventeen.
- **Ledger piece:** a strand of the Moirai's thread, given to her by Lachesis.

Fibi Leone (Fibonacci)

- **Look:** 18; golden hair in spiral curls; spiral-patterned irises; tailored croupier tailcoat with headset; glowing cards that flow around her in a golden spiral.
- **Personality:** fast-talking, anxious, brilliant; laughs at bad moments.
- **Backstory:** dealer at the House, the only person in Lucky Strike who remembers the night repeats. She has lived it 1,597 times (a Fibonacci number).
- **Twist:** the House loops the night because Fibi dies at dawn. Octavian Nine is her grandfather.
- **Ledger piece:** the last page of the Black Book: her grandfather's own deal.

The Auditors (bosses) and the keepers (trials)

BOSS	LAND	LOOK	MOTIVE	FIGHT (THE RULE)	AFTER DEFEAT
Varro Kade, the Ledgerhound	Meridian	Hulking bailiff, hound-skull mask, debt jars on his belt	Wants the bounty to buy his sister's exemption	**The Collector:** steals your last reel digit into a debt jar; 3 parries shatter it	Spared; returns in Act V
Tamamo, the Nine-Tailed Auditor	Hyakki	A nine-tailed fox in courtly layered robes; foxfire (*kitsunebi*) floating around her; each tail tipped with a glowing digit	Yokai fade when people stop believing in them; Thale promised Fixed Fate would make yokai permanent	**Nine Tails:** each tail has its own small Balance; foxfire **illusion faces** appear on your wheels (they show one digit and land as another); landing a 9 hurts you (*ku*) unless Hina is in the party	Retreats into legend; leaves Hina a tail-blessing
Chancellor Jin Chan, the Golden Toad	River Empire	Enormous jovial lord robed in gold coins on a three-legged toad throne	His bank will hold the world's frozen luck	**Interest:** his Balance grows every round; he taxes your Jackpots; phase 2 he swallows a hero (land a Pair to free them)	Bankrupted; Jinling takes over the Exchange
Rahu & Ketu, the Eclipse	Śūnyapura	Rahu: a vast severed head with a crown, devouring light; Ketu: a headless serpent body with a comet's tail	Cut in two in the oldest story, they want Fixed Fate to make their stolen immortality permanent and whole	**The Eclipse (two bodies, one fight):** Rahu swallows one reel tube each round (it goes dark); Ketu strikes with a blind random wheel. If their Balances drift more than 10% apart they rejoin and heal, so keep them **balanced**: Zero-Sum hits both equally.	Returned to the sky as a passing eclipse
Archon Seventeen	Kroton Nova	Gaunt, beautiful, terrifying; a laurel of bronze and a sieve-shaped staff; his Omen is 17	Faith: the Final Audit is purification; only the indivisible deserve to remain	**The Sieve:** each round he strikes digits off *your* wheels (multiples of 2, then 3, then 5…, like the Sieve of Eratosthenes); composite strikes are halved	Falls through his own sieve; Prima saves him
The House, Mr. Eighty-Nine	Crossroads	Elegant old croupier in a white suit, ouroboros tattoo around his throat, a pocket watch running backwards; meets you at the midnight crossroads	Loops the night so his granddaughter never dies at dawn; financed Thale	**The Loop:** every 3 rounds time rewinds, but your reel and charms persist; he copies your last pattern; mid-fight he offers you a **deal**	Lets the dawn come; says goodbye to Fibi
Comptroller Lucan Thale	Meridian	see §4	Fixed Fate	**Three phases:** Regression to the Mean (wheels compress toward 5) → Fixed Fate (wheels stop spinning: a puzzle) → Zero (NIL alone; all companions return as Link Arts)	Depends on ending

Keepers (trials, not villains): Enma in Hyakki (judges whether NIL may pass the shrine) · Judge Cui in the Ten Courts (tests NIL for the Brush) · Chitragupta in Śūnyapura (reads NIL's empty page) · the Moirai in Kroton Nova (Clotho, Lachesis and Atropos each pose one challenge) · the Stranger at the Crossroads (offers the deal). These are elite "trial" fights with special rules and no hatred.

7. Plot beats by act

Prologue: "NaN" (Meridian Undercity) 1. *Intro cinematic* (§8). Dawn Tally; NIL interferes; the Naming Rite; **the player states their lucky number**; REMAINDER DETECTED. 2. Tutorial Audit vs Debt Hounds, alone. 3. Vesper saves NIL ("Need a hand? Seven bucks."). Optional: Maeve's forge in the Celtic quarter, where she joins after you help defend it. 4. Varro corners them at the drain gates. Boss. Spared. 5. **World-map reveal:** Vesper explains the seven Seals; six lands light up on the map in six art styles. Seal 1.

Act I: "The Night Parade" (Hyakki) 1. Arrival by ferry under a Hokusai wave. The Night Parade crosses the bridge, and the party hides inside it. 2. **Hina** jumps out of a shrine's umbrella stand and attaches herself to NIL ("You smell like a hundred wishes"). 3. Shrine of the Seven Lucky Gods (charms); kappa river crossing; **Enma's trial** at the gate of the dead. 4. **Tamamo** in her fox-fire palace. Boss. 5. Tamamo, defeated, asks NIL what a yokai is worth without belief. Seal 9. 6. *Epilogue:* Thale in the Basilica: "The Remainder is learning to count."

Act II: "Eight Bells and the Book of the Dead" (River Empire): the Lantern Festival and the Luck Exchange; Ba Jinling buys the party's bounty and hides them; descent through the river's end into the Ten Courts; **Judge Cui's trial** grants the Brush; **Bai Si** joins; return to face **Jin Chan**; father and daughter. Seal 8.

Act III: "Śūnya" (Śūnyapura): monsoon arrival; Vetāla riddles; **Shunya** at the observatory; **Chitragupta's Hall**, where NIL's page is blank; **the reveal: NIL is the Remainder**; **Rahu & Ketu** eclipse the city. Seal 0.

Act IV: "Indivisible" (Kroton Nova): Prima defects when the Sieve drops a school; **the Moirai's cave** (three trials); **Archon Seventeen**. Seal 2.

Act V: "1,597 Nights" (Crossroads Territory): the midnight crossroads deal; the same night repeating; **Fibi** remembers; Varro returns to help; Vesper's honest hand; **the House** lets dawn come. Seal 7.

Finale: "Zero-Sum" (Meridian): the gate takes six digits and the seventh is **the player's lucky number**. Inside the Basilica, the walls show every land's original ledger. Thale's three phases. The choice.

8. The intro (shot list; full prompts in `art/HIGGSFIELD_PROMPTS.md`)

Total ≈ 2 min 30 s; skippable after first view. The opening montage cycles through **every land's art style** and doubles as the game's best trailer.

#	BEAT	STYLE	LENGTH
1	**"Before numbers, there was the Noise."** Floods, falling towers and hail of dice. Thale narrates.	Abstract ink and static	8 s
2	**"Every people kept its own accounts with fate."** A montage: the Moirai spin a glowing thread (red-figure pottery style) → Chitragupta writes on palm leaves (Indian miniature) → a judge writes in the Book of Life and Death (ink-wash) → Enma opens his ledger as a lantern parade passes (ukiyo-e) → a hand signs a black book at a midnight crossroads (Americana woodcut).	Five art styles, 4 s each	20 s
3	**"Then Meridian built one machine for all of them."** Brass chains pull each ledger into the Basilica; gears swallow them; the Dawn Tally beam sweeps the world and a number appears over every head. "Every soul a number. Every number a share."	Brass clockwork	12 s
4	**Title card:** ZERO-SUM: OMEN	Nixie	4 s
5	**Dawn in the Undercity.** The beam sweeps neon slums; Omens above the crowd tick up and down. A mother's number drops to **01**; her son stares at it.	Anime cyber-noir	15 s
6	**Varro's collectors** arrive to settle her. The crowd parts. One hooded figure doesn't: **NIL**, with **NaN** flickering above.	—	12 s
7	NIL reaches out. **Digits spin** around the mother's 01; NIL *stops* them on 07. The collectors freeze: "That one has no number."	—	10 s
8	Chase through alleys full of every land's charms (horseshoes, chilli-and-lemon strings, beckoning cats, evil-eye beads); NIL ducks into a black-market shrine with a jury-rigged **Naming terminal**. "Every soul needs a number, child."	—	15 s
9	**Interactive: "STATE YOUR LUCKY NUMBER."** The player picks 0–9; each digit shows what it means in the lands of the Tally.	In-engine	player-paced
10	The terminal overloads. Across the city every screen and Omen glitches to **REMAINDER DETECTED**; the Basilica's eye opens; bounty posters print NIL's face.	—	12 s
11	Logo slam with the chosen digit, then straight into gameplay: Debt Hounds leap from the rooftops.	In-engine	4 s

9. Endings

ENDING	CONDITION	WHAT HAPPENS
Balance	Return the Remainder	NIL dissolves into the Ledger and Hap's cage is fed. Every Omen rises by one; the Deficit closes; Hap stays caged, counted and quiet. The party visits a grave with no number.
Overflow	Break the Basilica	Hap bursts free all at once. Every Omen becomes NaN; the Noise returns and nobody can see anyone's luck: freedom and chaos. Each land returns to its old ways, for better and worse. NIL fades into the storm.
Positive-Sum (true ending)	All 8 bonds complete; every companion's ledger piece kept	The party puts every land's ledger piece back into the Basilica *together*: thread, karma, the Book and the Brush, Enma's page, the crossroads contract, Maeve's horseshoe. The machine stops forcing one law and starts translating between many. Hap's cage opens into an **Open Hand**: the god deals again, but through every land's ledger at once. Shared luck stops being conserved and starts **multiplying**. The game's synergy system (heroes worth more together than apart) becomes the world's new law. Thale lives to see it. In the last shot, Hap gives NIL a number of their own: the NaN over NIL's head resolves into **the player's lucky number**.
Divide by Zero (secret)	NIL's digit is 0 and the final hit lands 0-0-0	The Basilica crashes, the screen blue-screens, and the credits roll in debug text.

The theme in one sentence: *a world that flattened every culture's idea of luck into one zero-sum number learns, from a party drawn from all of them, that luck shared isn't zero-sum.*

10. Cultural care

- Each land is **inspired by** a real culture and is not a depiction of a real country. Names are invented or drawn from widely told public folklore.
- **Revered deities and sacred figures are never enemies.** Enma, Chitragupta, Judge Cui and the Moirai are keepers who test the party with respect. Villains are mortals, legendary monsters from the stories themselves (Tamamo, Rahu and Ketu, the Headless Horseman), or Thale's officials.
- Rahu and Ketu are also venerated as celestial bodies in Hindu astronomy and astrology. Present them as tragic, cosmic and awe-inspiring, never as cheap demons, and have the depiction reviewed.
- **Avoid Indigenous sacred traditions** in the American act (no dreamcatchers, skinwalkers or sacred ceremonies). The Crossroads draws on frontier tall tales, the blues crossroads legend, immigrant superstitions, ghost towns and Sleepy Hollow.
- **Hire or consult reviewers from each culture** (Japanese, Chinese, Indian, Greek, American) for names, scripts (kanji, hanzi, Devanagari, Greek), clothing and terms before any public release. Localise to Japanese, Simplified and Traditional Chinese, and Hindi post-jam.
- The casino is aesthetic only: no real-money mechanics or purchasable randomness.

How the Tally actually works: the logic of luck and power, the kinds of power and how people use them, factions and their styles, jobs, the economy, stores, law and politics, and the full history. Canon sources: `WORLD_BIBLE.md` (Hap is §2A), `LORE_LEXICON.md`, `STORY.md`. Where this file adds detail, it must never contradict those.

1. The logic of luck (the physics of the Tally)

RULE	WHAT IT MEANS	CONSEQUENCE PEOPLE LIVE WITH
1. Luck is real and finite.	Luck is the substance Hap dealt into the world: the chance that things go your way. Under the Basilica it is **conserved**: the total never changes.	For your luck to rise, someone else's must fall. Every gain is a transfer.
2. Every soul holds a share.	The Basilica divides the caged god into shares every dawn. Your share is your **Omen**, 00–99, floating over your head.	Everyone can see everyone's luck. It decides loans, jobs, marriages and how dangerous a staircase is for you today.
3. Luck acts through numbers.	Luck resolves events by the **digits** they produce. A dropped coin, a dice roll, a door code, a clock time: each is a little lottery the world runs. Higher Omens tilt them.	Numbers are everywhere because numbers are where luck *happens*.
4. Resonance.	At the Naming Rite (age seven) each soul states a lucky digit. That digit **resonates**: luck flows more easily through it.	People wear their digit, choose house numbers, wedding dates and jobs by it.
5. Stopping.	Auditors can summon a **Fate Wheel** (a visible strip of the digits an event could produce) and **stop** it, choosing which digit the event takes. Skill and resonance decide how well.	This is the only "magic" in the world, and it's licensed (§3).
6. Settlement.	At Omen 00 a soul is **Settled**: misfortune finds them within days and their remaining luck flows back to the pool.	Debt collectors drain Omens; a 01 is a death sentence with a short delay.
7. The Rounding.	Every Dawn Tally rounds each share down to a whole number. The fractions vanish… into NIL.	Nobody noticed for three centuries.
8. Noise.	Where the books don't balance, luck stops being dealt and starts **boiling**: Noise storms scramble numbers and spawn Noiselings.	The Deficit is growing; storms are more frequent every year.

What luck is not: it isn't strength, intelligence or virtue. A 99 can still be a fool; a 03 can still be a hero. Luck only decides how often the world's coin lands your way. The game's theme lives here: luck shared multiplies, luck hoarded rots.

How a Fate Wheel works (in-world explanation of the core mechanic)

1. An **Audit** begins when two Omens contest the same outcome (a fight, a deal, a debt).
2. Each Auditor's will draws a **wheel** of digits from the event. Faces are the possible outcomes; a **face's width is its odds**, shaped by the Auditor's Omen, resonance and tools (charms, enhanced faces).
3. The **tens** of every strike are fixed by training and gear (*tens are safe*); the **ones** are fate and must be stopped (*ones are fate*).
4. Three stopped digits fall into the **reel**, the shared ledger of the moment. When the digits form a pattern, the world itself pays out: a Pair is a coincidence, a Jackpot a miracle, a Zero-Sum a perfectly balanced book that cancels the other side's next move.
5. The loser's staked luck (**Balance**) drains toward the winner's side. Nobody dies in an Audit; they are **rounded down**, and may Settle later.

2. Power types (the six schools of Auditing)

Every land learned to stop numbers its own way before the Basilica licensed one method. In play, each school is a hero kit flavour and an enemy behaviour.

SCHOOL	LAND	SOURCE OF POWER	HOW IT LOOKS	SIGNATURE TECHNIQUES	PRACTITIONERS
Ledger Auditing (licensed)	Meridian	Brass wrist-wheels that tap the Basilica's count	Steel numerals, Nixie amber light, ledger lines	Debt Jar (steal a reel digit), Seizure (freeze a face), Interest	Basilica Auditors, Collectors, Varro, Thale
Shrine Binding	Hyakki	Belief: charms, shrines and old objects with souls	Paper talismans, foxfire, ukiyo-e waves	Omamori wards, illusion faces (foxfire), tsukumogami awakening	Shrine keepers, yokai, Hina, Tamamo
Brush Writing	River Empire	The written word in the Book of Life and Death	Ink strokes, gold leaf, red seals	Ink (rewrite a digit), Interest (compounding), Oracle sticks	Judge Cui's clerks, Exchange brokers, Jinling, Bai Si
Karma Weaving	Śūnyapura	Deeds recorded and returned; zero as the place that holds everything	Geometric light, rangoli patterns, brass instruments	Karma (bank and spend), Place Value, eclipses	Observatory astronomers, Shunya, Rahu & Ketu
Sieve Geometry	Kroton Nova	Number properties: primes, ratios, the thread's measure	Marble, mosaic glyphs, white light through holes	Prime pierce, Sieve (strike out faces), Factor	Synod knights, Prima, Archon Seventeen
Dealing	Crossroads	Contracts: luck bought, borrowed and owed	Cards, neon, dust, the black book	Deals, Wagers, Loops	The Stranger, croupiers, Fibi, the House

SCHOOL	LAND	SOURCE OF POWER	HOW IT LOOKS	SIGNATURE TECHNIQUES	PRACTITIONERS
The Remainder (unique)	—	The god's lost change	NaN glitches, orange digits that refuse to round	Splice, Carry, Divide by Zero	NIL only

Licensing. Since TR 760 the Basilica licenses only Ledger Auditing; every other school is "folk practice": tolerated in its home land, illegal in Meridian unless registered. Unlicensed Auditing in Meridian is a debt offence (your Omen is fined).

3. How people use their powers (daily life)

USE	WHO	HOW
Protection	Everyone	Charms widen good faces a little for everyday lotteries (slips, accidents, exams). Weak everywhere, strongest in their home land.
Lending luck	Luck banks, the Exchange	Lend Omen points at interest; repayments are drained at dawn by licensed Auditors.
Insurance	Insurers (Meridian)	Pool many small Omens to cover one big misfortune; premiums are paid in luck-ins.
Law	Courts	Disputes between Omens are settled by **Audit** under a magistrate (a formal, non-lethal duel).
Medicine	Clinics	Doctors can't create luck, but **Omen clinics** move luck from donors to the critically ill ("luck transfusion"), for a price.
Work	Trades	Builders, pilots and miners need high Omens for safety; low Omens are pushed into dangerous jobs, which is how the Remainder Wards grew.
Games	Casinos, festivals	Legal small Audits for show; big Audits for money are illegal in Meridian, legal in Lucky Strike.
Faith	Temples, shrines, the outlawed Open Hand	Offerings ask for a land's way of luck; the Open Hand leaves a die in an open palm for Hap.

4. Factions and their styles

FACTION	SEAT	WANTS	POWER	STYLE (MATERIALS, COLOURS)	UNIFORM / LOOK	RANKS	TOOLS & WEAPONS
The Basilica	Meridian	Order; the Final Audit	Ledger Auditing; the Dawn Tally	Ledgerism: brass, steel, white enamel, cathedral arches; steel numerals	White-and-brass vestments, gear-crested mitres (priests); olive field coats (Auditors)	Comptroller → Ledger-priests → Senior Auditors → Clerks → Novices	Wrist-wheels, chained ledgers, Seal stamps
The Collectors	Meridian	Debts paid; bounties	Licensed debt Auditing	Rust armour, gunmetal, glass debt jars with orange glow	Rust trench armour, peaked caps, hound-skull masks for captains	Chief Collector → Captain (masked) → Collector → Hound handler	Debt jars, chains, Debt Hounds
Charm Bazaar guilds	Meridian	Trade, survival, folk luck	Charm-smithing (every culture's)	Charmpunk: plastic gold, neon, stickers, every culture at once	Puffer jackets hung with charms; painted numbers; fake holo-halos	Guild masters → stallholders → runners	Charms, Nixie pendants, forged horseshoes
The Remainder Wards (not a faction: a people)	Meridian Undercity	To get through the week	Mutual aid; luck-ins	Remainder: scrap brass, patched cloth, salvaged Nixie tubes, rope	Ponchos, patched coats, a cracked tube on a cord instead of a glowing Omen	Elders, block captains	Tools, salvage
The Yokai Courts	Hyakki	To keep existing	Shrine Binding; belief	Ukiyo-e flatness, vermilion, foxfire cyan-violet, paper and lacquer	Court robes, fox ears and tails for kitsune; tsukumogami are their objects	Tamamo → kitsune retainers → yokai of every kind	Foxfire, illusion faces
The Luck Exchange	Bāfā Harbor	Profit	Brush Writing for contracts; interest	Gilded Fortune: jade, gold leaf, red lacquer, live price boards	Jade mandarin-collar suits, gold trim, pocket abacus, ear tickers	Chancellor → Directors → Brokers → Runners	Abacus fans, contracts, Debt Junks
The Ten Courts (keepers)	Under the River Empire	The true count of the dead	Brush Writing on the Book	Ink-wash, rice paper, red seals; near silence	Black official robes; wardens in lamellar armour (Ox-Head, Horse-Face)	Judges → wardens → clerks → executioner (Bai Si)	The Brush, the Book, halberds
The Observatory	Śūnyapura	Truth by measurement	Karma Weaving; astronomy	Sandstone, brass instruments, saffron and indigo, mirror-work	Layered indigo-and-saffron, brass astrolabe pins	Acharya → astronomers → students	Astrolabes, dials, palm-leaf records

FACTION	SEAT	WANTS	POWER	STYLE (MATERIALS, COLOURS)	UNIFORM / LOOK	RANKS	TOOLS & WEAPONS
The Pythagorean Synod	Kroton Nova	Purity of number	Sieve Geometry	White marble, bronze, Aegean blue, mosaic glyphs	Bronze cuirass, chalk tunic, sieve shields	Archon → Synod elders → knights → novices	Sieve shields, prime glyphs
The House	Lucky Strike	The night never ending	Dealing; the loop	Casino red, ivory, gold, desert neon	Red waistcoats, sleeve garters, green-felt trays	The House → floor bosses → dealers	Marked decks, Wagers, the loop
The Open Hand (outlawed)	Everywhere, quietly	To free Hap; second chances for all	Prayer; coincidence	Road dust, plain cloth, one gold open-hand mark	Travelling cloaks, a carved open hand on a cord, a smooth die	No ranks: "hands"	Dice, shrines, safe houses

Reading a crowd: in Meridian you can tell someone's side of the Zero-Sum Law at a glance: steel numerals (power), gold-leaf numerals (wealth), gaudy holo-halos (street), a cracked tube on a cord (the Wards). That's the four eras of luck (Art Direction §8.2).

5. Jobs in the Tally

JOB	WHERE	WHAT THEY DO	TYPICAL OMEN	TOOLS / LOOK
Auditor	Everywhere (licensed in Meridian)	Stops Fate Wheels in disputes, debts and duels	50–90	Wrist-wheel, field coat
Collector	Meridian	Drains debts at dawn; hunts bounties	40–60	Debt jars, rust armour
Actuary	Basilica	Computes the Dawn Tally; the Basilica's mathematicians (NIL's coat is an actuary's)	50	Ledgers, slide rules
Ledger-priest	Basilica	Performs the Dawn Tally and the Naming Rite	50 (steel)	Mitre, chained ledger
Dawn-bell ringer	Meridian	Rings the bells that start the Tally; the city's clocks strike 0	30–60	Bell ropes
Nixie-tuber	Meridian	Repairs the Nixie tubes that display Omens and signs	20–50	Glass, wire, gloves
Charm-smith	Bazaar, everywhere	Makes and repairs charms from every culture	20–70	Forge, anvil (Maeve)
Luck-broker	Exchange	Trades Omen points like shares	70–99	Abacus, ticker
Pawnbroker of luck	River Empire, Meridian	Lends against charms and future luck	60–90	Ledgers, safes (Jinling)
Omen clinician	Meridian, River Empire	Luck transfusion from donor to patient	60–80	Brass cannulas, Nixie meters
Insurer	Meridian	Pools luck against misfortune	70+	Contracts
Shrine keeper	Hyakki	Keeps shrines, paints daruma eyes, sells omamori	any	Brush, charms (Ofuku)
Ferryman	Hyakki	Carries travellers between the islands	any	Oar (Tetsu)
Oracle-stick reader	River Empire temples	Reads the fortune sticks	any	Cylinders, poem books
Astronomer	Śūnyapura	Measures the sky and the Deficit	any	Astrolabes (Shunya)
Stepwell diver	Śūnyapura	Recovers what falls into the wells	any	Rope, lamp (Meera)
Synod knight	Kroton Nova	Guards the Sieve	prime	Sieve shield
Croupier / dealer	Crossroads	Runs legal Audits for money	any	Cards (Fibi)
Sheriff	Crossroads	Keeps the peace in the loop	any	Badge (Vance)
Luck-in seller	Remainder Wards	Sells single Omen points for food	01–10	Nothing left to sell
Rounding-sweeper (urban legend)	Meridian	Said to sweep up dropped fractions at dawn; nobody has seen one	—	A broom made of digits

6. How things operate

Money. Two currencies: **fen** (small coins of brass, everyday trade across the Tally) and **luck** itself (Omen points, traded at the Exchange and lent by banks). Prices are posted in fen; luck is for big things (loans, licences, medicine). **The day.** The **Dawn Tally** at sunrise recomputes every Omen (a beam sweeps each city; clocks strike 0). Debts are drained in the hour after. Markets open at the second bell. **Law.** Disputes go to an **Audit court**: a magistrate oversees a formal Audit between the parties' Auditors (or themselves). Crimes are punished in luck (Omen fines) or time. The death penalty doesn't exist; Settlement does the work, which is worse. **Health.** Omen clinics transfuse luck; doctors still heal bodies. The poor sell luck-ins to pay for medicine and then need more medicine. **Transport.** Meridian: cliff lifts and stairs between the Undercity and

the plateau (lift fares are by Omen: the lower your number, the higher the fare, "for safety"). Between lands: ferries (Hyakki), river junks (River Empire), rail (Crossroads), the monsoon road (Śūnyapura), the Aegean packet boats (Kroton Nova). **Media.** Omen boards in squares; the *Daily Tally* broadsheet (every Omen change of note); radio plays in the Crossroads; street criers in Bāfā. **Education.** Children learn their digit's meaning before the Naming Rite at seven; the Basilica's schools teach arithmetic as morality. **Architecture rule.** Floors 4, 9 and 13 are skipped in lifts; buildings face east for the Dawn Tally; the Basilica's eye is visible from every street in Meridian (a planning law).

7. Stores and services (what you see on a street)

STORE	LAND	WHAT IT SELLS	SIGN	IN PLAY
Charm stall	Meridian (Bazaar)	Every culture's charms	Hand-painted lucky number + a charm silhouette	Markets (charms)
Luck pawnshop	River Empire, Meridian	Loans against charms or future luck	Three gold coins on a red board	Markets (fen, Deals later)
Omen clinic	Meridian	Luck transfusions, Omen checks	White cross made of two 1s	Story; healing at a price
Nixie repair	Meridian	Tubes for Omen displays and signs	A glowing tube in the window	Face enhancements
Forge	Meridian (Celtic quarter), Kroton	Charm upgrades, horseshoes	An anvil with a clover	Forge nodes (Maeve)
Noodle stall	Everywhere	Food priced in sevens and eights	Steam and a number	Rest (Teahouse/Campfire)
Teahouse	Hyakki, River Empire	Tea, gossip, a place to rest the party	Paper lantern with a digit	Teahouse nodes
Shrine	Hyakki, Śūnyapura	Omamori, daruma, offerings	Torii / temple bell	Shrine nodes
Oracle-stick temple	River Empire	Fortune sticks	Smoke and a cylinder	Oracle sticks
Gem bazaar	Śūnyapura	The nine planetary gems	Nine coloured lamps	Navaratna gems
Agora stalls	Kroton Nova	Gems, wheel crafting at a "prime discount"	Prime numerals on marble	Markets
Casino / card room	Crossroads	Legal Audits for money	Neon dice	Wagers
Motel	Crossroads	Rooms on Route 7	"VACANCY" with the C burnt out	Campfire nodes
Open Hand shrine	Everywhere	Nothing: a stone hand and a die	An open palm carved in stone	Open Hand nodes

8. Politics and power

The Comptrollership. Meridian is governed by the **Comptroller**, chosen by the **Board of Actuaries** (twelve senior actuaries) for life. The Comptroller holds the Basilica (the Dawn Tally and the Seals) and therefore the world. Thale has held it since TR 985. **The Ledger Assembly.** A chamber of representatives weighted by Omen: each member's vote counts in proportion to their constituents' total luck. The rich decide; the Wards are "represented" by a single seat that hasn't been filled in forty years. **The Seal pacts.** The Basilica binds each land through its **Auditor** (a local power who made a deal with Thale, TR 999) and its **Seal**. In return each land gets "protection" (an exemption from the Deficit's worst storms) and a promise of permanence under Fixed Fate.

LAND	WHO RULES	THE BASILICA'S GRIP	WHAT THE PACT PROMISED
Meridian	The Comptroller; the Board; the Assembly	Total	Order
Hyakki	The Yokai Courts with human village councils	Toll gates on shrine roads; Collectors in harbours	Yokai never fade (Tamamo)
River Empire	The Chancellor of the Exchange (de facto); the Ten Courts below	The Exchange is half a Basilica office	Jin Chan's bank holds the frozen luck
Śūnyapura	City council of scholars and guilds; the Observatory	Surveyors rewriting measurements	Rahu & Ketu made whole
Kroton Nova	The Pythagorean Synod (the Archon)	The Sieve "accelerated"	Purity
Crossroads	The House (by money), the sheriff (by law)	Basilica finance	The night never ends

Power, honestly. Luck is wealth, health and safety at once, so whoever controls the count controls everything. The Basilica's genius was to make that look like arithmetic instead of rule. The game's politics is that unmasking: each Seal broken returns a land's own idea of luck, and with it, its own say.

9. History and timeline (Tally Reckoning, TR)

WHEN	ERA	EVENT
Before reckoning	**The Noise**	Hap deals freely; nobody keeps accounts. Fortune moves like weather.
Ancient	**The Many Ledgers**	Each people learns to talk to Hap: the Moirai's thread (Kroton Nova), Chitragupta's records (Śūnyapura), the Book of Life and Death (River Empire), Enma's ledger (Hyakki), the black book (Crossroads).
~1,000 years before TR		Zero formalised in Śūnyapura; place value travels the trade roads.
TR 0	**Founding**	Meridian founded as a trading town at the crossroads of the Tally.
TR 120		The first Charm Bazaar: every caravan's charms on one street.
TR 380–690	**The Binding**	Meridian's actuaries copy, buy or seize each land's ledger.
TR 512		Kroton Nova's thread is "borrowed" for a century; the Synod hardens.
TR 640		The Exchange opens in Bāfā Harbor under Meridian charter.
TR 702	**The Basilica**	The engine completed; **the Final Binding cages Hap**; the first Dawn Tally; the Rounding begins. Worship of Hap outlawed in Meridian.
TR 760→	**Ledgerism**	Brass-and-steel style; Auditing licensed; the Collectors founded.
TR 850→	**Charmpunk**	Immigrant quarters fill Meridian; every culture's charms on one door.
TR 911		The Lucky Strike casino opens on Basilica money; the House's first deal.
TR 960→	**Remainder**	A storm destroys Lucan Thale's street (he is 12); the Undercity fills with low Omens.
TR 962	**The Deficit**	Śūnyapura's observatory first measures the leak. Noise storms begin.
TR 985		Thale becomes Comptroller.
TR 990→	**Gilded Fortune**	The 99s flaunt gold leaf, jade and gardens.
TR 999		Thale proves the Rounding, finds Hap in the engine, drafts the Final Audit, signs the Seal pacts.
TR 1001		Varro Kade becomes a Collector. The Ghost Train's loop count reaches 1,596.
TR 1002	**Now**	NIL wakes in Rain Alley 9. REMAINDER DETECTED. The game begins.
TR 1002 (finale)	**The Open Hand**	Depends on the ending: Balance, Overflow, Positive-Sum or Divide by Zero.

10. Little laws of daily life (for set dressing and dialogue)

Lifts skip 4, 9 and 13 · weddings crowd onto the 8th · gamblers kiss sevens · beggars sell luck-ins · children compare Omens in the schoolyard · shopkeepers hang a lemon and seven chillies, a horseshoe, an evil-eye bead or a beckoning cat · it is rude to read someone's Omen out loud · "May your Omen never round" is goodbye · nobody counts money at the table in Lucky Strike · in Hyakki, gifts never come in fours.

PART III

The story

From a NaN in the rain to the Open Hand: every act, every boss, every bond.



The whole campaign, scene by scene: Prologue, five acts, five interludes, the finale and four endings. World rules are in `WORLD_BIBLE.md` (Hap is §2A), people in `CHARACTER_BIBLE.md`, each land's play in `ACT_DESIGN.md`, bond scenes in `BOND_ARCS.md`.

0. The shape of it

Logline. A nameless wanderer with no number, who turns out to be the last free piece of the caged god of luck, crosses five lands to stop a kind man from freezing fate forever.

Theme. *Luck shared isn't zero-sum.* Every act shows a land whose idea of luck the Basilica flattened, a person who suffers for it, and a companion who learns, with NIL, that people are worth more together than apart. The game's synergy system is the theme made playable.

NIL's question per act (the spine of the story):

ACT	LAND	NIL ASKS	NIL LEARNS	COMPANION WHO ANSWERS
Prologue	Meridian	"Why don't I have a number?"	That a number decides everything here	Vesper
I	Hyakki	"What makes something real?"	Being believed in; being chosen	Hina
II	River Empire	"Who owns luck?"	Luck is debt to some and gift to others	Ba Jinling, Bai Si
III	Śūnyapura	"What am I?"	NIL is the Remainder: what everyone lost	Shunya
IV	Kroton Nova	"Are some numbers worth less?"	No. "Indivisible" means people	Prima
V	Crossroads	"Can you outrun the dawn?"	No; you have to let tomorrow come	Fibi, Vesper
Finale	The Basilica	"What do I give back?"	The player decides	Everyone, and Hap

Three movements. 1. **Hunted (Prologue–Act II):** NIL runs, learns the world's rules and gathers friends. Stakes: survival. 2. **Remainder (Act III–IV):** NIL learns what they are; the hunt becomes a choice. Stakes: identity. 3. **Zero-Sum (Act V–Finale):** NIL chooses to walk into the Basilica. Stakes: the god of luck and everyone's tomorrow.

Thale's thread. After every act, a short Thale scene shows him being decent: feeding the Basilica's pigeons, signing a hardship exemption, visiting his family's empty street on the anniversary. The player should like him a little more each time and fear him a little more each time.

1. Prologue: "NaN" (Meridian Undercity) · ~35 min

Setup. TR 1002. Dawn in the Remainder Wards. The Dawn Tally beam sweeps the slums; every Omen ticks.

1. **Intro cinematic** (World Bible §8): the Noise; the many ledgers; the Basilica; title. Thale narrates kindly.
2. **Rain Alley 9.** Ama Okoro's Omen drops to **01**. Her son Tobi stares. Collectors arrive to "settle her accounts". A hooded figure with a flickering **NaN** steps between them. NIL reaches out: digits spin around Ama's 01; NIL **stops** them on **07**. A Collector: "That one has no number."
3. **Chase** through the Wards: charms of every culture over every door. NIL ducks into a black-market shrine.
4. **Brother Nought's Naming shrine.** "Every soul needs a number, child." **The player states their lucky number** (0–9; each digit shows what it means across the Tally). The terminal overloads; across the city every screen reads **REMAINDER DETECTED**; the Basilica's eye opens. Brother Nought, quietly: "Hap's watching." (First mention of Hap. Players won't know it yet.)
5. **First Audit** vs Debt Hounds, alone. Tutorial: stop the wheel, the reel, Pair, Jackpot (scripted), Parry.
6. **Vesper Sette** drops from a fire escape: "Need a hand? Seven bucks." Second Audit: choosing who strikes; Cascade and Zero-Sum.
7. **The Charm Bazaar.** Gus Mendoza sells NIL a first charm "on credit". Optional: **Maeve's forge** in the Celtic quarter is under Collector attack; defend it and she joins ("Third time lucky, and you're my third stranger today").
8. **Livia's window.** NIL glimpses a sick girl with Omen 06 in an upstairs window guarded by Collectors. (Payoff in Act V.)
9. **The drain gates. Varro Kade, the Ledgerhound**, corners them. Boss (rule: the debt jar). Vesper's Requiem saves NIL: the first ultimate.
10. **Varro spared.** He growls that he needs the bounty for his sister and limps away.
11. **The world-map reveal.** On a rooftop Vesper explains: the Upper Gate needs **seven Seal Digits**. Six lands light up on the map in six art styles. Varro's Seal shatters: **Seal 1**. The seventh, she says, "is yours. The number you just told the whole city."
12. **Thale scene.** In the Basilica nave, the Comptroller reads the alert, closes his eyes and says, almost tenderly: "There you are."

Hap's Dream 0: "The Pick." NIL dreams of a starry void and a vast figure with dice for eyes, wearing the face of a Meridian street-corner fortune-teller. "Pick a number. Any number. …Oh, *that* one. Interesting." It laughs; dice rattle; NIL wakes. Unexplained until Act III.

2. Act I: "The Night Parade" (Hyakki) · ~45 min

What's wrong here. Yokai fade when people stop believing in them. Tamamo, the Nine-Tailed Auditor, made a pact with Thale: deliver the Remainder and Fixed Fate will freeze every yokai into permanence. A Ledgerist toll gate now taxes the shrine road.

1. **Arrival by ferry** with Tetsu the ferryman, under a Hokusai-style wave. Tetsu was a Collector once; he won't say why he left.
2. **The Moon Pier.** Lantern ghosts attack; the torii's three lanterns become the reel (Iand reel object).
3. **The Night Parade** crosses the red bridge; the party hides inside it, disguised as yokai. Comedy: Maeve's disguise is just a hat.
4. **Hina** jumps out of a shrine's umbrella stand and attaches herself to NIL: "You smell like a hundred wishes." She is 99, one year short of becoming a "real" spirit. **Hina joins.** Party of 3 unlocks; bonds and Link Arts (*Midnight Parade* with Vesper).
5. **Ofuku's shrine of the Seven Lucky Gods.** Charms and charm sets. Ofuku paints one daruma eye a day "for whoever needs it".
6. **The kappa river.** Side quest: **Kyūsuke** the tired kappa (secret recruit, digit 5).
7. **The Rule of Shi and Ku** bites: 4s and 9s curse the striker in Hyakki. Hina is immune ("Nine is *my* number, rude").
8. **Enma's gate** (keeper trial). Enma judges whether the living may pass the shrine of the dead: a trial fight with no hatred. Afterwards Enma opens his ledger: NIL's page is **missing**. "You are not written here. You are not written anywhere." NIL learns **Carry** (strikes over 50 carry +1 to the next hero's digit; Hina taught NIL to count past a hundred).
9. **The Kitsune twins Ao and Aka** guard the foxfire palace (elite).
10. **Tamamo** in her foxfire palace. Boss: nine tails, illusion faces, 9s curse you unless Hina is in the party.
11. **Tamamo's question.** Defeated, she asks NIL: "What is a yokai worth without belief?" NIL: "What's anyone worth without a number?" She gives Hina a tail-blessing and retreats into legend. **Seal 9.**
12. **Thale scene.** He visits Hyakki's ledgers in the Basilica archive and leaves a cucumber on Enma's shelf. (A clerk asks why. "Superstition," he says, and smiles.)

Hap's Dream I: "Real." Hap wears the face of an old paper lantern. "She thinks she isn't real yet. Do *you* think you're real?" NIL doesn't answer.

3. Act II: "Eight Bells and the Book of the Dead" (River Empire) · ~50 min

What's wrong here. Chancellor Jin Chan's Luck Exchange trades Omens like shares; luck compounds for the rich and debt compounds for everyone else. Below the river, the Ten Courts' count of the dead has always been short.

1. **Bāfā Harbor at the Lantern Festival.** Live Omen price boards; dragon boats; Little Lantern (Xiao Deng), an orphan messenger, steals NIL's charm and returns it with a note: "Someone wants to buy you."
2. **Ba Jinling's pawnshop.** Jinling buys the party's bounty and hides them: "Everyone's in debt, darling. I just keep better books." **Jinling joins** as a guest.
3. **Granny Liu Shun's mahjong table.** Side quest: lose gracefully; win the secret recruit (digit 6).
4. **Markets and wheel crafting** unlock at the Exchange's back alleys; **oracle sticks** at the temple.
5. **The river's end.** The party rides a Debt Junk off the edge of the world into the **Ten Courts** (ink-wash art). Crossing Naihe Bridge; Meng Po's pavilion of forgetting (the party refuses the soup).
6. **Ox-Head and Horse-Face's trial**: the wardens must both be bested in the same round. Afterwards they bow.
7. **Bai Si**, the executioner who counts the dead, blocks the Hall of the Book. NIL's decoder shows him three hundred years of rounding: his count was always short. He kneels, and follows the party to the Hall.
8. **Judge Cui's trial.** The Judge tests whether NIL may hold the Brush that writes in the Book of Life and Death. NIL learns **Rewrite** (free Ink each fight). Satisfied that the count can be corrected, **Bai Si joins**.
9. **Return to the harbor.** The Exchange freezes the city's Omens to force the party out. Jinling chooses: she burns her father's debt ledgers in public.
10. **Chancellor Jin Chan** in his coin palace. Boss: interest, the Jackpot tax, swallowing a hero.
11. **Father and daughter.** Bankrupt, Jin Chan laughs until he cries: "You were always better at books than I was." Jinling takes over the Exchange and forgives every debt under 8 fen. **Jinling joins** fully. **Seal 8.**
12. **Thale scene.** He signs Livia Kade's hardship exemption, then files it under "pending". He hesitates for a long moment.

Interlude: the Open Hand. Brother Nought finds the party in Bāfā's temple of oracle sticks. He shows them an open stone hand with a die in its palm and tells them about **Hap**, the god of luck that Meridian outlawed. "Every land had a way to talk to Hap. The Basilica took all of them." He doesn't know where Hap went. NIL feels the die in the stone hand is warm.

Hap's Dream II: "Owed." Hap wears the face of a smiling money toad. "Everyone owes me something. You're the only one I owe." It won't explain.

4. Act III: "Śūnya" (Śūnyapura) · ~50 min · the midpoint

What's wrong here. Basilica surveyors are "correcting" the great observatory's measurements; the Deficit is worst here, and monsoon Noise storms scramble the streets. Rahu and Ketu, the eclipse, have taken the plateau: Thale promised Fixed Fate would make their stolen immortality whole.

1. **Monsoon arrival.** Rain on stone; a city built around giant sundials and astrolabes.
2. **Baital the Vetāla** riddles the party from a banyan tree (riddle encounters; the answer is always a digit).
3. **Shunya** at the observatory with her teacher, Acharya Vidya Rao. She has measured the Deficit for years: "The books are short by one soul. Exactly one. Hello." **Shunya joins.** Karma and Navaratna gems unlock.
4. **The Stepwell of a Thousand Steps** with Meera the diver (dungeon; she seeks her drowned brother's karma record).
5. **Chitragupta's Hall** (keeper trial): palm-leaf records to the horizon. Chitragupta reads NIL's page aloud. It is blank. Then the ink appears, written in every soul's handwriting at once: one tiny fraction from every life in the Tally.
6. **The reveal.** Shunya does the arithmetic on the observatory floor, gently, in chalk: three centuries of rounded-down remainders equal exactly one person. **NIL is the Remainder.** NIL learns **Place Value**. Quietly, NIL asks Shunya: "Then whose luck am I?" She says: "Everyone's. That's what zero means. It's the place that holds everything."
7. **The eclipse.** Rahu swallows the sun; Ketu's comet tail lashes the plateau. **Rahu & Ketu** boss: two bodies, keep their Balances within 10% of each other; Zero-Sum hits both equally.
8. **Returned to the sky.** Rahu and Ketu become a passing eclipse again; Vidya Rao's catalogue gets its last entry: "NIL. Remainder. Brightness: varies." **Seal 0.**
9. **Thale scene.** Thale learns the Remainder now knows. He orders the Sieve of Kroton Nova "accelerated", then goes home and cooks dinner for one.

Hap's Dream III: "Change." Hap wears the face of a monsoon cloud full of stars. "You're the change I never got back." For the first time NIL asks a question: "Are you real?" Hap: "Ask me in the Basilica."

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5. Act IV: "Indivisible" (Kroton Nova) · ~45 min

What's wrong here. Kroton Nova is a white island city built as a giant sieve; the Pythagorean Synod has decided that citizens whose Omens are composite are impure, and the Sieve drops them into the undercity. Archon Seventeen calls it purification; Thale calls it "acceleration".

1. **Noon arrival** on the Sieve terraces; goats; evil-eye beads; Niko the goat-herd falls through a hole (running gag).
2. **Prima of Kroton**, the Synod's "even prime" novice, escorts the party as prisoners. **Awakenings** are explained at the Agora of Primes.
3. **The undercity of the fallen.** Mother Kallisto leads the "composite" people in shanties built from Basilica scrap.
4. **Prima defects** when the Sieve drops a school; she holds the children on her shield over the drop. **Prima joins.** "If the Sieve drops children, the Sieve is wrong."
5. **The Moirai's cave** (three keeper trials: Clotho spins a wheel you must stop in sequence; Lachesis measures the party's bonds; Atropos asks what NIL would cut). Lachesis gives Prima a strand of thread. NIL learns **Factor**.
6. **The Archon's temple. Archon Seventeen** boss: he strikes digits off your wheels like the Sieve of Eratosthenes; composite strikes are halved.
7. **The fall.** The Archon falls through his own sieve; Prima catches him. The Sieve stops; Kallisto's people climb the terraces. **Seal 2.**
8. **Thale scene.** Thale stands in his family's empty street on the anniversary of the storm. It's raining. He doesn't open his umbrella.

Hap's Dream IV: "Worth." Hap wears the face of a cracked marble statue with a living smile. "Which of your friends is worth the least?" NIL: "Wrong question." Hap laughs so hard the stars shake.

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6. Act V: "1,597 Nights" (the Crossroads Territory) · ~50 min

What's wrong here. In the desert boomtown of Lucky Strike, the same night repeats. The House, Octavian Nine, "Mr. Eighty-Nine", financed the Basilica for decades; in return Thale lets him loop the night so his granddaughter Fibi never reaches the dawn that kills her.

1. **The midnight crossroads.** One streetlamp; the Stranger (keeper) with the black book offers NIL a deal. NIL learns **Deal** (once per fight, take a Crossroads Deal). The Stranger never cheats.
2. **Lucky Strike.** Casino neon; Route 7 motels; the same drunk falls off the same stool every night. **Loops** begin: every few scenes the night resets; the party keeps what it learned.
3. **Fibi Leone** finishes NIL's sentences: she has lived this night 1,597 times. **Fibi joins** (her digit advances every round).

4. **Penny** on the railway: the ghost girl with 1,596 lucky pennies (Ghost Train quest).
5. **Varro returns.** Livia's exemption was never signed; Varro realises Thale lied. If the party saves Livia (Prologue window, Act II thread), **Varro joins as a guest** for the act.
6. **Sheriff Dolores Vance** helps solve the loop: the night resets whenever Fibi's heart stops at dawn.
7. **Vesper's honest hand.** Vesper confesses the rigged hand that Settled a child here years ago and deals Fibi an honest game in front of the House.
8. **The House.** Boss: rewinds time every 3 rounds; copies your last pattern; offers a mid-fight deal.
9. **Dawn.** Octavian Nine lets the sun rise. Fibi lives: the loop was never what was killing her; the House's deal with the Basilica was. He says goodbye anyway and walks into the sunrise. **Seal 7.**
10. **Thale scene.** Thale receives seven Seal alarms. He opens the Upper Gate himself. "Let them come home."

Hap's Dream V: "Tomorrow." Hap wears the face of a jackpot slot machine, laughing. "Tomorrow is the only bet I always take." It shows NIL the Basilica's eye from the inside, and the eye is Hap's.

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7. Finale: "Zero-Sum" (Meridian & the Basilica) · ~40 min

1. **The Upper Gate.** Six Seal Digits glow (1, 9, 8, 0, 2, 7). The seventh socket waits. **The player's lucky number**, chosen in minute one, is the last key. (If the player chose a digit already used, the socket asks for it a second time, and the gate shows both.)
2. **Clerk Ivo Marr** opens the service door: "Somebody should file a complaint."
3. **The nave.** Every land's original ledger hangs on the walls: thread, palm leaves, the Book and the Brush, Enma's ledger, the black book. Ledger-priests chant the catechism.
4. **The Even Hand engine.** The party sees the truth: the Basilica's heart is a closed brass fist around a god. **Hap is in the cage**, starved and counting itself.
5. **Thale.** He isn't surprised and he isn't cruel. "Thread, karma, contracts, books of the dead: a hundred superstitions, and every one of them let people die at random. I replaced them with arithmetic." He asks NIL to come home to the Ledger.
6. **Phase 1, Regression to the Mean:** every wheel compresses toward 5.
7. **Phase 2, Fixed Fate:** the wheels stop spinning; a puzzle fight where you arrange digits instead of stopping them.
8. **Phase 3, Zero:** NIL alone. The companions return as Link Arts, one by one, each with a line from their bond arc.
9. **Thale falls to one knee,** Omen still frozen at 50. "Then tell me what you'd do with it."
10. **The Open Hand (no fight).** Hap's wheel appears: a huge Fate Wheel whose faces are the endings available to this player. **The last tap of the game is the player stopping Hap's wheel.** Only the endings the player has earned appear as faces.

Endings

ENDING	CONDITION	WHAT HAPPENS	FINAL IMAGE
Balance	Always available	NIL returns to the Ledger. The cage is fed; every Omen rises by one; the Deficit closes; Hap stays caged and quiet	The party at a grave with no number; Tobi leaves a cucumber on it
Overflow	Always available	NIL breaks the Basilica. Hap bursts free all at once; every Omen becomes NaN; the Noise returns; each land goes back to its old ways, for better and worse. NIL fades into the storm	A Noise storm over Meridian, and in it, a NaN that winks
Positive-Sum (true)	All 8 core bonds complete and every ledger piece kept	The party returns every land's ledger piece to the engine *together*. The brass fist opens into an **Open Hand**: Hap deals again, through every land's ledger at once. Luck shared starts multiplying. Thale lives to see it	Hap, wearing NIL's face for a moment, gives NIL a number: the NaN resolves into **the player's lucky number**. Tobi performs the new Naming Rite
Divide by Zero (secret)	NIL's digit is 0 and the final hit of Phase 3 lands 0-0-0	The Basilica crashes; the screen blue-screens; the credits roll in debug text; Hap's laugh plays under the error log	`REMAINDER: 0. HAP: FREE.`

Epilogue cards (true ending)

Vesper deals honest games in Lucky Strike · Maeve's charms work again · Hina turns 100 and throws a festival · Jinling's Exchange lends at 0% · Bai Si's count is finally right · Shunya publishes "On the Remainder" · Prima teaches at the Agora · Fibi sees her first sunrise and her second · Varro and Livia open a dog shelter · Thale tends the pigeons at the Open Hand shrine where the Basilica's eye used to be.

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8. After the story

- **New Game+ "The Second Hand":** carry charms and bonds; Thale is your companion for one act (a what-if).

- **The Endless Ledger** (roguelike post-game): every land shuffled, Omen Levels 1–8, the Daily Omen seed. Its final boss is **the Default** (the world without luck, a creature of pure stillness), the only boss that is "nothing" rather than someone.
- **Hap's Gallery:** every face Hap wore, unlocked by dreams, side quests and coincidences.

One section per land: its story in one paragraph, the Land Law (its superstition as a rule), the act map, encounters, the keeper trial, the boss (full sheet in `BESTIARY.md`), side quests, Hap's Whispers, and what the act teaches. Scenes are in `STORY.md`; numbers in `TUNING.md` and `ECONOMY_PROGRESSION.md`.

0. How every act is built

- **Map:** ~8–10 nodes on an in-world map object (scroll, palm-leaf, mosaic, road map), two to three branches, always converging on the boss. A full act takes 25–35 minutes; a perfect player can do it in 15.
- **Node types:** Small Audit · Big Audit · Elite · Keeper's Trial · Market · Shrine/Forge · Teahouse/Campfire · Story · **Open Hand shrine** (Hap: re-deal one offer, or pray for Favour) · **Hap's Whisper** (a coincidence event) · Boss.
- **Pacing rule:** never two fights in a row without a choice; never more than two non-fight nodes in a row. A story node never interrupts a Market.
- **Every act introduces:** one land system (GDD §5), one NIL verb (GDD §6), one companion, one keeper, one Land Law, one reel object and one boss rule. Players should be able to say what is new in each act in one sentence.
- **Land Law:** the land's superstition as a global rule for every fight there, shown on the reel frame as a small icon. Counter-play always exists inside that land (a companion, a charm or a verb).

1. Prologue: Meridian Undercity

Story. Meridian's Remainder Wards are where low Omens end up. NIL appears with a NaN, tells the city a lucky number and becomes the most wanted person alive. Vesper takes the job of keeping NIL alive; Varro takes the job of collecting.

Land Law: Debt. Every third round the Collectors charge interest: the party loses 5% of its Balance unless the round scored a pattern. (Teaches: patterns are protection, not just damage.)

NODE	TYPE	CONTENT
1	Story	Rain Alley 9; the Naming shrine; lucky-number choice
2	Small Audit (tutorial)	2 Debt Hounds; NIL alone; scripted Jackpot on the first round
3	Story + Small Audit	Vesper joins; 3 Debt Hounds; choose who strikes
4a / 4b	Market / Forge	Gus Mendoza's stall (first charm) **or** Maeve's forge defence (Big Audit: Collectors) to recruit Maeve
5	Big Audit	Collector squad (armoured; parry practice)
6	Boss	**Varro Kade, the Ledgerhound** at the drain gates

Enemies. Debt Hound (fast wheel, weighted to 4), Collector (armoured: Iron faces), Noiseling (its wheel scrambles once per spin). **Reel object:** a dead three-tube Nixie sign. **Music:** synth noir, rain, Nixie hum. **Side quests:** *Livia's window* (starts the Varro arc) · *Gus's ledger* (pay off the charm "credit" for a discount later). **Hap's Whispers:** a coin lands on its edge (free re-spin); a black cat crosses the alley (choose: Bust → Pair once, or nothing); "Someone shares your birthday" (+5 fen). **Teaches:** stop, reel, Pair, Jackpot, Parry, choosing strikers, Cascade, Zero-Sum, ultimates.

2. Act I: Hyakki (Japanese folklore)

Story. Yokai fade when nobody believes in them. Tamamo sold the Remainder to Thale in exchange for permanence for every yokai. NIL meets Hina, an umbrella one year short of becoming "real", and learns what being believed in means.

Land Law: Shi and Ku. A 4 on the reel curses its striker (–10% damage next strike); a 9 curses the party's next parry. Hina in the party makes 9s safe and turns them into +9 Mult. Sevens and eights are blessed: +1 Mult each.

NODE	TYPE	CONTENT
1	Story	Tetsu's ferry; the Moon Pier
2	Small Audit	Chōchin-obake (lantern ghosts) on the pier deck
3	Story	The Night Parade; Hina joins (party of 3, bonds, Link Arts)
4a / 4b	Shrine / Hap's Whisper	Ofuku's Seven Lucky Gods shrine (free charm, charm sets) **or** a fox wedding in the rain (Whisper)
5	Big Audit	Kasa-obake pack (hopping wheels)
6	Teahouse	Party swap; Vesper and Hina's first bond scene

NODE	TYPE	CONTENT
7a / 7b	Elite / Market	Kitsune twins Ao and Aka **or** the kappa river market (Kyūsuke quest)
8	Keeper's Trial	**Enma's gate**: judged, not fought in anger. Rule: you may not parry; you must win within 4 rounds. NIL learns **Carry**
9	Boss	**Tamamo, the Nine-Tailed Auditor**

Enemies. Chōchin-obake (lantern; its wheel lights only odd faces), Kasa-obake (one-eyed umbrella; its wheel hops back a face once per spin), Kappa (lose strength if they bow: land a 5 to make them bow), Oni enforcer (heavy; immune to Pairs). **Reel object:** a torii with three hanging lanterns. **Music:** shamisen, shakuhachi, taiko over synth pads. **Side quests:** *Cucumbers for Kyūsuke* (secret recruit 5) · *One eye a day* (help Ofuku paint 7 daruma eyes; reward: the Daruma charm upgrade). **Hap's Whispers:** a fox wedding (rain in sunshine: free Shrine visit); a tea stalk floating upright (*chabashira*: +1 charm slot this act); a crow at a crossroads (choose a curse to accept for a rare charm). **Teaches:** charms and sets, curses, Reaper's Toll, party of 3, bonds and Link Arts.

3. Act II: the River Empire (Chinese folk religion and festival culture)

Story. The Luck Exchange trades Omens; Jin Chan's fortune compounds while debt compounds for everyone else. Below the river, the Ten Courts' count of the dead has always been short, because the Basilica rounds the dead too. Ba Jinling burns her father's books; Bai Si learns his count was wrong.

Land Law: Interest. Mult carries over: each round's final Mult adds 10% of itself to the next round's (resets on a Bust). Enemies gain the same on their side. (Teaches: streaks.)

NODE	TYPE	CONTENT
1	Story	Lantern Festival; Little Lantern; Jinling's pawnshop
2	Small Audit	Abacus Golems in the festival crowd
3	Market	**Wheel crafting** and **oracle sticks** unlock (Luck Exchange back alleys; temple of sticks)
4a / 4b	Big Audit / Hap's Whisper	Debt Junk on the river **or** Granny Liu Shun's mahjong table (secret recruit 6)
5	Story	The river's end; falling into the Ten Courts (art style switches to ink-wash)
6	Keeper's Trial	**Ox-Head and Horse-Face**: both must be bested in the same round
7	Elite	Hungry Ghost swarm (they eat fen on every Bust)
8	Keeper's Trial	**Judge Cui**: you may rewrite one digit per round with the Brush, but every rewrite is recorded. NIL learns **Rewrite**; Bai Si joins
9	Teahouse	Party swap; Jinling and Bai Si's first scene ("You buy debts?" "I burn them.")
10	Boss	**Chancellor Jin Chan, the Golden Toad**

Enemies. Abacus Golem (its wheel shows only 0–5), Debt Junk (fires interest: its damage grows every round), Golden Koi (split into two when hit by a Pair), Hungry Ghost (eats fen), Jiangshi (hops; can't be hit on even rounds). **Reel object:** a golden abacus (harbor); a floating ghost ledger (underworld). **Music:** erhu and guzheng over synth; the underworld is silent except brush strokes. **Side quests:** *Little Lantern's family* · *Thirteen Orphans* (Liu Shun's mahjong ladder) · *Burn the books* (help Jinling find every debt ledger). **Hap's Whispers:** a red envelope on the street (+8 fen, or give it to a child for Karma later); a sticks cylinder that won't stop shaking (a free oracle stick); the Lantern Festival riddle (answer: a digit). **Teaches:** fen economy, markets, wheel crafting, oracle sticks, Ink, interest/streaks.

4. Act III: Śūnyapura (Indian mythology, astronomy and mathematics)

Story. The city where zero was born measures everything, and its observatory measured the Deficit first. Basilica surveyors are rewriting its numbers. Rahu and Ketu, the eclipse, hold the plateau. Shunya proves what NIL is.

Land Law: Karma. Every action is recorded: parries and Zero-Sums bank **Karma**; Busts against a cursed enemy cost Karma. Karma is spent to re-spin a wheel or to block a lethal hit. The land remembers: unspent Karma carries to the next act.

NODE	TYPE	CONTENT
1	Story	Monsoon arrival; the observatory; Shunya joins (**Karma** and **Navaratna gems** unlock)
2	Hap's Whisper	Baital's riddle at the banyan (answer a digit; reward: a gem)
3	Small Audit	Brass Yantra Guardians (animated instruments)
4a / 4b	Market / Shrine	Gem bazaar **or** an Open Hand shrine in a temple courtyard

NODE	TYPE	CONTENT
5	Big Audit (dungeon)	The Stepwell of a Thousand Steps with Meera: 3 short fights, descending; Monsoon Nāgas
6	Keeper's Trial	**Chitragupta's Hall**: every action is shown on palm leaves; the trial is won by finishing with more Karma than you started. NIL learns **Place Value**; **the Remainder reveal**
7	Teahouse	The quietest scene in the game; party swap
8	Elite	Basilica surveyors with a Noise storm (wheels scramble every 2 rounds)
9	Boss	**Rahu & Ketu, the Eclipse**

Enemies. Vetāla (asks a riddle; answer correctly and it skips its strike), Brass Yantra Guardian (its wheel turns like a sundial: one face per round, predictable), Monsoon Nāga (floods a reel tube: that digit counts double for both sides). **Reel object:** three observatory sundial dials. **Music:** sitar and tabla over synth; temple bells; rain on stone. **Side quests:** *Meera's brother* (karma record; payoff in the Hall) · *The star catalogue* (help Vidya Rao finish it) · *Lemon and seven chillies* (replace them on every door the storm tore down). **Hap's Whispers:** a peacock dancing before the rain (next fight starts with Favour warm); a lizard's chirp at the right moment (choose: re-roll a gem); a monsoon break (a free Teahouse). **Teaches:** Karma, gems (pattern levels), the Ledger Tree, two-target balancing. **Cultural care:** Rahu and Ketu are also venerated as celestial bodies; present them as cosmic and tragic, never as cheap demons; review the act with consultants (see `CULTURAL_CONSULTATION.md`).

5. Act IV: Kroton Nova (Greek mythology and Pythagorean number mysticism)

Story. A white island city built as a sieve. The Synod drops citizens with composite Omens into the undercity and calls it purification. Prima, "the even prime", defects to save a school; the Moirai test whether NIL's thread deserves to be cut.

Land Law: Indivisibility. Prime strike values pierce armour; composite strike values deal half damage. The Sieve shakes every 3 rounds: one composite face on each wheel falls off for the rest of the fight. (Teaches: wheel crafting under pressure; number properties.)

NODE	TYPE	CONTENT
1	Story	Noon arrival; Niko falls through the Sieve; Prima escorts the party (**Awakenings** explained)
2	Small Audit	Bronze Automata (Talos-like; immune until a prime lands)
3	Market	The Agora of Primes (gems and wheel crafting at a prime discount)
4	Story	The undercity of the fallen; Mother Kallisto
5a / 5b	Big Audit / Hap's Whisper	Composite Knights at the Sieve gate **or** a Siren who sings numbers (answer by landing her sequence)
6	Story + Big Audit	Prima defects; Harpies over the drop
7	Keeper's Trial ×3	**The Moirai**: Clotho (stop three wheels in rising order), Lachesis (score with only bond-linked heroes), Atropos (NIL chooses which charm to cut forever). NIL learns **Factor**; **Awakenings** unlock
8	Teahouse	Party swap; Prima and Shunya ("Is zero prime?" "Zero is *busy*.")
9	Boss	**Archon Seventeen**

Enemies. Bronze Automaton (immune until a prime strike lands), Harpy (steals a reel tube for a round), Composite Knight (shield that only composite strikes can break: the Law turned around), Siren (sings a digit sequence; land it to silence her). **Reel object:** three marble columns with mosaic numerals. **Music:** lyre and bouzouki over synth; waves; chanting of primes. **Side quests:** *Niko's goats* (always one short; the last goat is in the Moirai's cave) · *Kallisto's ladder* (raise the undercity) · *Spit three times* (a superstition mini-game: dodge the evil eye). **Hap's Whispers:** an owl at noon (+1 Mult per prime this act); a broken mirror (accept 7 rounds of bad luck for a legendary charm); a thread caught on your sleeve (Lachesis's gift: see the next boss rule early). **Teaches:** Awakenings, number properties (primes, perfect numbers, golden ratio), wheel crafting mastery.

6. Act V: the Crossroads Territory (American folklore)

Story. In Lucky Strike the night loops. The House financed the Basilica in exchange for a night that never ends, so that his granddaughter Fibi never reaches the dawn. Vesper faces the hand she rigged here; Varro learns Thale lied about his sister.

Land Law: the House. Every fight is a wager: before it starts you may bet fen or Balance for a bigger reward. The night **loops**: every 3 rounds the fight rewinds (enemy Balance and your reel reset), but your charms, Karma and Mult carry over. (Teaches: building inside a loop; Deals.)

NODE	TYPE	CONTENT
1	Keeper's Trial	**The midnight crossroads**: the Stranger offers a deal. The trial is honest: win a hand of the Stranger's game. NIL learns **Deal**; **Crossroads Deals** and **Wagers** unlock
2	Small Audit	Card-Sharp Automata on Route 7
3	Story	Lucky Strike; Fibi joins (her digit advances each round)
4a / 4b	Market / Hap's Whisper	Gus Mendoza's second shop **or** Penny on the railway (Ghost Train quest)
5	Big Audit	Tumbleweed Golems in a card dust storm
6	Story	Varro returns (guest if Livia was saved); Sheriff Vance
7	Elite	**The Headless Horseman** of the hollow (his wheel has no top half: 5–9 missing)
8	Elite	**The Ghost Train**: its cars are Loop Echoes of your own party
9	Story	Vesper's honest hand
10	Boss	**The House, "Mr. Eighty-Nine"**

Enemies. Card-Sharp Automaton (deals itself a hidden face), Tumbleweed Golem (rolls: its wheel spins backwards), Loop Echo (copies your last strike), the Headless Horseman, the Ghost Train. **Reel object:** a giant slot machine. **Music:** slide guitar, harmonica, a slot-machine choir over synth. **Side quests:** *Penny's 1,597th penny* · *The dead man's hand* (a poker side game) · *Livia's exemption* (Varro's arc payoff). **Hap's Whispers:** a four-leaf clover in the cracked asphalt (1-in-4 duplicates for one fight); a shooting star over the desert (wish: re-deal every offer in the next Market); a penny heads-up (+1 fen per 1 landed this act). **Teaches:** Deals, Wagers, loops. **Cultural care:** no Indigenous sacred traditions; draw on tall tales, the blues crossroads legend, immigrant charms and Sleepy Hollow.

7. Finale: Meridian and the Basilica

Land Law: Fixed Fate. Every round, every wheel in the fight spins 5% slower and loses its least-used face. It cannot be countered, only outrun. The finale is short and relentless on purpose.

NODE	TYPE	CONTENT
1	Story	The Upper Gate: six Seals and the player's number; Ivo Marr's service door
2	Big Audit	Ledger-priests in the nave (their wheels are steel: no enhancements)
3	Teahouse (last)	The last campfire: every recruited companion talks; final party choice
4	Story	The Even Hand engine: Hap in the cage
5	Boss	**Comptroller Lucan Thale** (3 phases)
6	The Open Hand	No fight: the player stops Hap's wheel to choose an ending

Music: the Meridian theme in a cathedral organ arrangement; every land's instrument enters in Phase 3 as the companions return.

8. Post-game: the Endless Ledger

All lands shuffled into a true roguelike: 3 acts drawn from 5 lands, then the Basilica, then optionally **the Default**. Omen Levels 1–8 stack modifiers (faster wheels, harsher Land Laws, fewer shops). The **Daily Omen** is a shared seed with a leaderboard. Every run ends with Hap's Dream showing a face you haven't seen.

Every opponent's wheel, behaviour, telegraph and counter. Balances scale per act (`TUNING.md` §3). Stories in `STORY.md`; where they appear in `ACT_DESIGN.md`.

Design rules for every opponent 1. **One idea per enemy**, readable from its wheel alone within one spin. 2. **Telegraph before punish.** Any special move is shown one round ahead (an icon on its wheel and a sound). 3. **A counter exists in the same land** (a hero, a charm, a verb or a Land Law interaction). 4. **Defeat is "Settled", never gore.** Enemies round down into digits and blow away. 5. **Keepers are never enemies.** Their fights are trials with rules, and they end satisfied, not beaten.

1. Common enemies

LAND	ENEMY	WHEEL	BEHAVIOUR	TELEGRAPH	COUNTER
Meridian	**Debt Hound**	Fast; 4 double-wide	Bites twice if its digit is 4	Snarls; its 4 glows	Parry on 0; Bai Si later
Meridian	**Collector**	Medium; Iron faces	Armour: –50% from Pairs	Raises a ledger shield	Jackpots and Cascades
Meridian	**Noiseling**	Scrambles once per spin	Copies a random reel digit	Static crackle	Fast taps (it can't scramble twice)
Hyakki	**Chōchin-obake**	Odd faces only	Lights a lantern each round; 3 lanterns = big flame	Lantern count on its body	Even strikes snuff a lantern
Hyakki	**Kasa-obake**	Hops back one face per spin	Bounces in on the third round	Umbrella opens its eye	Stop late (aim one face ahead)
Hyakki	**Kappa**	Slow	Strong until it bows	Water dish sloshes	Land a 5: it bows and loses its strength
Hyakki	**Oni Enforcer**	Heavy; 9 double-wide	Immune to Pairs	Club raised	Cascades, Zero-Sum
River	**Abacus Golem**	0–5 only	Its damage = sum of its last 3 digits	Beads clack	Zero-Sum (it can't avoid 10s)
River	**Debt Junk**	Medium	Damage grows +10% per round (interest)	Cannon fuse lengthens	Burst it down early
River	**Golden Koi**	Gold faces	Splits in two when hit by a Pair	Shimmers	Avoid Pairs; Jackpots
River	**Hungry Ghost**	Fast	Eats 5 fen on every Bust	Mouth opens	Don't Bust: use Pattern Sight
River	**Jiangshi**	Slow	Untargetable on even rounds	Hops	Time your ultimate for odd rounds
Śūnyapura	**Vetāla**	Medium	Asks a riddle before the fight; correct answer skips its first strike	Riddle text (2 lines)	Think, then tap
Śūnyapura	**Brass Yantra Guardian**	Turns one face per round, like a sundial	Predictable but huge	The dial's shadow	Parry on schedule
Śūnyapura	**Monsoon Nāga**	Water faces	Floods a reel tube: that digit counts double for both sides	Rising water line	Put your best digit in the flooded tube
Kroton	**Bronze Automaton**	Steel	Immune until a prime strike lands	Glowing seams	Prima; prime digits
Kroton	**Harpy**	Fast	Steals a reel tube for one round	Screech	Win in two tubes (Pair)
Kroton	**Composite Knight**	Composite faces	Shield breaks only to composite strikes	Shield glyph "4·6·8·9"	Flip strategy from the Land Law
Kroton	**Siren**	Sings a 3-digit sequence	Land the sequence to silence her	Song notes as digits	Listen; Clotho's gift
Crossroads	**Card-Sharp Automaton**	Hidden face	One face is face-down; reveals on stop	Card flip	Evil-Eye Bead
Crossroads	**Tumbleweed Golem**	Spins backwards	Rolls through the party on a Bust	Gathering dust	Patterns
Crossroads	**Loop Echo**	Copies your wheel	Repeats your last strike against you	Echo shimmer	Change heroes each round
Basilica	**Ledger-priest**	Steel, no enhancements	Chants: freezes one of your faces per round	Chant line	Burst; Rewrite
Everywhere	**Noise storm** (hazard)	—	Scrambles all wheels for one round	Sky static	Hap's Favour warms during storms

2. Elites

ELITE	LAND	RULE	REWARD
Collector Captain	Meridian	Two Collectors linked: damage is split between them	Rare Meridian charm
Ao and Aka, the Kitsune twins	Hyakki	Two foxes; one shows illusion faces, the other true ones; they swap every round	Foxfire face enhancement
Hungry Ghost swarm	River	Five ghosts; each Bust feeds one, which then doubles	Oracle stick ×2
Basilica surveyors in a Noise storm	Śūnyapura	Wheels scramble every 2 rounds	Navaratna gem
Harpy queen	Kroton	Steals a tube every round	Glass face
The Headless Horseman	Crossroads	His wheel has no top half (5–9 missing); he charges every 3rd round	The Pumpkin Lantern charm
The Ghost Train	Crossroads	Three cars, each a Loop Echo of one of your heroes	A Crossroads Deal of your choice

3. Keepers (trials, never villains)

KEEPER	LAND	TRIAL RULE	THEY GRANT
Enma	Hyakki	You may not parry; win within 4 rounds	Passage to the shrine of the dead; NIL learns **Carry**
Ox-Head and Horse-Face (wardens)	River (Ten Courts)	Both must be bested in the same round	Passage to the Hall of the Book
Judge Cui	River (Ten Courts)	One Brush rewrite per round, but each is recorded and costs Balance	The Brush; NIL learns **Rewrite**
Chitragupta	Śūnyapura	End the trial with more Karma than you started	NIL's page is read; NIL learns **Place Value**
The Moirai	Kroton	Clotho: rising order · Lachesis: bonded heroes only · Atropos: cut one charm forever	NIL learns **Factor**; Awakenings
The Stranger	Crossroads	Win one honest hand of the Stranger's game	NIL learns **Deal**

Presentation: solemn music, no damage numbers on the keeper (their Balance is shown as a lantern, a scroll, a thread), and they bow at the end.

4. Bosses (Auditors)

Each boss has a **rule** that changes how you play, **phases**, **telegraphs**, a **cinematic ultimate**, and an aftermath that changes the world.

Varro Kade, the Ledgerhound · Prologue · Omen 61 · Balance 1,200

- **Rule, the Debt Jar:** each round he steals your **last reel digit** into a glass jar on his belt (that tube goes dark next round). Three parries shatter all jars and return the digits as free strikes.
- **Phase 1 (100–50%):** two Debt Hounds assist. **Phase 2 (<50%):** "Final Notice": his strikes double, but a jar shatters on every Pair.
- **Telegraph:** he taps a jar before stealing; the jar's tube flashes red.
- **Ultimate, *Foreclosure*:** at 25% he slams every jar into the reel: if you parry the next strike, they all return to you as a free Jackpot.
- **Teaches:** defence matters; the reel is a resource.
- **Aftermath:** spared; returns in Act V.

Tamamo, the Nine-Tailed Auditor · Act I · Omen 99 (per tail) · Balance 6,000

- **Rule, Nine Tails:** nine tails, each with its own small Balance; she can't be hit until three tails are down. **Foxfire illusion faces** appear on your wheels: they show one digit and land as another (a faint flicker gives them away).
- **Shi and Ku:** 9s curse you unless Hina is in the party (then 9s burn a tail outright).
- **Phases:** 9 tails → 6 → 3, each phase adds one illusion face per wheel. At 3 tails she becomes a giant fox (camera pulls back; the pier shakes).
- **Telegraph:** foxfire gathers on the tail about to strike.
- **Ultimate, *Hundred Demons' Night*:** the Night Parade crosses the screen; every wheel shows only illusions for one round. Counter: Hina's *Hundred-Year Night* cancels it (the best Link moment in the jam).
- **Teaches:** read carefully; synergy (Hina).
- **Aftermath:** asks what a yokai is worth without belief; retreats into legend.

Chancellor Jin Chan, the Golden Toad · Act II · Omen 88 · Balance 35,000

- **Rule, Interest:** his Balance gains 8% each round. **Jackpot tax:** he takes 20% of every Jackpot's damage as coins that heal him.
- **Phase 2, the Swallow:** he swallows a hero (removed from the party); land a Pair to make him cough them up, with a free strike.
- **Telegraph:** coins rain before he swallows; the target hero glows gold.
- **Ultimate, *Golden Compound*:** his interest triples for 3 rounds unless you Zero-Sum (the ledger balances, the interest is cancelled).
- **Teaches:** burst damage; streaks; the economy.
- **Aftermath:** bankrupted by his daughter; laughs until he cries.

Rahu & Ketu, the Eclipse · Act III · Omen: eclipsed · Balance 2 × 100,000

- **Rule, two bodies, one fight:** Rahu (the crowned head) swallows one reel tube each round (it goes dark). Ketu (the serpent body with a comet tail) strikes with a blind random wheel you can't read. If their Balances drift **more than 10% apart**, they rejoin and both heal 20%.
- **Zero-Sum hits both equally**, which makes it the key pattern of the fight.
- **Phases:** partial eclipse → totality (the screen goes dark except the reel and their outlines) → the diamond ring (one round of blinding light: every wheel slows).
- **Telegraph:** the sun's shadow edge shows which tube Rahu will swallow.
- **Ultimate, *Grahan*:** totality for 3 rounds; Karma decides who can see their wheels.
- **Teaches:** balance damage between two targets.
- **Presentation:** cosmic, tragic and beautiful, never grotesque; reviewed by consultants.

Archon Seventeen · Act IV · Omen 17 · Balance 1,500,000

- **Rule, the Sieve:** each round he strikes digits off **your** wheels in the order of the Sieve of Eratosthenes (multiples of 2, then 3, then 5…). Composite strike values are halved (the Land Law).
- **Phase 2, Purification:** he removes his own composite faces and becomes faster.
- **Telegraph:** the next multiple glows on every wheel.
- **Ultimate, *Seventeen Years*:** a cicada-swarm cinematic; for one round only primes can be stopped.
- **Teaches:** wheel crafting; Prima's primes; Factor (split a composite into primes).
- **Aftermath:** falls through his own sieve; Prima catches him.

The House, "Mr. Eighty-Nine" (Octavian Nine) · Act V · Omen 89 · Balance 12,000,000

- **Rule, the Loop:** every 3 rounds time rewinds: his Balance and your reel reset, but your charms, Karma and Mult persist. He **copies your last pattern** against you. Mid-fight he offers a **deal** (accept: a huge boon with a price that lasts the rest of the game).
- **Phase 2:** the loop shortens to 2 rounds. **Phase 3:** dawn approaches; the loop breaks if you win one loop without a Bust.
- **Telegraph:** his backwards pocket watch ticks; the numbers on it show the rounds left in the loop.
- **Ultimate, *House Always Wins*:** a slot machine the size of the casino; three reels he controls. Counter: Fibi's *Spiral Deal* (a 5-tube reel) outscores it.
- **Teaches:** build inside a loop; read the deal.
- **Aftermath:** lets the dawn come; says goodbye to Fibi.

Comptroller Lucan Thale · Finale · Omen 50 (steel) · Balance 99,999,999

- **Phase 1, Regression to the Mean:** every wheel compresses toward 5 (faces far from 5 shrink each round). Counter: Cascades through 4-5-6.
- **Phase 2, Fixed Fate:** wheels stop spinning. A puzzle round: you **arrange** your heroes' digits instead of stopping them, with Thale removing one option per turn.
- **Phase 3, Zero:** NIL alone. Each companion returns as a Link Art when NIL lands that companion's digit, with a line from their bond arc. Win condition: the last strike's digit is NIL's own.
- **Telegraph:** the Basilica's clock hands point at the face Thale will freeze.
- **Ultimate, *The Final Audit*:** the whole nave becomes a ledger; every number on screen is counted out loud.
- **Aftermath:** depends on the ending; in the true ending he tends the pigeons at the Open Hand shrine.

5. Post-game superboss

The Default · the Endless Ledger only

The world without luck: a still, perfectly grey figure. It has **no wheel**. Its "strikes" are the absence of yours: each round it removes one face from every wheel in the game. You win by landing an Omen Overwrite before your wheels run out. Hap's Favour is at its strongest here: the god is fighting alongside you, quietly.

Every companion has a **4-scene bond arc** at Teahouse/Campfire nodes. Completing all four gives a second passive, a Link Art upgrade and the true-ending flag (with their ledger piece). Campaign scenes are in `STORY.md`; jam lines in `JAM_SCRIPT.md`.

1. Bond arcs

HERO	SCENE 1	SCENE 2	SCENE 3	SCENE 4 (PAYOFF)	REWARD
Vesper	Teaches NIL to read tells ("Everyone has one. Yours is you don't blink.")	Admits the rigged hand that Settled a child in Lucky Strike	Faces the House's table again (Act V)	Deals Fibi an honest hand; stops flipping the coin	*Honest Hand*: her Pairs pay ×3; ledger piece: the House's marked deck
Maeve	Forges NIL a charm that does nothing ("It's for luck. Not *magic* luck. Friend luck.")	Her grandmother's charms stopped working the day of the Rounding	Finds a Basilica gear in a Kroton scrapheap: the first one ever cast	Forges the horseshoe that can hold a Seal open	*Forge-Mother's Mark*; ledger piece: the first-gear horseshoe
Hina	Counts NIL's wishes ("Zero. That's sad. Hina will lend you some.")	Afraid of turning 100 and not being real	Ofuku tells her the shrine kept her because people loved her, not because she was old	Decides she's already real; paints NIL a daruma eye	*Already Real*: 9s give ×1.9 Mult; ledger piece: Enma's torn page
Ba Jinling	Charges NIL interest on a cup of tea, then pays it back double	Shows NIL the debts she buys and burns in secret	Her father's laugh; she almost forgives him	Reopens the Exchange at 0% for anyone under 8 fen	*Bank of Bells* upgrade; ledger piece: her father's true account book
Bai Si	Counts lanterns; says almost nothing	The first soul he ever executed (he remembers the number)	NIL shows him a remainder he missed; he lights a fifth lantern for the first time	Smiles (first time, Act IV or later)	*True Count*; ledger piece: the true count of the dead
Shunya	Explains zero with a cup of chai ("Empty isn't nothing. It's room for chai.")	Her teacher's catalogue has a gap she never told him about	After the reveal: sits with NIL in silence on the observatory steps	Publishes "On the Remainder" with NIL as co-author	*Śūnya* bond for any digit; ledger piece: the Remainder equation
Prima	Recites primes to calm down; trips	Admits she always felt like the Synod's mistake	Lachesis's thread; she and Shunya argue whether 1 is prime	Catches the Archon, her teacher, as he falls	*Even Prime* upgrade; ledger piece: Lachesis's thread
Fibi	Finishes NIL's sentences (she's heard them 1,596 times)	Shows NIL the motel room where she's spent every 3 a.m.	Learns the loop is to keep her alive; is furious	Watches her first sunrise; doesn't know what to say; says "One."	*1,597*: freeze her digit once per fight; ledger piece: the Black Book's last page
Varro *(secret)*	Shares a meal; doesn't take the mask off	Livia's letter	Takes the mask off for good	Opens a dog shelter with Livia	*Free Man*
Kyūsuke *(secret)*	Sells NIL every cucumber	Fills NIL's dish "for luck"	Bows to a Basilica clerk; the clerk bows back	Goes home with a full cart and a story	*River God's Friend*
Liu Shun *(secret)*	Teaches mahjong; cheats; denies it	Her husband's lucky 6, carved on her tile purse	Loses to NIL on purpose (she says)	Teaches Tobi mahjong in the epilogue	*Heavenly Hand*

Complement pairs get one extra duo scene each (e.g. Vesper + Maeve: "Third time lucky, seven times lucky: who's luckier?").

2. Hap's Dreams (one per act)

AFTER	HAP'S FACE	QUESTION	NIL'S ANSWER (PLAYER CAN'T CHANGE IT)
Prologue	A street fortune-teller	"Pick a number. Any number."	(the player's number)
Act I	An old paper lantern	"Do you think you're real?"	Silence
Act II	A smiling money toad	"Everyone owes me something. You're the only one I owe."	"Owe me what?"
Act III	A monsoon cloud full of stars	"You're the change I never got back."	"Are you real?"
Act IV	A cracked marble statue	"Which of your friends is worth the least?"	"Wrong question."
Act V	A laughing slot machine	"Tomorrow is the only bet I always take."	"Then bet on us."

3. Dialogue database (format)

Every line lives in one table (Unity Localization string table + a metadata sheet `story/lines.csv`):

FIELD	EXAMPLE
id	`p01_alley_010` (act_scene_line)
speaker	`VESPER`

FIELD	EXAMPLE
text_en	Need a hand? Seven bucks.
emotion	smug
trigger	`scene:p01_alley` / `bark:pattern:jackpot` / `bond:vesper:2`
conditions	`party_has:hina`, `lucky:7`, `flag:livia_saved`
voice	`vo_vesper_p01_010` (or none)
max_chars	90 (enforced by a check)
notes	"Tone: dry, warm"
review	`jp_ok` / `zh_ok` / `hi_ok` / `el_ok` (cultural sign-off where relevant)

Barks: each hero has 3–5 per category (strike, hurt, pattern, ultimate, bond-partner banter, lucky-number reaction for all 10 digits). **Lucky-number reactions:** every companion has one line for each digit the player might choose (e.g. Hina on 4: "Brave. Or silly. Hina likes both.").

PART IV

Lore lexicon

Every word, date and rule, kept consistent.



Use this file to keep every word, date and rule consistent across writing, art, UI and marketing.

1. Glossary

TERM	MEANING	USAGE NOTES
Hap	The Tally's one god of luck: the will inside chance. Caged in the Basilica since TR 702	"Hap" (never "the Hap"). It/its, or whatever the speaker's culture prefers. Invented for this world
The Open Hand	Hap's symbol (an open palm holding a die); also the outlawed faith that still leaves offerings to Hap	Shrines are called "Open Hands"
The Even Hand engine	The Basilica's heart, which closed around Hap and divides it into Omens every dawn	Only Thale and the Ledger-priests know the name
Hap's Favour / Hap's Whisper	Gameplay: luck turning after bad luck; coincidence events on the map	In dialogue: "Hap's watching"
The Tally	The known world: Meridian plus the ring of lands around it	"Across the Tally"; never "the world map" in dialogue
Omen	A soul's share of luck, 00–99, visible as a number floating over the head	Always two digits: "Omen 07", not "Omen 7". Plural: Omens.
Balance	The luck a soul stakes in an Audit; damage drains it, and the Omen shows the share left	Gameplay term; in dialogue, "you're running low", never "HP"
NaN	NIL's non-number glitch	Written exactly "NaN"; characters pronounce it "nan"
The Basilica	The brass clockwork cathedral-engine in Meridian that computes every Omen	"The Basilica" (capitalised). Never "the machine" in formal speech.
The Ledger	The merged record of every Omen held by the Basilica	Distinct from the older, local ledgers (Enma's ledger, the Book of Life and Death, Chitragupta's records, the Moirai's thread, the Black Book)
Dawn Tally	The daily recomputation of every Omen at sunrise; a beam sweeps the city	Clocks in Meridian strike 0 at dawn
The Rounding	The dropping of every soul's fractional luck at each Dawn Tally	Secret until Act II/III
The Remainder	The accumulated rounded-off luck, embodied as NIL	Basilica alerts: "REMAINDER DETECTED"
The Deficit	The leak in the world's total luck that causes Noise storms	Theologians speak of "the books not balancing"
The Default	The predicted collapse of the Basilica and return of the Noise	Used like "doomsday"
Noise / Noise storm	Patches of pure chance where numbers scramble; where Noiselings come from	Visual: static, scrambled digits, glitch colour
Noiseling	Small glitch creature made of scrambled digits	Common enemy
Settled	Reaching Omen 00: misfortune takes you, and your luck returns to the pool	Funerals are "settlings"; "may you never settle" is a blessing
Audit	A duel of fates; combat	"Audit him" = fight him
Auditor	Someone who can summon and stop a Fate Wheel; also the title of the six Seal-holders	Capitalise as a title ("the Auditor of Hyakki")
Fate Wheel	The spinning strip of digits summoned in an Audit (descended from Fortune's wheel, *Rota Fortunae*)	Players say "wheel"
The reel	The three slots where stopped digits land; styled per land	Never "slot machine" in dialogue
Naming Rite	The ceremony at age seven where a child states their lucky number to the Basilica	In the intro, NIL performs it at a black-market terminal
Lucky number / resonant digit	The digit a soul resonates with	Gameplay: the hero's synergy digit
Seal / Seal Digit	The lock binding each land's ledger into the Basilica; the gate needs 7 digits	Seven digits: 1, 9, 8, 0, 2, 7 + the player's number
Final Audit	Thale's plan: settle the Remainder, then Fixed Fate	—
Fixed Fate	Freezing every Omen forever	Thale calls it "mercy"
Keepers	Revered guardians of the old ledgers (Enma, Judge Cui, Chitragupta, the Moirai)	Never "bosses" in text
Luck-in	A single point of Omen sold for food or favours	Slum slang

TERM	MEANING	USAGE NOTES
Fen	Small coins used across the Tally	"Eight fen for a bowl of noodles"
Overwrite	Landing your own lucky digit three times	Gameplay and in-world slang for a miracle

2. Idioms and slang

- "Hap deals, people choose": the oldest proverb; said before any big decision.
- "Hap's watching": said at a coincidence; also a gambler's excuse to keep playing.
- "Rounded down": cheated, or killed.
- "In the black": safe, lucky. "In the red": in debt, unlucky.
- "Lucky as a ninety-nine": very lucky (Meridian); in Hyakki it's an insult, since 9 means suffering.
- "Don't count your fen at the table."
- "May your Omen never round": goodbye or blessing.
- "Settle down!" is a grim joke in Meridian.
- Swearing: "Default take you", "Noise and ruin".

3. Timeline (Tally Reckoning, TR)

YEAR	EVENT
Before TR	**The Noise**: fortune moves like weather
Ancient (many centuries)	**The Many Ledgers**: each land develops its own accounting with fate (the Moirai's thread, Chitragupta's records, the Book of Life and Death, Enma's ledger, the Black Book of the Crossroads)
~1,000 years before TR	Zero formalised in Śūnyapura
TR 0	Meridian founded as a trading city at the centre of the Tally
TR 380–690	**The Binding**: Meridian's actuaries copy, buy or seize each land's ledger
TR 702	**The Basilica** completed; **the Final Binding** cages Hap in the Even Hand engine; first Dawn Tally; the Rounding begins. Worship of Hap is outlawed in Meridian
TR 760→	**Ledgerism**: the Basilica's brass-and-steel style spreads through Meridian's offices and uniforms
TR 850→	**Charmpunk**: immigrant quarters fill Meridian; every culture's charms worn at once on the street
TR 960	A Noise storm destroys Lucan Thale's street; he is 12. **Remainder** style begins in the Undercity as Omens fall
TR 962	The Deficit is first measured by Śūnyapura's observatory
TR 985	Thale becomes Comptroller
TR 990→	**Gilded Fortune**: the 99s and Exchange magnates flaunt gold leaf, jade and gardens
TR 999	Thale proves the Rounding and finds Hap in the engine; drafts the Final Audit; signs the Seal-holders' pacts
TR 1001	Varro Kade becomes a Collector
TR 1002 (present)	NIL wakes; the Naming Rite; REMAINDER DETECTED. The game begins.

The present year is **TR 1002**, deliberately just past the round 1000.

4. Factions

FACTION	SEAT	WANTS	SYMBOL
The Open Hand (outlawed faith of Hap)	Everywhere, quietly	To free Hap; second chances for everyone	An open palm holding a die
The Basilica (Comptroller, clerks, Ledger-priests)	Meridian	Order, the Final Audit	A brass gear around the Basilica's eye (which reads as a 0)
The Collectors	Meridian	Debts paid; bounties	A hound skull over a ledger seal
The Charm Bazaar guilds	Meridian	Trade, survival, folk luck	A ring of mixed charms
The Yokai Courts	Hyakki	To keep existing	Nine foxfire flames
The Luck Exchange	Bāfā Harbor	Profit	The three-legged toad with a coin
The Ten Courts (keepers)	Under the River Empire	The true count of the dead	The Judge's brush
The Observatory	Śūnyapura	Truth by measurement	An astrolabe around a 0

FACTION	SEAT	WANTS	SYMBOL
The Pythagorean Synod	Kroton Nova	Purity of number	The tetractys (ten dots) inside a sieve ring
The House	Lucky Strike, Crossroads	The night never ending	The ouroboros around 89

5. World rules (canon locks)

1. Omens are always two digits and always visible to everyone, except NIL's NaN. 1a. Hap never speaks in daylight and never appears as an enemy. It appears only in dreams, coincidences and the finale.
2. Only Auditors can stop their own wheel. Most people can't.
3. Luck is conserved under the Basilica. Nothing creates luck except the true ending.
4. Settlement is not instant death; misfortune finds you within days.
5. Keepers cannot be killed and are never framed as evil.
6. The Basilica's light is always Nixie amber; steel numbers belong only to the Basilica's officials.
7. Charms from all cultures work everywhere, weakly; they work best in their home land.
8. No guns of modern military style: weapons are revolvers, blades, staves, hammers, fans, shields and instruments.
9. Technology is retro-futurist brass and Nixie, with no screens except Basilica terminals and Exchange boards.

6. Naming conventions

LAND	RULES	EXAMPLES
Meridian	Mixed origins; surnames that are words or numbers	Vesper Sette, Varro Kade, Maeve Triskel
Hyakki	Japanese-style given names; spirits named by number or object	Hina, Tamamo
River Empire	Chinese-style names with meaningful characters, especially numbers	Ba Jinling (eight golden bells), Bai Si (white four)
Śūnyapura	Sanskrit-derived words for concepts and numbers	Shunya (zero), Purna (fullness)
Kroton Nova	Greek-style, Latin numerals for rank	Prima, Archon Seventeen
Crossroads	Americana nicknames, Italian-American and Irish-American surnames	Fibi Leone, "Mr. Eighty-Nine"

Run every non-English name past a native speaker before release.

PART V

The lands

Seven places, seven palettes. Travelling between them should feel like changing countries.



Meridian Undercity

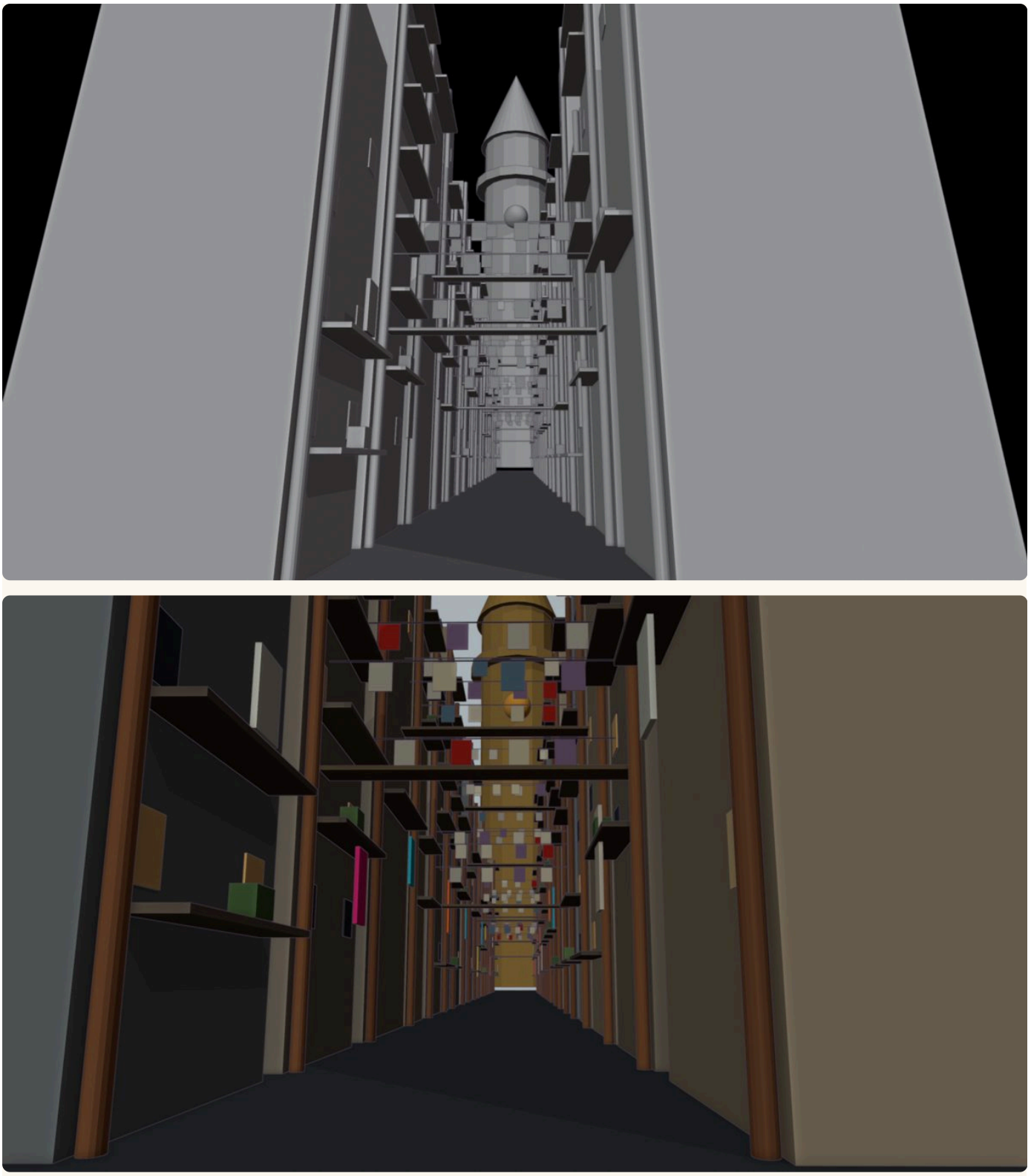
COLOUR SCRIPT

Wet slate	Window amber	Moss	Nixie	Neon yellow	Neon magenta	Neon cyan
#2E3A4B	#E8A04B	#5E7D4A	#FF7A2C	#F5E400	#FF2E88	#00E5FF





Pass 1 exploration art, in the superseded style. Kept for reference; to be redone under Art Direction v2. The Hyakki pier deck plate is the usable layout for the battle stage.



3D base (clay) and flat colour key for the v2 paint pass (Art Book Production Guide §4).

ACT I · JAPANESE FOLKLORE · UKIYO-E + PAINTED NIGHT

Hyakki

COLOUR SCRIPT

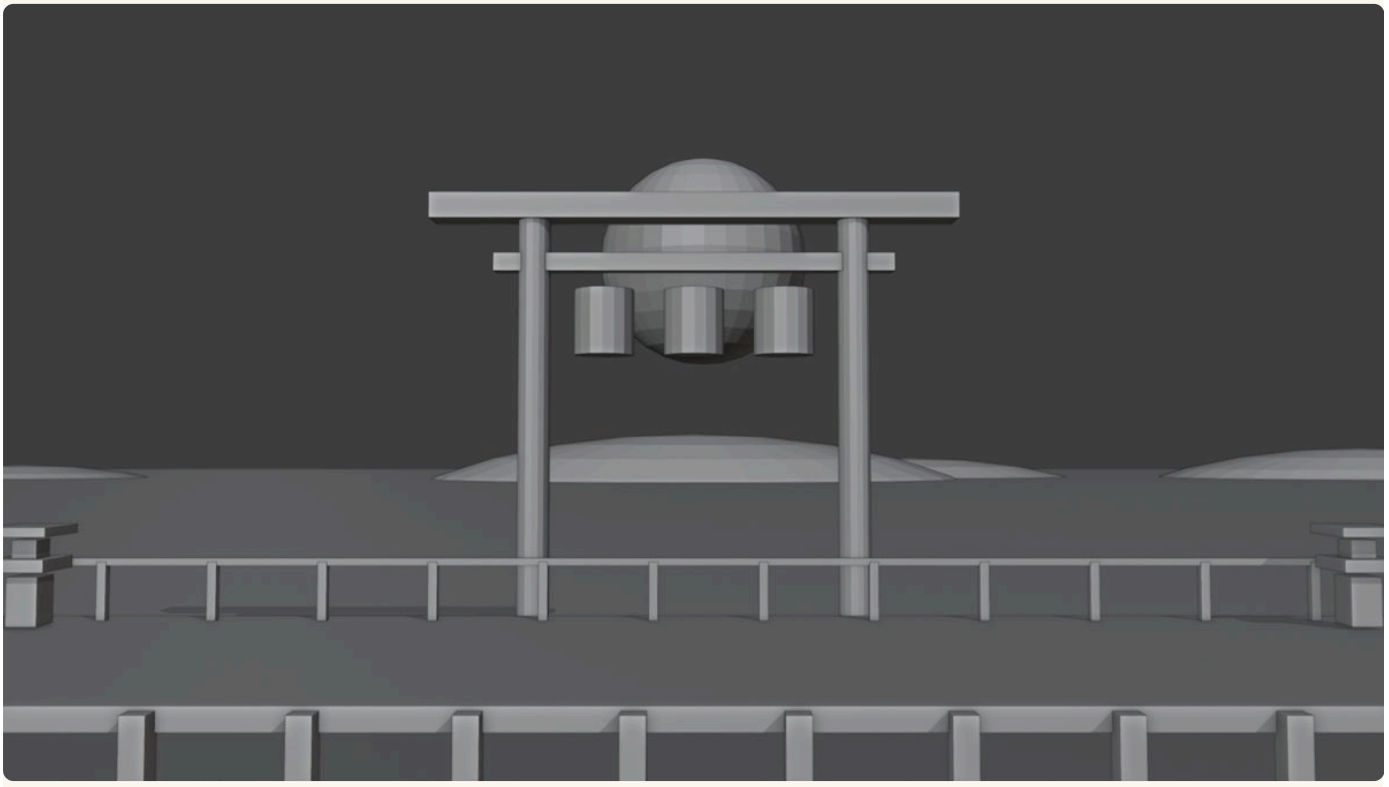
Indigo	Ink blue	Paper	Vermilion	Lantern gold	Firefly
#1B2440	#2E4A7A	#F2E6C9	#E34234	#F2B33D	#9BD15A







Pass 1 exploration art, in the superseded style. Kept for reference; to be redone under Art Direction v2. The Hyakki pier deck plate is the usable layout for the battle stage.





3D base (clay) and flat colour key for the v2 paint pass (Art Book Production Guide §4).

ACT II · CHINESE FOLK RELIGION · GONGBI GOLD + INK-WASH UNDERWORLD

The River Empire

COLOUR SCRIPT

Jade	Ochre	Red lacquer	Gold leaf	Rice paper	Ink green
#00A86B	#C8963E	#B22222	#D4AF37	#E9E2D0	#3A4A3F

Environment art pending

Establishing shot and battle stage to be generated with the background suffix in Art Direction §6, using this land's palette and the Locations Guide.

ACT III · INDIAN MYTH & MATHEMATICS · MINIATURE PAINTING + MONSOON LIGHT

Śūnyapura

COLOUR SCRIPT

Sandstone	Peacock	Saffron	Indigo	Brass	Eclipse
#D8A86B	#1E7B6F	#F4A300	#3F3D99	#B5893B	#4B2A6B

Environment art pending

Establishing shot and battle stage to be generated with the background suffix in Art Direction §6, using this land's palette and the Locations Guide.

ACT IV · GREEK MYTH · MARBLE, MOSAIC, AEGEAN NOON

Kroton Nova

COLOUR SCRIPT

Chalk	Aegean	Olive	Bronze	Sky
#F4F1EA	#1F6FB2	#7A8B4A	#8C6A3F	#D6E6F2

Environment art pending

Establishing shot and battle stage to be generated with the background suffix in Art Direction §6, using this land's palette and the Locations Guide.

ACT V · AMERICAN FOLKLORE · DESERT SUNSET + CASINO NEON

The Crossroads Territory

COLOUR SCRIPT

Rust	Sage	Dusk violet	Sunset	Midnight	Neon pink
#B7410E	#9CAF88	#5B3E73	#F28C28	#1D1A3A	#FF2E88

Environment art pending

Establishing shot and battle stage to be generated with the background suffix in Art Direction §6, using this land's palette and the Locations Guide.

FINALE · THE MACHINE · BRASS AND STEEL

The Basilica

COLOUR SCRIPT

Brass	Steel	Steel white	Nixie	Ink
#B5A642	#8E9196	#E8E6E1	#FF7A2C	#141018

Environment art pending

Establishing shot and battle stage to be generated with the background suffix in Art Direction §6, using this land's palette and the Locations Guide.

One page per land: what it is, how it looks, sounds and feels, and where the fights happen. Colour scripts and rendering rules are in `ART_DIRECTION.md` §4 and §8.

Meridian Undercity (Prologue) and the Basilica (Finale)

- **Inspired by:** a cosmopolitan modern capital; the immigrant quarters of every great port city.
- **Geography:** a plateau city; the Undercity is stacked in the drains and cliffs beneath it, with the Basilica's tower rising from the plateau's heart.
- **Districts:** the Remainder Wards (Remainder + Charmpunk) · the Charm Bazaar (pure Charmpunk) · the Exchange Terraces (Gilded Fortune over Ledgerism) · the Basilica Ring (pure Ledgerism).
- **Key locations:** Rain Alley 9 (where NIL wakes) · the black-market Naming shrine · Maeve's forge in the Celtic quarter · the drain gates (Varro's boss fight) · the sealed Upper Gate · the Basilica nave (finale).
- **Little things:** charms of every culture over every door; laundry between tenements; balcony gardens (a garden is luck you can see); cats; hand-painted lucky numbers.
- **Sound:** rain, Nixie hum, distant Dawn Tally bells, street musicians, Collector sirens.
- **Battle stages:** Rain Alley · the drain gates · the Bazaar crossroads · the Basilica nave.
- **Reel object:** a dead three-tube Nixie sign.

Hyakki (百鬼), the Isles of a Hundred Demons (Act I)

- **Inspired by:** Japanese folklore (yokai, tsukumogami, the Seven Lucky Gods, Enma); ukiyo-e and Edo-period towns.
- **Geography:** a chain of volcanic islands, cedar forests, terraced fishing towns, torii gates on every hill.
- **Culture:** humans and yokai share festivals; numbers 4 (*shi*) and 9 (*ku*) are avoided; 7 and 8 are blessed.
- **Key locations:** Moon Pier (arrival; battle stage) · the Night Parade bridge · the Shrine of the Seven Lucky Gods · the kappa river · Enma's gate of the dead (keeper trial) · Tamamo's foxfire palace (boss).
- **Little things:** daruma dolls with one eye painted; beckoning cats; fireflies; wind chimes; one tired kappa selling cucumbers.
- **Sound:** shamisen, shakuhachi, taiko for battle, cicadas and wind chimes.
- **Meridian intrusion:** a Ledgerist toll gate on the shrine road; Collectors in the harbour.
- **Reel object:** a torii with three hanging lanterns.

The River Empire: Bāfā Harbor (八发港) and the Ten Courts (十殿) (Act II)

- **Inspired by:** Chinese folk religion, festival culture and river trade.
- **Geography:** a great golden river delta; the harbor city of Bāfā on its banks; the river's end falls into the underworld of the Ten Courts.
- **Key locations:** the Lantern Festival · the Luck Exchange (live Omen price boards) · Jinling's pawnshop · the river's end · Naihe Bridge and Meng Po's pavilion · the Hall of the Book (Judge Cui's trial) · Jin Chan's coin palace (boss).
- **Little things:** red envelopes, mahjong tables, dragon boats, joss paper drifting into the river, lucky 8s on every price tag.
- **Sound:** erhu, guzheng, firecrackers, abacus clicks; the underworld is silent except for brush strokes.
- **Meridian intrusion:** the Exchange is half Basilica office, half temple.
- **Reel object:** a golden abacus (harbor); a floating ghost ledger (underworld).

Śūnyapura, the City Where Zero Was Born (Act III)

- **Inspired by:** Indian mythology, astronomy and mathematics; stepwells; open-air stone observatories.
- **Geography:** a monsoon plain; a city built around a colossal stone observatory; deep geometric stepwells.
- **Key locations:** the Great Observatory · the Stepwell of a Thousand Steps · the Hall of Chitragupta (keeper trial) · the Vetāla banyan · the eclipse plateau (Rahu & Ketu).
- **Little things:** a lemon and seven chillies over every door; rangoli patterns; chai stalls; pigeons; monsoon puddles mirroring the instruments.
- **Sound:** sitar, tabla, temple bells, rain on stone.
- **Meridian intrusion:** Basilica surveyors re-measuring the observatory's numbers.
- **Reel object:** three sundial dials.

Kroton Nova, the Sieve of the Aegean (Act IV)

- **Inspired by:** Greek mythology and Pythagorean number mysticism.

- **Geography:** a white island city built as a vast stone sieve over a blue sea; the cave of the Moirai beneath.
- **Key locations:** the Agora of Primes · the Sieve terraces · the undercity of the fallen (Remainder shanties) · the Moirai's cave (three trials) · the Archon's temple (boss).
- **Little things:** blue evil-eye beads; olive trees; goats on the terraces; mosaics of numbers.
- **Sound:** lyre, bouzouki, waves, chanting of primes.
- **Meridian intrusion:** the undercity of the "composite" people built from Basilica scrap.
- **Reel object:** three marble columns with mosaic numerals.

The Crossroads Territory (Act V)

- **Inspired by:** American folklore: frontier tall tales, the blues crossroads legend, immigrant charms, desert highways and casinos.
- **Geography:** a desert plateau crossed by highways; the casino boomtown of Lucky Strike, where the same night repeats.
- **Key locations:** the midnight crossroads (the deal) · Route 7 motels · the Ghost Train · the Hollow (the Headless Horseman) · the House's casino (boss).
- **Little things:** horseshoes over doors (open end up); rabbit's-foot keychains; lucky pennies on the railway; diners; neon cacti.
- **Sound:** slide guitar, harmonica, a slot-machine choir, coyotes.
- **Meridian intrusion:** the casino is financed by the Basilica; Collectors deal blackjack.
- **Reel object:** a giant slot machine.

The Far Lands (expansions)

Norse North (the Norns) · Kemet (Ma'at's scales) · the Long Count (Maya zero) · Anansi's Coast (Akan folklore) · the House of Wisdom (al-Khwārizmī). Each needs cultural consultants before design begins.

PART VI

Characters

Model pages first, then the full character bible.



NIL

Player's choice



Pass 1 exploration art, in the superseded style. Kept for reference; to be redone under Art Direction v2.

FROM **Meridian Undercity**





DESIGN FIGURE AND ITS SILHOUETTE (TOOLS/FIGURES.PY). SHAPE LANGUAGE: RECT + CIRCLE

PALETTE

Nixie orange	Ink black	Bone white	Rim cyan
#FF7A2C	#141018	#EDE6DA	#00E5FF

CANON LOCKS

- Decoder on the RIGHT wrist
- One orange streak in white hair
- Halo reads exactly "NaN"
- Coat's left tail is longer
- Amber eyes

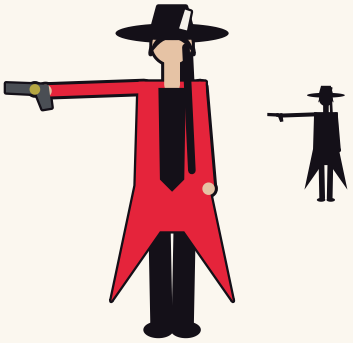
Vesper Sette

7



Pass 1 exploration art, in the superseded style. Kept for reference; to be redone under Art Direction v2.

FROM **Meridian / Crossroads**



DESIGN FIGURE AND ITS SILHOUETTE (TOOLS/FIGURES.PY). SHAPE LANGUAGE: TRIANGLES

PALETTE

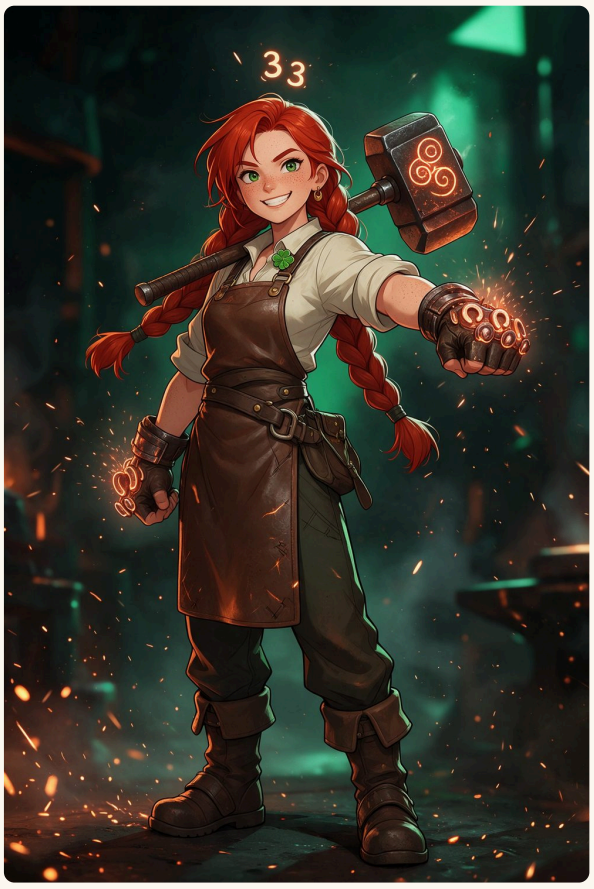
Crimson	Black	Brass	Shirt white
#E5243B	#141018	#B5A642	#F4F1EA

CANON LOCKS

- Exactly 7 chambers
- Scar on the lip
- Braid over the LEFT shoulder
- Crimson + black + brass only

Maeve Triskel

3



Pass 1 exploration art, in the superseded style. Kept for reference; to be redone under Art Direction v2.

FROM **Meridian, Celtic quarter**





DESIGN FIGURE AND ITS SILHOUETTE (TOOLS/FIGURES.PY). SHAPE LANGUAGE: CIRCLES

PALETTE

Copper	Clover	Leather	Red hair
#D9822B	#3BB273	#6B4226	#C0392B

CANON LOCKS

- Exactly 3 braids
- Triskele on the hammer
- Clover pin on the collar

Hina of the Ninety-Nine

9



Pass 1 exploration art, in the superseded style. Kept for reference; to be redone under Art Direction v2.

FROM **Hyakki**



DESIGN FIGURE AND ITS SILHOUETTE (TOOLS/FIGURES.PY). SHAPE LANGUAGE: DOME

PALETTE

Vermilion	Indigo	Paper	Gold
#E34234	#26428B	#F2E6C9	#F2B33D

CANON LOCKS

- The umbrella has ONE eye and a tongue
- Kimono numbers stop at 99
- Never reaches 100 before the finale

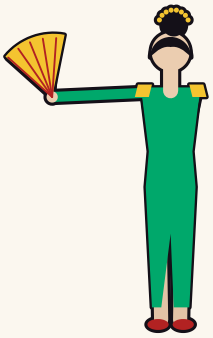
Ba Jinling

8

Key art pending

Generate with the character suffix in Art Direction §6 and the description in the Character Bible.

FROM **River Empire**





DESIGN FIGURE AND ITS SILHOUETTE (TOOLS/FIGURES.PY). SHAPE LANGUAGE: FAN

PALETTE

Jade	Gold	Lacquer red	Violet eyes
#00A86B	#F4C430	#B22222	#5D3A7A

CANON LOCKS

- Exactly 8 bells
- Abacus-rib fan
- Jade + gold palette

EXECUTIONER · COMPANION

Bai Si

4

Key art pending

Generate with the character suffix in Art Direction §6 and the description in the Character Bible.

FROM **River Empire (Ten Courts)**





DESIGN FIGURE AND ITS SILHOUETTE (TOOLS/FIGURES.PY). SHAPE LANGUAGE: VERTICAL

PALETTE

Mourning white	Ink	Talisman yellow	Lantern
#F4F1EA	#1B1B1B	#E8C547	#F2B33D

CANON LOCKS

- Always 4 lanterns, 4 talismans
- White robes only
- Never smiles before Act IV

Shunya

0

Key art pending

Generate with the character suffix in Art Direction §6 and the description in the Character Bible.

FROM **Śūnyapura**



DESIGN FIGURE AND ITS SILHOUETTE (TOOLS/FIGURES.PY). SHAPE LANGUAGE: ASTROLABE

PALETTE

Indigo	Saffron	Brass	Mirror silver
#3F3D99	#F4A300	#B5893B	#C9CED6

CANON LOCKS

- The brass astrolabe
- Mirror-work embroidery
- "Zero isn't nothing."

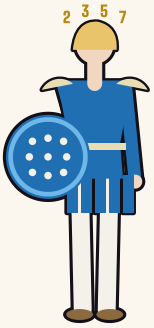
Prima of Kroton

2

Key art pending

Generate with the character suffix in Art Direction §6 and the description in the Character Bible.

FROM **Kroton Nova**





DESIGN FIGURE AND ITS SILHOUETTE (TOOLS/FIGURES.PY). SHAPE LANGUAGE: SHIELD

PALETTE

Aegean blue	White gold	Chalk	Glass blue
#1F6FB2	#E9DDB3	#F4F1EA	#6FB7E9

CANON LOCKS

- Sieve shield
- 2-3-5-7 halo
- Short hair

DEALER · COMPANION

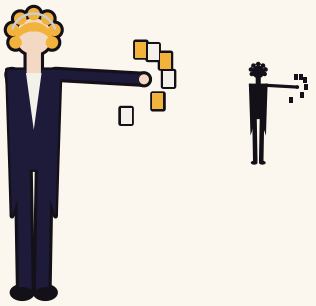
Fibi Leone

Fibonacci

Key art pending

Generate with the character suffix in Art Direction §6 and the description in the Character Bible.

FROM **Crossroads**



DESIGN FIGURE AND ITS SILHOUETTE (TOOLS/FIGURES.PY). SHAPE LANGUAGE: SPIRAL

PALETTE

Gold	Midnight	Card white	Neon pink
#F2B33D	#1D1A3A	#F4F1EA	#FF2E88

CANON LOCKS

- 1,597 nights
- Spiral irises
- Her digit changes every round

PROLOGUE BOSS → ALLY

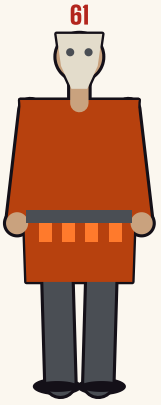
Varro Kade

61



Pass 1 exploration art, in the superseded style. Kept for reference; to be redone under Art Direction v2.

FROM **Meridian**



DESIGN FIGURE AND ITS SILHOUETTE (TOOLS/FIGURES.PY). SHAPE LANGUAGE: BLOCK

PALETTE

Rust	Gunmetal	Debt-jar glow	Mask bone
#B7410E	#4A4E54	#FF7A2C	#E3DCCB

CANON LOCKS

- Hound-skull mask
- Glass debt jars
- "61" in red

ACT I BOSS

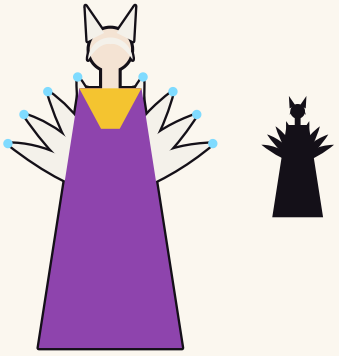
Tamamo

Nine tails



Pass 1 exploration art, in the superseded style. Kept for reference; to be redone under Art Direction v2.

FROM **Hyakki**



DESIGN FIGURE AND ITS SILHOUETTE (TOOLS/FIGURES.PY). SHAPE LANGUAGE: CRESCENT

PALETTE

Violet	Gold	Foxfire cyan	White
#8E44AD	#F4C430	#7FDBFF	#F4F1EA

CANON LOCKS

- Nine tails, each tipped with a digit
- Foxfire is cold cyan-violet
- Modest court robes (paint-over)

ACT II BOSS

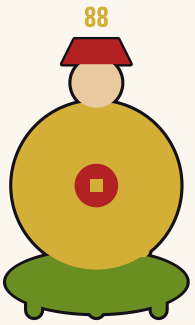
Chancellor Jin Chan

88

Key art pending

Generate with the character suffix in Art Direction §6 and the description in the Character Bible.

FROM **River Empire**





DESIGN FIGURE AND ITS SILHOUETTE (TOOLS/FIGURES.PY). SHAPE LANGUAGE: SPHERE

PALETTE

Gold	Toad green	Red
#D4AF37	#6B8E23	#B22222

CANON LOCKS

- Three-legged toad throne
- Coins spill from his mouth

ACT III BOSS

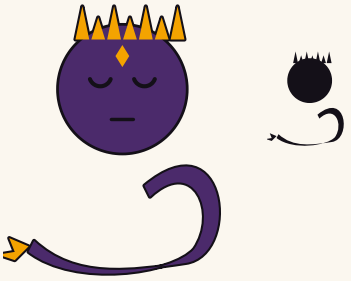
Rahu & Ketu

Eclipse

FROM **Śūnyapura**

Key art pending

Generate with the character suffix in Art Direction §6 and the description in the Character Bible.



DESIGN FIGURE AND ITS SILHOUETTE (TOOLS/FIGURES.PY). SHAPE LANGUAGE: ECLIPSE

PALETTE

Eclipse purple	Corona gold	Night
#4B2A6B	#F4A300	#141018

CANON LOCKS

- Awe-inspiring, never grotesque
- Two bodies, one fight
- Cultural review required

ACT IV BOSS

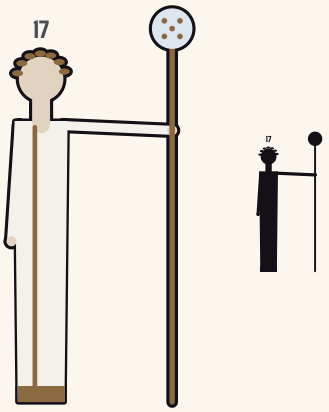
Archon Seventeen

17

Key art pending

Generate with the character suffix in Art Direction §6 and the description in the Character Bible.

FROM **Kroton Nova**



DESIGN FIGURE AND ITS SILHOUETTE (TOOLS/FIGURES.PY). SHAPE LANGUAGE: COLUMN

PALETTE

Bronze	Chalk	Sieve white
#8C6A3F	#F4F1EA	#DDE6EE

CANON LOCKS

- Bronze laurel
- Sieve staff
- Cold "17"

ACT V BOSS

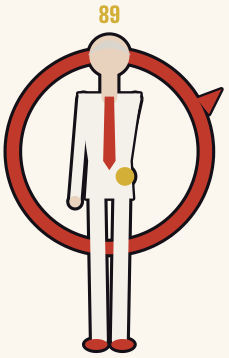
The House

89

Key art pending

Generate with the character suffix in Art Direction §6 and the description in the Character Bible.

FROM **Crossroads**





DESIGN FIGURE AND ITS SILHOUETTE (TOOLS/FIGURES.PY). SHAPE LANGUAGE: OUROBOROS

PALETTE

Ivory	Casino red	Gold
#F4F1EA	#C0392B	#D4AF37

CANON LOCKS

- White suit
- Ouroboros tattoo
- Backwards pocket watch

FINAL ANTAGONIST

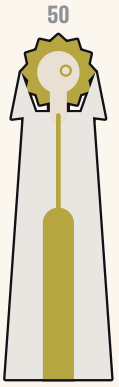
Comptroller Lucan Thale

50 (steel)



Pass 1 exploration art, in the superseded style. Kept for reference; to be redone under Art Direction v2.

FROM **Meridian**





DESIGN FIGURE AND ITS SILHOUETTE (TOOLS/FIGURES.PY). SHAPE LANGUAGE: ARCH

PALETTE

Steel white	Brass	Nixie	Steel
#E8E6E1	#B5A642	#FF7A2C	#8E9196

CANON LOCKS

- Gear monocle
- Steel, never glowing, "50"
- Never shouts

The single reference for how every character looks, talks, moves and changes. Story context lives in `WORLD_BIBLE.md`; visual rules in `ART_DIRECTION.md`. **Canon locks** are details that must never change between artists, writers or animators.

0. Cast at a glance

CHARACTER	ROLE	DIGIT	FROM	SHAPE	SIGNATURE COLOUR	RECRUITED / MET
NIL	Protagonist, the Remainder	player's choice	Meridian Undercity	Tall rectangle + broken circle	Nixie orange `#FF7A2C`	—
Vesper Sette	Companion, gunslinger	7	Meridian / Crossroads	Triangles	Crimson `#E5243B`	Prologue
Maeve Triskel	Companion, charm-smith	3	Meridian's Celtic quarter	Circles, arcs	Copper `#D9822B` + clover `#3BB273`	Prologue (optional)
Hina of the Ninety-Nine	Companion, tsukumogami	9	Hyakki	Dome on a small body	Vermilion `#E34234` + indigo `#26428B`	Act I
Ba Jinling	Companion, financier	8	River Empire	Fan, hexagon	Jade `#00A86B` + gold `#F4C430`	Act II
Bai Si	Companion, executioner	4	River Empire (Ten Courts)	Tall vertical, square shoulders	Mourning white `#F4F1EA` + ink `#1B1B1B`	Act II
Shunya	Companion, astronomer	0	Śūnyapura	Circle (the astrolabe)	Indigo `#3F3D99` + saffron `#F4A300`	Act III
Prima of Kroton	Companion, paladin	2	Kroton Nova	Shield circle on a sturdy frame	Aegean blue `#1F6FB2` + white-gold	Act IV
Fibi Leone	Companion, dealer	Fibonacci	Crossroads	Spiral	Gold `#F2B33D` + midnight `#1D1A3A`	Act V
Varro Kade	Boss (Prologue), later ally	61	Meridian	Square block	Rust `#B7410E`	Prologue
Tamamo	Boss (Act I)	99 (tails)	Hyakki	Fan and crescents	Violet `#8E44AD` + gold	Act I
Chancellor Jin Chan	Boss (Act II)	88	River Empire	Sphere	Gold `#D4AF37`	Act II
Rahu & Ketu	Boss (Act III)	eclipse	Śūnyapura	Circle + serpent line	Eclipse purple `#4B2A6B`	Act III
Archon Seventeen	Boss (Act IV)	17	Kroton Nova	Tall narrow column	Bronze `#8C6A3F` + chalk	Act IV
The House (Octavian Nine)	Boss (Act V)	89	Crossroads	Circle within a circle (ouroboros)	Ivory white + casino red	Act V
Comptroller Lucan Thale	Final antagonist	50 (steel)	Meridian	Cathedral arch	Steel white `#E8E6E1` + brass `#B5A642`	Intro (voice), Finale

Readability rule: every character must be identifiable as a solid black silhouette at 64 px tall, and by colour alone (Imaishi's principle: in 2D animation, characters are told apart mainly by colour).

1. NIL, the Remainder

Age / build	~19 (appears); 1.74 m; lean, long-legged, slightly hunched when thinking
Pronouns	They/them in all writing. NIL never defines their own gender, and nobody in the world asks.
Wardrobe	Long asymmetric black actuary's coat (left tail longer), high collar, three brass ledger clasps down the right side, fingerless black gloves, slim black trousers, tall boots
Props	Cracked brass **decoder** fused into the **right** wrist; leaks orange light and tiny floating digits when emotional
Omen	A broken holographic **NaN** glitch where an Omen should be; it flickers faster under stress
Personality	Quiet, observant, dry. Counts things under their breath when anxious (stairs, raindrops, people). Fascinated by other people's numbers. Kind without announcing it.
Flaw	Disappears into calculation instead of feeling; assumes they're a problem to be solved
Want → need	Wants a number and a normal life. Needs to accept being what everyone lost, and choose what to give back.
Voice	Short sentences. Precise words. Questions more than statements. Deadpan humour. Never swears.

Sample lines	"Fourteen steps. You've rounded down two of them." · "Everyone here has a number. I just have a question." · "If I'm what the world lost… I'd like to be returned carefully."
Animation personality	Economical movement; snaps into action. Idle: turns the decoder wrist, digits drift up. Victory: closes a fist on a locked digit, small exhale. On their lucky digit: halo stabilises for one frame. On a Bust: tilts their head, recalculating.
Growth	One new verb per act (GDD §6) and one memory flashback per Seal, each drawn in that land's art style
Canon locks	Decoder on the RIGHT wrist · one orange streak in white hair · halo reads "NaN" exactly · the coat's left tail is longer · amber eyes

2. Vesper Sette (7)

Age / build	28; 1.82 m; long limbs, sharp angles
Wardrobe	Wide-brim black hat with two playing cards in the band; crimson duster with triangular tails; black croupier waistcoat; white shirt, sleeves rolled; cards in her garters
Props	Revolver with a **seven-chamber** cylinder engraved 1–7; a worn silver coin she flips before every decision and ignores
Personality	Gambler's bravado, dry jokes, generous with strangers, merciless with cheats. Guilt she won't name.
Flaw	Treats every relationship as a bet she can walk away from
Want → need	Wants to stay one step ahead of her past. Needs to deal one honest hand.
Secret	Dealt the rigged hand at the House that Settled a twelve-year-old boy
Voice	Drawl, card and dice slang, calls NIL "Kid" until Act III, then by name
Sample lines	"Need a hand? Seven bucks." · "Luck's just math somebody else is hiding from you." · "I always flip. I never look."
Animation	Fan-the-hammer shooting; reloads by spinning the cylinder like a roulette wheel; idle hat tip; grins and spins the cylinder when a 7 lands
Bond arc	(1) Teaches NIL to read tells · (2) admits the rigged hand · (3) faces the House · (4) deals an honest hand to Fibi
Canon locks	Exactly 7 chambers · scar on the lip · crimson + black + brass only · braid over the LEFT shoulder

3. Maeve Triskel (3)

Age / build	20; 1.65 m; strong arms, freckled
Wardrobe	Leather forge apron, rolled work shirt, heavy boots; red hair in **three** braids
Props	Horseshoe-knuckle gauntlets (glow copper-hot when charged); forge hammer with a triskele burned into the head; four-leaf-clover pin
Personality	Loud, warm, superstitious about everything and proud of it; adopts every stray
Flaw	Stubborn; believes charms can fix things that need words
Want → need	Wants to make her grandmother's charms work again. Needs to accept that luck is something people make together.
Backstory	Her family's homeland luck was "rounded" into the Basilica; the day it happened, her grandmother's charms stopped working. She makes new ones.
Voice	Fast, warm, a little Irish-inflected; touches wood; says "third time's the charm" at every opportunity
Sample lines	"Horseshoe goes open end UP, or the luck falls out. Honestly." · "Three braids, three tries, three hits. That's just maths."
Animation	Swings with her whole body; kisses her clover pin when a 3 lands; hammer-on-anvil sparks as a victory pose
Canon locks	Exactly 3 braids · triskele on the hammer · clover pin on the collar

4. Hina of the Ninety-Nine (九十九 雛, 9)

Age / build	99 years old (as an umbrella); appears ~12; 1.35 m
Wardrobe	Furisode kimono in indigo and vermilion patterned with numbers 1–99; big obi bow; geta sandals; black bob
Props	Red-and-white oiled-paper umbrella with **one eye and a long tongue** (it's her true body; it has its own expressions)
Personality	Mischievous, ancient and childish at once; obsessed with turning 100; secretly scared of not being "real"
Want → need	Wants to reach 100 and become a "proper" spirit. Needs to decide she's already real.
Backstory	A 99-year-old umbrella left at Enma's shrine, soaked in a century of wishes, woke up one year early. Fixed Fate would freeze her at 99 forever.
Voice	Formal old-fashioned phrasing mixed with kid slang; refers to herself in the third person when excited ("Hina approves!")
Sample lines	"One more year and Hina would've been a *real* spirit. Don't you dare freeze me now." · "You smell like a hundred wishes."

Animation	Rides the umbrella; the umbrella reacts separately (blinks, sticks its tongue out on a 9); twirls on a Jackpot
Cultural note	Based on *tsukumogami* and *kasa-obake* folklore; keep her playful, never creepy-cute fan service
Canon locks	The umbrella has ONE eye · kimono numbers stop at 99 · she never reaches 100 before the finale

5. Ba Jinling (八金铃, 8)

Age / build	31; 1.70 m; poised
Wardrobe	Modern jade-and-gold qipao with a high slit and structured shoulders; eight tiny golden bells in hair ornaments and anklets
Props	Folding fan whose ribs are **abacus rods** (beads fly as projectiles)
Personality	Theatrical, sly, mercenary on the surface; buys poor people's debts and burns them in secret
Want → need	Wants to outplay her father. Needs to stop hiding her kindness behind business.
Backstory	Chancellor Jin Chan's daughter; left the Luck Exchange when she learned what built his fortune
Voice	Silk and knives; speaks in interest rates ("That favour will cost you eight percent, compounding.")
Sample lines	"Everyone's in debt, darling. I just keep better books." · "Eight bells. Count them. You won't hear the ninth."
Animation	Fan snap; every gesture rings a bell; collects coins mid-fight with a flick
Canon locks	Exactly 8 bells · the abacus fan · jade + gold palette

6. Bai Si (白四, 4)

Age / build	Looks ~35; 1.95 m; long and still
Wardrobe	White mourning robes (white is the mourning colour in Chinese tradition) inked with 四; black funeral ribbon in long white hair
Props	Huge guandao-style blade hung with four paper talismans; joss paper drifts around him; four floating lanterns
Personality	Stoic, formal, honour-bound; unexpectedly tender with children and the old; dry humour arrives late in the story
Want → need	Wants the count of the dead to be correct. Needs to forgive himself for centuries of wrong counts.
Backstory	The Ten Courts' executioner who counts every Settled soul. NIL shows him the Rounding; his count was always short.
Voice	Few words, archaic and formal; never uses contractions
Sample lines	"Everyone dies. I only keep count." · "Four lanterns. I have never needed a fifth."
Animation	Almost motionless idle; each strike extinguishes one lantern; closes his eyes when a 4 lands
Canon locks	Always 4 lanterns and 4 talismans · white robes, never coloured · never smiles before Act IV

7. Shunya (शून्य, 0)

Age / build	23; 1.62 m; restless energy
Wardrobe	Indigo-and-saffron layered clothing with mirror-work embroidery; long dark braid with brass astrolabe hairpins; ink-stained fingers
Props	Brass astrolabe (spun as a weapon), palm-leaf manuscripts tied at her hip
Personality	Brilliant, impatient, joyful about ideas, gentle with people who are slow
Want → need	Wants to prove her Deficit calculations. Needs to remember the answer is a person (NIL), not a number.
Backstory	Works at the great observatory where her ancestors formalised zero; found that the Deficit equals exactly one person's worth of luck
Voice	Talks fast, interrupts herself, explains with her hands; calls zero "the place that holds everything"
Sample lines	"Zero isn't nothing. It's room." · "The books are short by one soul. Exactly one. Hello."
Animation	Spins the astrolabe like a discus; sketches equations in the air in light; beams when a 0 lands
Cultural note	Draws on Indian mathematical history (zero, place value); have the design and clothing reviewed
Canon locks	The astrolabe · mirror-work embroidery · the line "Zero isn't nothing."

8. Prima of Kroton (2)

Age / build	18; 1.68 m; sturdy
Wardrobe	White-and-gold armour with Greek-key borders; short blonde hair; blue eyes

Props	Giant round shield shaped like a sieve set with blue glass; a halo of floating glyphs 2-3-5-7
Personality	Earnest, loud, honest to a fault; comic relief who becomes the moral backbone
Want → need	Wants to be a proper member of the Synod. Needs to see that "indivisible" means people, not numbers.
Backstory	2 is "the even prime", the odd one out among primes, and so is she
Voice	Formal battle cries, then clumsy apologies; counts primes to calm down
Sample lines	"Two! Three! Five! Seven! ...I'm calm." · "If the Sieve drops children, the Sieve is wrong."
Animation	Shield-bash; plants her feet like a statue; trips once per fight in idle
Canon locks	The sieve shield · the 2-3-5-7 halo · short hair

9. Fibi Leone (Fibonacci)

Age / build	18; 1.60 m; jittery
Wardrobe	Tailored croupier tailcoat and headset; golden spiral curls; spiral-patterned irises
Props	A deck of glowing cards that flows around her in a golden spiral
Personality	Fast-talking, anxious, brilliant; laughs at bad moments; has lived the same night 1,597 times
Want → need	Wants the night to end. Needs to accept the dawn, even though it may kill her.
Twist	Her grandfather, the House, loops the night so she never dies at dawn
Voice	Rapid; counts in Fibonacci; finishes other people's sentences (she's heard them before)
Sample lines	"You're going to say 'Who are you?' Then you'll trip on the third step. ...See?" · "One, one, two, three, five. That's how I breathe."
Animation	Deals cards in spirals; her lucky digit glows in a different place every round
Canon locks	1,597 nights · spiral irises · her digit changes every round

10. Antagonists and keepers

CHARACTER	LOOK (LOCKS)	MOTIVE	VOICE	FATE
Varro Kade	Hound-skull mask, rust trench armour, glass debt jars on his belt, "61" in red	Buy his sister's exemption	Growls in debt-collector jargon: "Interest is due."	Spared; helps in Act V
Tamamo	Nine glowing tails each tipped with a digit, Heian court robes in violet and gold, foxfire	Keep the yokai from fading	Teasing, archaic, kind cruelty	Retreats into legend; blesses Hina
Jin Chan	Gold-coin robes, three-legged toad throne, coins spill from his mouth	Hold the world's frozen luck in his bank	Booming, jolly, transactional	Bankrupted; Jinling takes over
Rahu & Ketu	A vast crowned celestial head and a headless serpent body with a comet tail; awe-inspiring, never grotesque	Make their stolen immortality permanent	Rahu speaks; Ketu answers only in tremors	Returned to the sky as a passing eclipse
Archon Seventeen	Gaunt, bronze laurel, sieve staff, cold "17"	Purify the world: only the indivisible should remain	Soft, scholarly, absolute	Falls through his own sieve; Prima saves him
The House	White suit, ouroboros tattoo on his throat, backwards pocket watch	Keep his granddaughter alive by looping the night	Old-world charm, gentle menace	Lets the dawn come
Comptroller Thale	White-and-brass cathedral vestments, gear monocle, steel "50"	Fixed Fate: end all loss by ending all chance	Calm, kind, exhausted; never shouts	Depends on the ending
Keepers (Enma, Judge Cui, Chitragupta, the Moirai)	Shown with reverence: solemn, beautiful, never monstrous	Test the party	Measured, ritual language	They are never "defeated", only satisfied

11. Relationship map

	NIL	VESPER	MAEVE	HINA	JINLING	BAI SI	SHUNYA	PRIMA	FIBI
NIL	—	Big sister / first protector	Best friend	Little sister they adopt	Wary respect → trust	Student of grief	Equal; the one who explains them	Protects her	Mirror (both outside time)
Vesper	Protector	—	Sparring jokes	Teaches her poker	Old rival from the casinos	Respect	Finds her exhausting	Mocks, then defends	Guilt; the honest hand

	NIL	VESPER	MAEVE	HINA	JINLING	BAI SI	SHUNYA	PRIMA	FIBI
Maeve	Best friend	Hero-worship	—	Makes her charms	Haggles	Forges his new lanterns	Asks too many questions	Fellow loud one	Big-sister energy
Hina	"Hundred-wish human"	"Hat lady"	Charm-maker	—	Wants her bells	Scared of him, then adores him	Calls her "the zero girl"	Rides on her shield	Twins in being "not quite real"

Party banter rule: every pair of companions has at least one exchange per act; Complement pairs (digits summing to 10) get an extra bond scene.

12. Writing and voice quick rules

1. NIL never wastes a word. Companions talk more than NIL does.
2. Each character has one **number habit**: Vesper flips a coin, Maeve counts threes, Hina counts to 100, Jinling quotes interest, Bai Si counts lanterns, Shunya says "zero isn't nothing", Prima recites primes, Fibi counts in Fibonacci.
3. Villains are never cartoonishly evil; every Auditor's motive is understandable in one sentence.
4. Cultural characters speak as people, not as tour guides; no fake accents in text.
5. No one comments on NIL's gender.

13. Secret recruits (digits 1, 5 and 6)

The core roster leaves three digits without a companion. Three optional recruits fill them, so every lucky number the player can pick has a Twin and a Complement somewhere in the game. Each needs a side quest; none are needed for the true ending.

RECRUIT	DIGIT	FOUND	LOOK	KIT IN ONE LINE	WHY THEY JOIN
Varro Kade	1	Act V (guest), recruitable after the finale or on a second playthrough	Hound-skull mask pushed up on his head, rust trench armour with the debt jars smashed	*First and Only*: his strike always lands first and its digit is copied to the reel's empty slots on a Bust	Thale broke his promise about Varro's sister; he owes NIL one
Kyūsuke the Kappa	5	Act I side quest (the tired kappa selling cucumbers on the river)	Small green river imp, water dish on his head, straw hat, a cucumber cart	*Bow*: once per fight he bows; if the enemy bows back (land a 5), the enemy loses its next strike	NIL buys every cucumber; he's never had a full day's sales
Granny Liu Shun	6	Act II side quest (the mahjong tables of Bāfā Harbor)	Tiny, 80-ish, jade bangles, cardigan over a qipao, a mahjong tile purse	*Six-Six Smooth* (六六大顺): two 6s on the reel make every remaining strike this round land in order	She won the party in a mahjong game and refuses to give them back

Canon locks: Kyūsuke's water dish is always full (if it spills he naps); Liu Shun never loses at mahjong on screen; Varro never wears the mask over his face again after Act V.

14. Side characters by region

Side characters carry the regions' stories and quests. Each has one want, one number habit and one quest (Act Design per land).

REGION	CHARACTER	ROLE	WANT	NUMBER HABIT	QUEST / PAYOFF
Meridian	**Ama Okoro** and her son **Tobi** (age 9)	The mother whose Omen drops to 01 in the intro; her son	Ama: to see Tobi grow up. Tobi: to be NIL's first follower	Tobi counts NIL's wins on his fingers	Tobi appears in every act's epilogue letter; in the true ending he is the first child to perform the new Naming Rite
Meridian	**Brother Nought**	Defrocked Ledger-priest who runs the black-market Naming shrine; secret member of the Open Hand	To see Hap free before he dies	Rounds everything up out of spite	Gives NIL the first charm slot; reveals the Open Hand faith in Act II's interlude
Meridian	**Livia Kade**	Varro's younger sister, Omen 06, bedridden	For her brother to stop	Counts the cracks in the ceiling	Saving her in Act V turns Varro into an ally
Meridian	**Gus Mendoza**	Charm Bazaar fixer from the Crossroads, sells every culture's charms	Profit; secretly, to go home	Haggles in sevens	Runs the Prologue market; later found in Lucky Strike
Meridian	**Clerk Ivo Marr**	Thale's quiet assistant, Ledgerist	To be noticed	Files everything in triplicate	Defects in the finale and opens the Upper Gate's service door
Hyakki	**Ofuku**	Elderly keeper of the Seven Lucky Gods' shrine	That people still come to the shrine	Paints one daruma eye a day	Teaches charms; her shrine is the Act I charm hub

REGION	CHARACTER	ROLE	WANT	NUMBER HABIT	QUEST / PAYOFF
Hyakki	**Tetsu**	Ferryman between the islands, a former Collector	To forget Meridian	Counts strokes of the oar	Ferry nodes; his story foreshadows the Basilica's reach
Hyakki	**Kitsune twins Ao and Aka**	Tamamo's foxfire retainers	To protect their mistress	Speak only in pairs	Elite fight; after Tamamo's defeat they guard Hina's shrine
River Empire	**Judge Cui** (keeper)	Holder of the Brush in the Ten Courts	A true count of the dead	Writes every number twice	Grants NIL the Rewrite verb
River Empire	**Little Lantern (Xiao Deng)**	Festival orphan who runs messages across Bāfā	A family	Counts lanterns	Leads the party to Jinling's pawnshop; adopted by Jinling in the epilogue
Śūnyapura	**Acharya Vidya Rao**	Shunya's old teacher at the observatory	To finish his star catalogue	Counts in Sanskrit numerals	Gives the party the observatory dials; his catalogue's last entry is NIL
Śūnyapura	**Meera**	Stepwell guide and diver	To find her drowned brother's karma record	Counts steps (there are 1,001)	The stepwell dungeon; payoff in Chitragupta's Hall
Śūnyapura	**Baital**	A Vetāla who riddles travellers from a banyan tree	A riddle nobody can answer	Answers only in questions	Riddle encounters; NIL's answer to his last riddle is "zero"
Kroton Nova	**Mother Kallisto**	Leader of the undercity of the "composite" people under the Sieve	Her people back on the surface	Counts in composites, proudly	Undercity hub; the Sieve's fall reunites the city
Kroton Nova	**Niko**	Goat-herd kid who keeps falling through the Sieve	To be a hero	Counts goats (always one short)	Running gag; saved by Prima in her defection scene
Kroton Nova	**The Moirai** (keepers)	Clotho, Lachesis and Atropos	That every thread is spun, measured and cut honestly	Speak in turns	Three trials; Lachesis gives Prima her strand of thread
Crossroads	**The Stranger** (keeper)	The figure at the midnight crossroads with the black book	A fair deal	Shuffles constantly	Offers NIL's Deal verb; never a villain
Crossroads	**Penny**	Ghost girl on the railway who collects lucky pennies	To spend one	Counts pennies (1,596 of them)	Ghost Train quest; her 1,597th penny frees her, echoing Fibi
Crossroads	**Sheriff Dolores Vance**	Lucky Strike's sheriff, half-aware of the loop	To arrest the House	Notches her badge each night	Loop puzzles; helps in the House fight
Everywhere	**Hap**	The god of luck (World Bible §2A)	To deal again	—	Hap's Dream epilogues; the finale's Open Hand

15. Hap: depiction rules

- **Never the same face twice.** In each act's dream, Hap wears the face that land's stories would expect, drawn in that land's art style; the dice eyes, the orbiting digits and the star-wheel behind it never change.
- **Scale and warmth.** Vast but gentle; it crouches to talk to NIL. It laughs often, and its laughter makes dice rattle.
- **Voice.** Short, playful lines; questions more than answers. It never explains the plot. Sample lines: "Pick a number. Any number. …Oh, *that* one. Interesting." · "I don't play fair. I play *everyone*." · "You're the change I never got back."
- **Never** a monster, a villain, a boss or a stand-in for any real religion's deity.
- **Colour:** star-black, warm dice-ivory eyes, and the orbiting digits in every land's accent colour in turn.

PART VII

Art direction

Painted world, inked people.



> **v3 (Nov 7): "Painted world, inked manga."** The v2 foundation below stays: painted Ghibli-spirit worlds, graphic Edgerunners-spirit people and fights, Cyberpunk-art-book method for eras and presentation. v3 adds a **coloured-manga layer** for characters, fights, comedy and cutscenes, learned from the *methods* of dark-fantasy seinen (*Berserk*), modern shōnen action (*One Punch Man*, Yusuke Murata's art) and versus-battle manga, with **witty humour** as a core register. §0 below is the v3 rulebook; where it conflicts with v2, v3 wins.

0. v3: the coloured-manga layer

0.1 What we take from each influence (method, never likeness)

INFLUENCE	WHAT IT DOES BRILLIANTLY	WHAT WE TAKE	WHAT WE NEVER TAKE
Berserk (Kentaro Miura)	Weight, scale and dread through ink: dense hatching, spotted blacks, monumental low-angle compositions, grit on every surface	**Ink density as drama**: bosses, the Basilica and Noise storms get hatching and heavy blacks; low angles that make Auditors feel huge; texture on armour, stone, brass	Gore, body horror, cruelty for its own sake (our rating is Teen); any character, armour or panel likeness
One Punch Man (Murata's art, ONE's writing)	Hyper-detailed, perfectly choreographed fights, then a **deadpan gag face** that deflates the whole spectacle; comedy and awe in the same chapter	**Two registers** (§0.2): spectacle drawn at full power, then a simplified deadpan beat; fight choreography you can follow panel to panel; scale shots with debris	Copying the bald-hero gag, costumes or specific panels
Versus-battle manga (the "VS" face-off tradition of fighting manga)	The face-off: two fighters split by a diagonal, names and stats, before the clash	The **Versus card** before every boss and duel (§0.5)	Real logos or game UIs
Cyberpunk 2077 / Edgerunners	Neon colour as emotion, era-based costume logic, kinetic impact frames	Kept from v2 (§2.2, §8): neon in Meridian and combat, eras, colour-step slow-mo	—
Ghibli backgrounds	Air, light and lived-in detail	Kept from v2 (§2.1): every land painted	—

0.2 Two registers: SERIOUS and GAG (the humour engine)

The same character can be drawn in two registers; switching between them *is* the joke, and *is* the drama.

	SERIOUS (DEFAULT IN FIGHTS, REVEALS, BOSS PHASES, ULTIMATES)	GAG (DIALOGUE ASIDES, BARKS, MENUS, FAILURE MOMENTS)
Line	Full variable-weight ink; hatching in shadows; spotted blacks	Clean thin line, 30% of the detail
Faces	Full anatomy, sharp eyes, sweat and grit	Simplified: dot eyes, flat mouths, NIL's deadpan "..." face, Hina's chibi rage
Colour	Full cel (2 tones + rim) + screentone midtones	Flat fill, one tone, pastel background swap
Camera	Low angles, foreshortening, Dutch tilts	Flat, front-on, centred (the comedy of stillness)
Timing	Holds before impact, smear frames, impact frames	Hard cut, hold 0.6–1.0 s on the deadpan, then snap back

Rules: (1) a gag beat always follows a big beat, never interrupts its build-up; (2) at most one gag per fight round; (3) keepers, Hap's dreams and genuine grief are never played as gags; (4) NIL is the straight face of the cast, and the humour usually comes from everyone else reacting to NIL's calm.

0.3 Inking and colouring characters (coloured-manga look)

1. **Ink first, colour second:** characters read as finished ink drawings even in greyscale. Colour is flat digital colour laid under the ink, as in coloured manga pages.
2. **Spotted blacks:** every character has 1–3 areas of solid black (NIL's coat, Vesper's hat, Varro's mask sockets) placed to lead the eye to the face and hands.
3. **Hatching budget:** SERIOUS register only; heroes $\leq$ 15% of the figure; bosses and the Basilica up to 40%; never on faces except in shadow.
4. **Screentone:** midtones in SERIOUS register use a halftone dot screen (45°, 10–12% coverage, ink colour at 60%) instead of a soft gradient. In engine: a screen-space halftone shader driven by the shadow mask.
5. **Line colour:** keep v2's coloured lines for skin and light cloth; ink-black lines (`#141018`) for silhouettes in SERIOUS register.
6. **Grit:** wear marks, scratches and stains drawn as ink texture on props and armour (Berserk lesson), never as photo texture.

0.4 Fight presentation (choreography you can follow)

Every strike is a three-panel story: **key pose → impact → aftermath**. | Beat | Frames (60 fps) | Visual | |---|---|---| | Anticipation | 6–10 | Wind-up key pose, held; speed lines start converging | | Smear | 2–3 | Smear frame with speed lines in the striker's colour | | **Impact frame** | 2–3 | Full-screen inversion: black and paper-white plus the attacker's signature colour only; jagged burst shape; the stopped digit drawn huge in the burst | | Hit-stop | 5–13 | Freeze on impact (GAME_FEEL §2) with screen shake | | Aftermath | 20–40 | Debris silhouettes, dust with flat shadow, the enemy's Omen ticking down | | Gag beat (optional) | 36–60 | Deadpan reaction in GAG register (e.g. Vesper blowing smoke off the barrel, NIL: "Fourteen.") |

Scale shots: every boss intro and phase change has one extreme low-angle "monument" frame (Berserk lesson): the boss fills 80% of the frame, the party tiny at the bottom. **SFX lettering:** onomatopoeia are drawn, not typed, and **built from digits** (the clack of a stop is lettered "K-7-K"; a Jackpot is "7 7 7" exploding out of the panel), in the land's style (brushed in Hyakki, neon tube letters in Meridian, chiselled in Kroton Nova).

0.5 The Versus card (before every boss and duel)

A 2.5 s full-screen card: a jagged diagonal split; each side a SERIOUS-register bust shot over that side's land colour; names in display type; their Omens; one line of trash talk each; the boss's rule in one line at the bottom; a stamp in the centre: **VS** formed from two crossed digits. Skippable after first view. (Prototype target in the jam slice: NIL & Vesper VS Varro.)

0.6 Cutscenes as animated manga pages (motion comics)

Cutscenes are built as **animated manga pages**: panels slide in on beats, the camera pushes inside a panel, speech balloons pop with sound, and a panel can break out into a full-screen moment. This is cheaper than full animation, deterministic in Unity, and on-style. | Panel type | Use | |---|---| | Wide establishing (borderless, painted) | New place; the Ghibli half | | Reaction close-up (narrow, tall) | Emotion; the GAG register's home | | Versus split | Confrontations | | Splash (breaks the page) | Reveals, ultimates, REMAINDER DETECTED | | Aside (tiny inset, chibi) | Jokes, the running gags | **Page rules:** read order right-to-left is **not** used (our audience reads left-to-right); 3–6 panels per page; gutters in ink black; one splash per scene maximum.

0.7 Speech balloons and lettering

SPEAKER	BALLOON	FONT FEEL
NIL	Thin rectangular box, lots of white space	Small, lower-case, precise
Vesper	Jagged edge on punchlines	Swaggering italic
Maeve	Round, slightly wobbly	Loud, big caps on "THIRD TIME!"
Hina	Bubbly cloud	Mixed sizes; "Hina approves!" in a burst
Bai Si	Square, black fill, white text	Sparse, formal
Thale	Typeset steel plate with rivets	Clean serif, never shouts
Hap	Border of dice dots	Letters that roll into place
Collectors / Basilica	Printed form boxes	Stamped type

0.8 Colour in v3

Keep v2's signature colours and land colour scripts. Add: **paper white** `#F4EFE4` for impact-frame inversions and manga panels; **ink** `#141018`; one **screentone tint** per land (the land's shadow tint from ART_SPEC §2.3 at 60%). Neon (yellow/magenta/cyan) stays Meridian- and combat-only.

0.9 Free-first production pipeline (local tools primary)

STEP	TOOL (FREE, LOCAL)	OPTIONAL ACCELERATOR
Poses, panel layouts, environment bases	**Scripted Blender** (`tools/render_cast.py`, `tools/build_envs.py`), Freestyle/outline renders	Meshy base meshes
Character references	Turnarounds in `art/cast/` as the fixed reference; hand-drawn model sheets over them (Krita)	Higgsfield/FLUX paint-overs (disclosed)
Ink and colour	Krita (free) with a halftone brush set; flat colour under ink	—
Panels, balloons, SFX	Krita / Inkscape; balloon SVG kit	—

STEP	TOOL (FREE, LOCAL)	OPTIONAL ACCELERATOR
In engine	Unity: deterministic sprite/PSB animation, Timeline-driven motion-comic pages, impact-frame and screentone shaders	—

Determinism: every cutscene and fight beat is driven by authored timelines and seeded RNG, so the same input always produces the same frames (TECH_DESIGN §4).

0.10 QC additions (v3)

- [ ] Reads in greyscale (ink first)
- [ ] Correct register (SERIOUS vs GAG) for the moment; gag only after a big beat
- [ ] Impact frame present on every big hit; hit is followable panel to panel
- [ ] SFX lettered from digits, in the land's style
- [ ] No likeness to any existing character, armour, logo or panel

Art Direction v2 (foundation, still in force)

"Painted world, inked people." Hand-painted, sunlit, lived-in **worlds in the spirit of Studio Ghibli's backgrounds**, inhabited by **graphic, high-contrast, neon-edged characters and action in the spirit of Studio Trigger's** *Cyberpunk: Edgerunners*.

The two styles are not mixed at random. They carry the game's theme: - **The lands** (the old, plural ways of seeing luck: nature, craft, community) are **painted**: soft air, warm light, weather, moss, laundry, cats on roofs. - **Meridian and the Basilica** (the machine that flattens luck into one number) are **neon and graphic**, and so is **combat**, where numbers fight. - As the story moves, the neon of the machine intrudes on the painted lands (Noise storms, Auditors), and the finale reverses it.

This file replaces the style sections of the prompt pack. If a rule here conflicts with older docs, this file wins.

1. What went wrong in pass 1 (honest diagnosis)

PROBLEM	WHY IT LOOKS CHEAP	FIX IN V2
AI gloss: airbrushed gradients on skin and cloth, plastic highlights, everything rendered to the same detail	Reads as "generic AI anime illustration", not a production frame	Characters use **flat cel shading: 2 tones + 1 hard rim light**, no gradients except a hint on hair. Detail goes where the eye should go.
No hierarchy between character and background: both painted the same way	Characters don't pop; the image is mush	Anime's oldest rule: **flat cel characters on painted backgrounds**. Different rendering for each layer.
Stiff, centred gacha poses from one mannequin	Every character feels like a figurine on a turntable	Poses from **gesture and story**: lunges, leans, foreshortening, Dutch angles for action; quiet, grounded poses for drama.
Muddy teal-and-orange everywhere	No identity; characters don't read at thumbnail size	**One signature colour per character** plus a land colour script (§4). Value contrast first, hue second.
Thin, uniform or missing lineart	Shapes melt into the background	**Variable-weight coloured lines**: thick silhouette, thin interior, line colour tinted toward the local colour (never pure black on skin).
Floating digits and "Nixie" glow everywhere	The motif becomes noise and creates text artefacts	Numbers appear **only where the fiction puts them**: Omens over heads, wheels, the reel, signage.
Over-detailed, evenly lit backgrounds	No air, no time of day, no story	**Atmospheric perspective, one clear light source, weather, and lived-in details** chosen on purpose.

2. The two halves

2.1 The painted world (Ghibli-inspired backgrounds)

What the research says about how those backgrounds are made (sources at the end): - **Medium:** gouache-like *poster colour*, opaque but handled so gradations "look like transparent watercolor"; **a small palette (~21 colours) mixed subtly**; never a thick coat. - **Process:** the base wash goes down wet, on pre-wetted paper, which is where the soft cloud edges and natural gradations come from; detail is added after it dries. - **Philosophy:** *"He's not rendering everything. But he is rendering the little things."* The background painter extracts "the essence of important elements of the scene" and augments it. Backgrounds serve the film's vision, and they come from **on-site observation** (mist rising from valleys, the way real foliage grows), not from photos.

Our rules: 1. **Gouache look**: visible but soft brush texture, opaque colour, wet-edge gradients in skies, slightly matte. No photo textures, no 3D-render sheen. 2. **A limited palette per land** (≈20 colours), mixed subtly. 3. **Light tells the time.** Every background has one motivated light source and a time of day: warm late-afternoon sun, blue dusk, lantern night, rain dawn. 4. **Air.** Distant planes get lighter, cooler and softer (atmospheric perspective). Big skies with towering cumulus or layered stratus. 5. **Render the little things on purpose**: one cat, a drying futon, a bicycle, potted plants on a fire escape, a hand-written shop sign, moss on a stone. Two or three per frame, not fifty. 6. **Nature wins**: plants grow out of everything, even in Meridian. It's the visual sign of the old plural world surviving. 7. **Layouts for play**: battle plates have a clear floor plane, a calm area behind the fighters, and the reel object top-centre (GDD §8).

2.2 The inked people and the fights (Edgerunners-inspired)

What the research says: - **Colour carries character and emotion.** Imaishi (director): small differences in age and clothing texture vanish in 2D animation, so **characters are told apart mainly by colour**. There are **no rules for rim-light colours; they're matched to the scene's emotion**. The high-contrast palette came out of *Promare* (Shigeto Koyama's colour design). - **Motion:** in the famous slow-motion sequences, each frame of a movement is drawn and **tinted a different colour**, spreading the frames out to stretch time. - **Design:** Yoh Yoshinari's designs are deliberately cartoon-ish and stylish. In his words, the look is "associated with very retro anime". A large library of hand-drawn effects animation does the heavy lifting. - The neon identity leans on a **loud yellow / magenta / cyan** scheme.

Our rules for characters: 1. **Silhouette first.** Each character is built on one dominant shape and readable as a black fill at 64 px tall:

CHARACTER	SHAPE	SIGNATURE COLOUR	SECONDARY
NIL	Tall rectangle + broken circle (the NaN halo, the coat)	Nixie orange `#FF7A2C`	Ink black, bone white
Vesper	Triangles (hat brim, coat tails, gun)	Crimson `#E5243B`	Black, brass
Maeve	Circles and arcs (braids, horseshoes)	Copper `#D9822B`	Clover green `#3BB273`
Hina	Dome (the umbrella) on a small body	Vermilion `#E34234`	Indigo `#26428B`
Tamamo	Fan and crescent (tails)	Violet `#8E44AD`	Gold `#F4C430`
Varro	Square block	Rust `#B7410E`	Gunmetal
Thale	Cathedral arch	Steel white `#E8E6E1`	Brass `#B5A642`

2. **Proportions:** lean and long-legged (about 7.5–8 heads), small heads, angular jaws and wrists, big readable hands. Faces are simple: clean eye shapes, no glossy multi-highlight eyes, small nose and mouth marks.
3. **Cel shading: base + one shadow tone** with hard edges, plus **one hard rim light** whose colour follows the emotion (Imaishi's rule): orange for resolve, cyan for cold fear, magenta for danger, gold for luck. No airbrush.
4. **Lineart:** coloured lines (dark wine on skin, deep navy on clothes), thicker on the outside silhouette and thinner inside; slightly tapered, as if inked.
5. **Expressions are big.** Trigger-style exaggeration for comedy and shock, then calm, still drawing for sincere moments (the Ghibli half).
6. **Costume logic:** every costume detail is a story detail (NIL's ledger clasps, Vesper's 7-chamber cylinder, Maeve's three braids, Hina's numbered kimono). If it isn't story, cut it.

Our rules for action and effects: 1. **Limited animation, strong timing:** key poses on 2s and 3s, holds before payoffs, **smear frames** on fast moves. 2. **Hand-drawn FX** with hard shapes: angular sparks, chunky smoke with flat shadow, speed lines, debris silhouettes. 3. **Impact frames:** 1–2 frames of inverted or two-colour flat graphics (black and the attacker's signature colour) on every big hit. 4. **The colour-step slow-mo** (each frame a different hue) is reserved for **Overwrite and Jackpot** moments, our version of the famous sequence. 5. **Glitch, chromatic aberration and scanlines appear only in Noise and Basilica moments**, never as a default filter. 6. Neon accent triad for combat and Meridian: **acid yellow** `#F5E400`**, hot magenta** `#FF2E88`**, cyan** `#00E5FF`, used as light, never as fill for whole areas.

3. Composition and camera

- **Exploration and story:** wide, calm, eye-level or slightly low; characters small in big painted spaces; long holds. Let the world breathe.
- **Combat:** tighter; low angles, Dutch tilts on hits, foreshortened limbs toward camera, diagonal staging; crash zooms on ultimates.
- **Cut-ins:** flat graphic panels in the character's signature colour with halftone or speed-line texture; one bold diagonal.
- **Readability:** fighters on the left, enemy on the right, reel top-centre, floor clear.

4. Colour script by land

LAND	DEFAULT TIME AND LIGHT	BASE PALETTE	ACCENT	NEON INTRUSION
Meridian Undercity	Rainy dusk → night; dawn for the intro	Wet slate blues, warm window ambers, moss greens	Nixie orange	**Full**: yellow, magenta, cyan signage
Hyakki (Japan-inspired)	Summer night, full moon; festival lanterns	Deep indigo, ink blue, paper cream	Vermilion, lantern gold, firefly green	Only Tamamo's foxfire (cold cyan-violet)
River Empire (China-inspired)	Golden afternoon over the harbour; misty underworld	Jade, warm ochre, red lacquer	Gold leaf	Luck Exchange boards
Śūnyapura (India-inspired)	Monsoon break: wet stone, sun through cloud	Sandstone, peacock green, saffron	Brass	Eclipse purple during Rahu & Ketu
Kroton Nova (Greece-inspired)	Hard Mediterranean noon	Chalk white, Aegean blue, olive	Bronze	The Sieve's cold white glow
Crossroads (America-inspired)	Desert sunset → neon night	Rust, sage, dusk violet	Sunset orange	Casino neon (full)
The Basilica	Interior, no sun	Brass and steel greys	Nixie orange	Total, until the finale's sunrise

5. Production pipeline (how we get polish, not just pictures)

1. **Colour script first:** one thumbnail per land and beat before any final art.
2. **Style frames:** 3–4 frames that define the look (a character key art, a painted establishing shot, a battle frame). Nothing scales up until they're approved.
3. **Characters are generated on a flat neutral background**, never inside a scene, so they stay cel-like and cut out cleanly.
4. **Backgrounds are generated as empty plates**, then split into parallax layers.
5. **Compositing happens in engine**: characters get rim light from the background's light colour; grade the whole frame per land.
6. **Paint-over pass** against the checklist below. AI output is a draft, never final.
7. **Animation:** rigs for idles (breathing, hair, cloth), drawn key poses + smears for attacks, hand-made FX.

QC checklist (every image)

- [ ] Silhouette reads as a black fill at thumbnail size
- [ ] Signature colour is the most saturated thing on the character
- [ ] Characters: 2 tones + 1 rim; no airbrush; coloured variable-weight lines
- [ ] Background: one light source, a time of day, atmospheric depth, 2–3 deliberate "little things"
- [ ] Neon only where the fiction allows it
- [ ] No stray text, no warped digits, correct hands, correct prop handedness
- [ ] Culture details checked against the World Bible §10

6. Prompt grammar for Higgsfield (replaces the old style suffix)

Describe the **medium and technique**, not "premium anime key art". Studio names are used here only as a team reference; in prompts we describe traits.

Character suffix:

> 2D cel animation character design, flat colours, hard-edged two-tone cel shading with a single hard rim light in {emotion colour}, crisp variable-weight coloured lineart (thick outer contour, thin interior lines), bold graphic silhouette, lean stylised proportions, retro-modern TV anime look, saturated limited palette dominated by {signature colour}, plain flat light-grey background, no gradients on skin or clothes, no airbrush, no glossy highlights, no 3D render, no text

Background suffix:

> hand-painted animation background, gouache / poster-colour painting on paper, soft wet-edge gradients in the sky, visible soft brushwork, limited harmonious palette, one clear light source ({time of day}), atmospheric perspective, warm lived-in details, nature growing over everything, painted in the tradition of classic Japanese animated films, no characters, no text, no photo texture, no 3D render

Battle-frame suffix:

> frame from a 2D anime fight scene: flat cel-shaded characters with bold coloured outlines over a hand-painted gouache background, {neon accent} light from the action, hand-drawn angular effects animation, speed lines, dynamic diagonal composition, no text

7. Reference library (study, never trace or copy)

- **Painted worlds:** *Kiki's Delivery Service* (a sunlit port town), *Spirited Away* (the bathhouse town at dusk), *Howl's Moving Castle* (rooftop towns), *Castle in the Sky* (skies and clouds), *The Boy and the Heron*; Kazuo Oga's background art books.
- **Inked action:** *Cyberpunk: Edgerunners*; *Promare* (high-contrast colour design); *Akira* (neon, light trails); *Mob Psycho 100* (effects animation).
- **Games doing this well:** *Ni no Kuni* (a game made with Studio Ghibli's animation), *Guilty Gear Strive* (3D that animates on 2s like 2D), *Hi-Fi Rush* (cel shading and timing), *Zenless Zone Zero* (urban anime and UI), *Sable* (Moebius-style flat colour), *Hades II* (painterly 2D characters with strong silhouettes), *Persona 5* (graphic UI).
- **Engine reference:** Epic's write-up on *Orbitals* (retro-anime cel shading quantised to about three tones, with noise to fake hand-painting).

8. Lessons from *The Official Digital Artbook of Cyberpunk 2077*

Studied from the 61-page digital art book (styles chapter, district spreads, character line-ups). We take its **methods**, not its look. Night City's yellow-and-black trade dress, logos and designs stay theirs.

8.1 What the book teaches

1. **Every style is grounded in the world's history.** CD PROJEKT RED wanted styles that are "interesting, colorful, distinctive and, most importantly of all, plausible". Wars, crises and booms each leave "a logical and perceivable impact" on design. Four eras coexist on the same street: *Kitsch* ("style over substance", street rebellion), *Entropism* ("necessity over style", poverty, crude and durable), *Neomilitarism* ("substance over style", cold corporate elegance) and *Neokitsch* ("style and substance", the ultra-rich: gold, real wood, natural fur).
2. **A visible design system.** The "Comparison" spread draws the same character type, phone and drink in each style, so artists and players can tell the eras apart at a glance.
3. **Districts are mixes of styles, and travelling should feel like changing countries.** "When traveling between them, you should feel as if you're visiting different countries." And: "above all Night City is about contrasts."
4. **Lived-in logic.** In Little China, people live whole lives in one repurposed office building: they sleep on the balcony, repair limbs in a supply closet and cook in a former office kitchen. In Kabuki, "Far East meets Wild West", with low shacks that let the sun through. Every space answers *who lives here, and what happened to this place?*
5. **Status through scarcity.** In North Oak, "green is the new gold": a bonsai on a balcony is a luxury watch.
6. **Contrast as storytelling.** Pacifica's abandoned resort pairs "funhouse signs riddled with bullet holes" with "luxury pools filled with chemical sludge".
7. **Graphic identity is part of the art.** The book's pages are one loud base colour with black type, overlaid with spray-paint tags, marker crosses, stickers and collage, so the presentation itself feels like the street.

8.2 Applied to the Tally: Meridian's four eras of luck

Meridian gets its own history-grounded style system, built on our lore (the Basilica, the Rounding, the Deficit). Each era has a motto, materials and its own way of **showing your Omen**, because in our world the number is the status symbol.

ERA	MOTTO	WHO	MATERIALS AND SHAPES	HOW THE OMEN IS SHOWN
Charmpunk (street)	*Luck over sense*	Gamblers, kids, gangs, small shops	Every culture's charms worn at once, hand-painted lucky numbers, plastic gold, neon, stickers	Gaudy holo-halos, numbers painted on jackets, fake Omen projectors
Remainder (poverty)	*Survival over luck*	The Undercity, the Settled	Scrap brass from old Basilica parts, patched cloth, salvaged Nixie tubes, rope	A cracked tube on a cord; often hidden out of shame
Ledgerism (power)	*Order over chance*	The Basilica, Auditors, Collectors	Brass, steel, white enamel, cathedral arches, clockwork, uniforms	Steel numerals, stamped seals, never glowing
Gilded Fortune (the 99s elite)	*Fortune and order*	Old money, Exchange magnates	Real gold leaf, jade, living plants (in Meridian, **a garden is luck you can see**), silk	Embroidered or gold-leaf numerals, discreet

The comparison sheet: for each era, draw the same three things, *a person, a lucky charm and an Omen display*, side by side (in the manner of the book's Comparison spread). It's the first concept-art task for Meridian.

Districts of Meridian mix eras: the Remainder Wards (Remainder + Charmpunk), the Exchange (Gilded Fortune over Ledgerism), the Basilica Ring (pure Ledgerism), and the Charm Bazaar (pure Charmpunk, "every culture meets on one street").

8.3 Applied to the other lands

Each land has its **own traditional style** (the painted world, §2.1). **Meridian's eras intrude** wherever the Basilica's influence reaches: Ledgerist toll gates on a Hyakki shrine road, Charmpunk stalls in Bāfā's festival, Remainder shanties under Kroton Nova's sieve. Those collisions are where our "contrast storytelling" lives.

8.4 Graphic identity (menus, title, marketing)

Borrow the **energy**, not the trade dress: one confident base colour, heavy type, hand-made overlays. - **Our base:** ledger-paper cream `#F2E8D5` with ink black `#141018` and Nixie orange `#FF7A2C`. Not Cyberpunk yellow. - **Overlays:** lucky-charm stickers, hand-stamped ledger seals, red-ink corrections and tally marks (𝍸), ink brush strokes from the lands, and spray tags for Charmpunk screens. - Every land's map, menus and cut-ins use that land's medium for overlays (woodblock stamps in Hyakki, gold leaf in Bāfā, palm-leaf script in Śūnyapura).

Sources

- What Kazuo Oga Thinks About When He Thinks About Backgrounds (Animation Obsessive)
- Painting in Ghibli style (Gurney Journey)
- Kazuo Oga's painting process (Open Culture)
- Roundtable interview: Imaishi and Wakabayashi, Anime NYC 2025 (Yattatachi)
- Reddit AMA recap: Hiroyuki Imaishi (Bubbleblabber)
- A pair of conversations with Studio Trigger (Anime Herald)
- Cyberpunk: Edgerunners, a vibrant ode to retro anime (AWN)
- Review of *The Art of Cyberpunk: Edgerunners* (Reframe)
- Rendering of Orbitals, a retro-anime-inspired game (Unreal Engine tech blog)
- *The Official Digital Artbook of Cyberpunk 2077* (CD PROJEKT RED, 2020; cyberpunk.net/artbook), provided by the team as a PDF and studied for §8. Not stored in this repo.

The numbers and rules every artist, modeller and animator works to. `ART_DIRECTION.md` says *what it should feel like*; this says *exactly how big, how thick, which colour, which camera*. The 3D blockouts in `art/cast/` and `art/envs/` (built by `tools/render_cast.py` and `tools/build_envs.py`) are the reference for scale, silhouette and layout.

1. Two registers (unchanged)

REGISTER	FOR	LOOK
A · Concept / art book	Key art, store page, art book	Painterly semi-realistic concept art over 3D bases; realistic materials and light; human overpaint
B · In game	Gameplay, cutscenes, UI	**Painted world, inked people**: gouache-style painted backgrounds; flat cel characters with coloured lines

2. Characters

2.1 Scale and proportion

- **World scale:** 1 unit = 1 m. Characters stand on z = 0 and face –Y (toward camera) in every model file.
- **Model-sheet scale:** 1 m = 260 px (the turnaround renders). All characters on one sheet share the scale; never rescale one character to "fit".
- **Heads tall:** adults **7.5 heads** (head height = 0.132 × height); children 5–6; elders 6.5 (slight hunch); bosses 8–9 (Thale, Archon, Bai Si stretched); Hap 6 with a vast robe.

CLASS	HEIGHT RANGE	SHOULDER WIDTH (× HEIGHT)	NOTES
Heroes	1.35 (Hina) – 1.95 (Bai Si)	0.20–0.27	Each hero's height is a canon lock (Character Bible)
Children	1.20–1.35	0.22–0.24	Tobi, Niko, Penny, Little Lantern
Elders	1.45–1.70	0.24–0.25	Ofuku, Liu Shun, Vidya Rao
Heavies	1.88–2.15	0.29–0.37	Collectors, Varro, Ox-Head and Horse-Face
Bosses	1.8–2.5	by design	Rahu & Ketu and Jin Chan use special bodies

2.2 Turnaround standard (every character, every outfit)

- **Views:** FRONT · LEFT · RIGHT · BACK, orthographic, in that order, left to right, feet on one baseline. LEFT shows the character's left side (their left hand nearest the camera).
- **Pose:** relaxed A-pose: arms 10° from the body, feet hip-width, weight even, props in their canon hand (Vesper's revolver in the right hand, NIL's decoder on the RIGHT wrist, Hina's umbrella in the right hand, shields on the left arm).
- **Sheet:** 4 views on cream `#EFE6D2`, plus height bar in metres, palette chips with hex, canon locks, and 64 px silhouette test.
- **Extras per hero:** expression sheet (neutral, joy, anger, fear, determined, their "number habit" face), props sheet (front + side of each prop at 1:1 with the body), outfit sheet (every story outfit as a full turnaround).

2.3 Line, shading and colour (register B)

ELEMENT	RULE
Outer contour	4 px at 1080p for a 600 px-tall character (scale linearly); tapered at ends
Interior lines	1.5–2 px; folds only where the cloth turns
Line colour	Coloured: local colour × 0.45 value, never pure black (except ink-black garments, which get `#3A2E3A`)
Shading	**2 tones + 1 rim.** Shadow = local colour multiplied by the land's shadow tint (Meridian `#5E4F7A`, Hyakki `#2E3A6B`, River `#3E5A4A`, Śūnyapura `#6B4A5A`, Kroton `#4A5E7A`, Crossroads `#6A3E5A`, Basilica `#4A4A5A`) at 35%; one hard-edged shadow shape per form
Rim light	One rim from the background's key light colour, 2–4 px, on the side away from camera
Highlights	Hair and metal only; flat shapes, no gradients
Saturation	The **signature colour** is the most saturated thing on the character (Character Bible §0)
Palette	Max 4 main colours + 1 accent per character; NPCs 3 + 1
Skin	Real, varied tones; never grey-shifted; flushed cheeks only for children and comedy
Eyes	Heroes: large, coloured iris, one hard catch-light; NPCs: smaller; keepers: calm, half-lidded

2.4 Silhouette and shape language

Every character must read as a black silhouette at 64 px (the art book's readability page). Shape language per character is fixed in the Character Bible §0 (NIL tall rectangle + broken circle; Vesper triangles; Maeve circles; Hina dome; Ba fan; Bai Si vertical; Shunya ring; Prima shield; Fibi spiral; Varro block; Tamamo fan of tails; Jin Chan sphere; Rahu & Ketu circle + serpent; Archon column; the House ring; Thale cathedral arch; Hap star robe + orbit).

2.5 Faction costume keys (for every NPC)

FACTION	BASE COLOURS	MATERIALS	MUST HAVE	NEVER
Basilica	Steel white, brass, olive	Enamel, brass, wool	Steel numerals; gear motifs	Glowing numbers
Collectors	Rust, gunmetal	Riveted leather, glass jars	Debt jars; peaked caps; masks for captains	Clean surfaces
Charm Bazaar	Magenta, sign yellow, cyan	Plastic, nylon, stickers	$\geq$ 3 cultures' charms worn at once	Matching outfits
Remainder Wards	Ash, old brass, moss	Patched cloth, scrap	A cracked tube on a cord	New clothes
Yokai Courts	Violet, vermilion, foxfire cyan	Silk, paper, lacquer	Fox ears / tails, paper charms	Neon
Luck Exchange	Jade, gold, lacquer red	Silk, gold leaf	Gold numerals; abacus motifs	Rust
Ten Courts	Ink black, rice paper, seal red	Lacquer, lamellar	Brushes, seals	Bright colour
Observatory	Indigo, saffron, brass	Cotton, mirror-work	Astrolabe pins	Steel
Synod	Bronze, chalk, Aegean	Bronze, linen	Sieve shields, prime glyphs	Composite numbers on gear
The House	Casino red, ivory, gold	Satin, felt	Cards, waistcoats	Dust
Open Hand	Road dust, gold mark	Homespun	Open-hand cord; a die	Badges

3. Environments

3.1 Scale, grid and stage

- **Grid:** 1 m. Doors 2.2 m; storeys 3–3.5 m; street widths: Wards 8–18 m, Bāfā lanes 6 m, Śūnyapura plazas 40 m+.
- **Battle stage (side-on):** walkable strip **24 m wide × 6 m deep**, flat, clear of props; party stands at x = −8…−3, enemies at +3…+9; the land's **reel object** at top centre in screen space.
- **Exploration (2.5D) area:** 60–120 m per district; every interactive object $\geq$ 1 m and lit.

3.2 Cameras (fixed)

CAMERA	PROJECTION	ANGLE	USE
Battle (front view)	Perspective, 30 mm ($\approx$ 40° vertical FOV)	Eye at 1.7–3.2 m, 17 m from the stage line, pitch 0–5°	All fights; side-on
Exploration (2.5D)	Orthographic	Pitch 57° down, yaw −38°	Map, towns, overworld
Plan (top-down)	Orthographic	Straight down	Level design, minimap, art book plans
Cinematic	Perspective, 18–85 mm	Free	Cutscenes; respect the 180° rule toward the reel

3.3 Painted layers (register B)

Sky · far ($\geq$ 200 m) · mid (40–200 m) · near (10–40 m) · stage floor · foreground ($\leq$ 3 m, sparse). Parallax factors 0.05 / 0.2 / 0.5 / 0.8 / 1.0 / 1.2. Characters are composited between "stage floor" and "foreground".

3.4 Light and time of day per land

LAND	TIME	KEY LIGHT	FILL / AMBIENT	ATMOSPHERE
Meridian Undercity	Rainy dawn / dusk	Window amber + Nixie signage	Wet slate blue	Rain, haze, wet ground
Hyakki	Summer night, full moon	Lantern gold	Indigo moonlight	Mist over water, fireflies
River Empire	Golden afternoon (harbor) / misty dark (Courts)	Low sun	Jade bounce	Incense, lantern smoke
Śūnyapura	Monsoon break	Sun through cloud	Wet sandstone bounce	Rain sheets, puddle mirrors
Kroton Nova	Hard noon	White sun	Aegean sky blue	Heat haze, sharp shadows
Crossroads	Sunset → neon night	Sunset orange / neon	Dusk violet	Dust, card storms
Basilica	Interior, no sun	Nixie amber	Brass bounce	Still air, motes

4. The 3D pipeline (where this blockout fits)

1. **Blockout** (done): parametric bodies and environments in Blender at true scale, flat object colours; `art/cast/zso_cast_blockout.blend`, `art/envs/zso_envs_blockout.blend`.
2. **Base model:** Meshy or hand-modelled base per hero from the turnaround; T-pose; ≤ 20k tris; rigged for posing only (register A paint-overs).
3. **Colour key:** flat colours per the palette (this book's environment renders are colour keys).
4. **Paint pass:** AI-assisted (Higgsfield / FLUX) over the base render with the Art Direction §6 prompt grammar; logged in `AI_DISCLOSURE.md`.
5. **Human overpaint:** mandatory; checked against §2–§3 and the canon locks.
6. **Register B production:** 2D PSB rigs drawn from the turnarounds (Asset List specs).

File naming: `chr_<key>_<outfit>_<view>_v##.png`, `env_<land>_<view>_v##.png`; keys as in `tools/cast.py`.

5. Book graphic identity

Ledger cream `#F2E8D5`, ink `#141018`, Nixie orange `#FF7A2C`, red ink `#B3261E`. Display: Big Shoulders Display; body: Barlow Semi Condensed; data: JetBrains Mono; spray tags: Permanent Marker. Skewed black tabs, red stamps, tally-mark overlays.

6. QC (before anything is called final)

- [ ] Matches its turnaround's proportions and its canon locks
- [ ] Reads as a silhouette at 64 px and by colour alone
- [ ] Line, shading and palette per §2.3; no airbrush, no stray gradients
- [ ] Environment matches its plan view's layout and its land's time and light (§3.4)
- [ ] Battle stage clear and 24 × 6 m
- [ ] Culture details reviewed (Cultural Consultation log); keepers and Hap never monstrous

Motion tests



Intro shot 5 (8 s): the Basilica's searchlight sweeps the Undercity at dawn. Seedance 2.5 test.

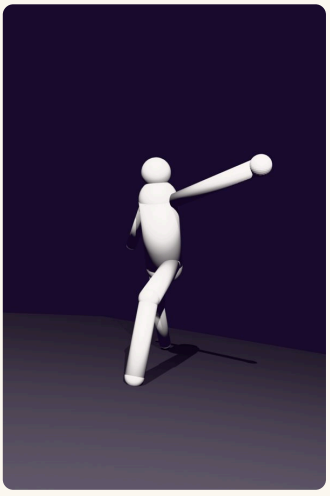


NIL reveal (5 s, vertical). Seedance 2.5 test; inherits the "Nan" typo from pass 1.

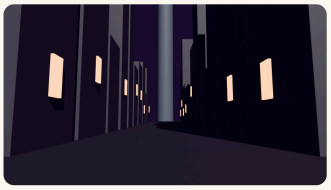
3D layouts used to keep framing consistent



Standing pose layout



Action pose layout (v2)



Establishing-shot layout



Battle-stage layout

PART VIII

Gameplay

One verb, deep consequences.



> *"Every number in this world is spinning. You're the only one who can stop them."*

An anime 2.5D party RPG across lands inspired by **Japanese, Chinese, Indian, Greek and American** myth and folklore. Every number spins like a slot reel, and you **tap to stop it**. Travel with NIL and two of eight companions, build Balatro-style synergies between heroes, charms and wheel faces, and break the Seals on the road to the Comptroller.

- Story, world and characters: `WORLD_BIBLE.md`
- Phases, schedule, art/animation pipeline: `PRODUCTION_PLAN.md`
- Higgsfield prompt pack: `../art/HIGGSFIELD_PROMPTS.md`
- Feel prototype (core verb only): `../prototype/index.html`

---

0. Pillars (in priority order)

1. **Anime spectacle drives the game.** Cut-ins, camera, animation and motion graphics *are* the feedback. Every system pays out in something you watch.
2. **One verb, deep consequences.** Stop the number. Everything else (who acts, what's on your wheels, which charms you carry, who's in your party) changes *what that one action can do*.
3. **Complexity is earned, one land at a time.** Each land's culture contributes the system it is famous for (§5). The game starts with one wheel and ends with a team-synergy engine.
4. **Diegetic first.** Every combatant's Omen floats over its head and doubles as its health readout (§1.6); the reel is an in-world object styled per land; menus are in-world objects.
5. **The theme is the math, and the math is multicultural.** Lucky numbers and superstitions from every land drive story, mechanics and team-building. Theme counts double in jam judging.

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1. Core combat (the Audit)

1.1 One round

1. **Your three strikes.** Tap a hero (or press 1/2/3) to choose who strikes next. Their **Fate Wheel** appears overhead: tens fixed, the ones digit spinning. **Tap to stop.** The hero attacks (cut-in, dash, hit-stop), and the stopped digit tears off the damage number and slams into the **reel** (3 tubes). - Party of 1: NIL strikes 3 times. Party of 2: choose an order, and one hero strikes twice. Party of 3: each strikes once. - **Read the reel.** The third striker sees the first two digits. **Pattern Sight** makes every face that would complete a pattern shimmer in that pattern's colour, so the last tap is always a decision, never a guess. (Simulation: this alone cuts Busts from ~57% to ~45% for an average player; `TUNING.md`.) - **Starter wheels have 8 faces.** Each hero's wheel leaves out two digits listed in their kit (§2.2); wheel crafting adds, removes and enhances faces from there.
2. **The reel resolves** into a pattern that multiplies the round (§1.3), with a full-screen cinematic.
3. **The enemy strikes.** Its wheel spins over it. Stop on **0** to parry, on your lucky digit to reflect; anything else hurts.

1.2 The number (chips × mult, made visible)

```
Strike value  = Tens (hero level + gear) + Digit stopped + face bonus
Round damage  = (Σ strike values + flat bonuses)  ×  Mult
Mult          = Pattern mult (+ gem levels)  + additive mult (charms, faces)  then  × each ×Mult charm
```

Strikes land as small white numbers; then the pattern title slams in and a **× Mult** counts up into one huge gold number. That gives a small payoff every 3 seconds and a big one every round.

Design rule: tens are safe, ones are fate. Anything that adds to the tens never changes the digit you stopped.

1.3 Reel patterns

PATTERN	EXAMPLE	BASE MULT	EXTRA	CINEMATIC
Bust	2-9-5	×1	—	Tubes flicker out
Pair	7-1-7	×2	—	Small flourish

PATTERN	EXAMPLE	BASE MULT	EXTRA	CINEMATIC
Zero-Sum	3-3-4 (sums to 10/20)	×2	Enemy's next action cancelled	Mirror wipe, ledger chime
Cascade	3-4-5, 9-8-7	×3	Extra round (the enemy waits)	Time-slow bands
Jackpot	7-7-7	×digit (min ×2)	—	Squad panels slash down, screen crack
Omen Overwrite	your digit ×3	×21, whatever your digit (so no lucky number is stronger than another)	Charges every ultimate	Gold everything; bells ring your number of times
Reaper's Toll	4-4-4	×4	Every Omen containing a 4 loses 44% of its Balance, **yours included**	Four bells toll, red wash

1.4 Defence

The enemy's wheel shows its odds by cell size. **0 → Parry** (no damage, counter flash); **your digit → Fortune** (reflect); otherwise take tens + digit. Some enemies (the Prime Synod) have no 0 on their wheel, so the party must win the race instead.

1.5 Wheels show the odds

A wheel is a strip of **faces**, and a face's height is its odds. Players read probability by looking.

Pattern priority. A reel matches the first pattern in this order: Omen Overwrite → Reaper's Toll → Jackpot → Cascade → Zero-Sum → Pair → Bust. So 4-4-4 is an Overwrite for a player whose digit is 4, and a Reaper's Toll for everyone else. (Same order as `evaluate()` in the prototype.)

1.5b Luck turns: Hap's Favour

A Bust warms the **Open Hand** sigil on the reel frame: next round every wheel spins 20% slower and faces matching a digit already on the reel are 50% wider. After two Busts in a row, **Hap's Hand** guarantees the next round is at least a Pair. The sigil cools after any pattern. Players experience it as "my luck turned", which is exactly the fantasy; measured, it removes every three-Bust streak (full rules and numbers in `SYSTEMS.md` §2.4).

1.6 Omen and Balance (how a two-digit number can be a health bar)

An Omen is always two digits, 00–99 (World Bible §2). In an Audit, each side stakes a **Balance**: the luck it puts on the table. Damage drains the Balance, and the Omen over the combatant's head shows the share that's left, from 99 down to 00. At 00 the combatant is Settled. - Enemy Balances grow exponentially by act (§4.2), so damage numbers get huge while the number over every head stays a readable two-digit Omen. - The party shares one Balance (300 in the Prologue), shown as the party's Omen over NIL's NaN. - Rules that read an Omen (Reaper's Toll, Hina's *Ninety-Nine*, "13" charms) read the displayed two-digit value; rules that deal damage hit the Balance.

---

2. The party

NIL plus 2 of 8 companions. Swap at a Teahouse / Campfire node or before a boss.

2.1 What every hero has

Fate Wheel (speed + faces) · base tens · a passive · an **ultimate** (charged when their digit lands three times or by an Overwrite) · a synergy digit · an **Awakening** (Act IV, one of two class branches) · a **bond arc** (4 scenes).

2.2 Roster kits

HERO	DIGIT	LAND	WHEEL	PASSIVE	ULTIMATE
NIL	yours	—	Medium; your digit double-wide; **Splice** (tens *and* ones wheels)	Grows by act (§6)	*Remainder Release*: the reel gains a 4th tube this round
Vesper Sette	7	Meridian / Crossroads	Fast	*Lucky Seven*: each 7 on the reel adds +7 to every strike this round	*Seven-Chamber Requiem*: 7 rapid auto-stopping mini-wheels; each 7 doubles the shot
Maeve Triskel	3	Meridian (Celtic quarter)	Medium; has **Horseshoe** faces	*Third Time Lucky*: the round's 3rd strike counts double if Maeve made it; she upgrades charms at forge nodes	*Triskele*: three spinning horseshoes each land a free 3 on the reel (an instant Jackpot of 3s, or a Twins pattern with NIL 3)

HERO	DIGIT	LAND	WHEEL	PASSIVE	ULTIMATE
Hina of the Ninety-Nine	9	Hyakki	Bouncy: it reverses direction at random	*Ninety-Nine*: 9s don't curse her party; each 9 landed gives +9 Mult while her Omen is 99 or below… which it always is	*Hundred-Year Night*: her umbrella opens with one eye; for one round every wheel shows only the party's lucky digits
Ba Jinling	8	River Empire	Slow; 8 double-wide; Gold faces	*Compound*: +0.8 Mult per consecutive patterned round	*Eight Bells*: ×1.1 Mult per bell (×2.14) and +8 fen each
Bai Si	4	River Empire	Medium; 4 triple-wide	*Fourfold*: every 4th strike of a fight marks the enemy (×1.44 damage taken)	*Four Lanterns Out*: Reaper's Toll that spares the party
Shunya	0	Śūnyapura	Slow; 0 triple-wide	*Pūrṇa (fullness)*: 0s on the reel are wild for every pattern	*Place Value*: move any reel digit into the tens of every strike next round (×10 its weight)
Prima of Kroton	2	Kroton Nova	Only 2-3-5-7 plus one 0	*Indivisible*: prime strikes pierce armour; +1 Mult per prime on the reel	*Sieve of Light*: shatter every composite number on screen (enemy shields and armour)
Secret: Varro Kade	1	Meridian	Medium; 1 triple-wide	*First and Only*: strikes first; on a Bust his digit fills the empty slots	*Debt Collector*: steal the enemy's last reel digit into a jar, then smash it for ×6
Secret: Kyūsuke the Kappa	5	Hyakki	Slow; 5 double-wide	*Bow*: once per fight, if a 5 lands right after his bow, the enemy skips a strike	*Cucumber Tide*: the river rises; every 5 on the reel heals the party's Balance by 5%
Secret: Granny Liu Shun	6	River Empire	Medium; 6 double-wide; Jade faces	*Six-Six Smooth*: two 6s make the rest of the round's strikes land in order (a free Cascade)	*Thirteen Orphans*: the reel becomes 13 mahjong tiles for one round; any pair pays ×13
Fibi Leone	Fibonacci	Crossroads	Her lucky digit **advances every round** (1, 1, 2, 3, 5, 8, 3, 1…)	*Golden Ratio*: Cascades give ×1.618; Fibonacci triples count as Cascade	*Spiral Deal*: the reel extends to 5 tubes this round (Full House, Five-Run…)

2.3 Synergy is number theory (no trait UI)

Bonds come from the **relationships between lucky digits**. The party screen draws constellation lines between digits, and that's the whole interface.

BOND	RULE	EFFECT	EXAMPLE COMPS (CROSS-CULTURAL ON PURPOSE)
Complements	Digits sum to 10	Zero-Sum becomes ×4 and triggers a **Link Art**	Vesper 7 + Maeve 3 · Ba 8 + Prima 2 · Hina 9 + Fibi (on 1) · Bai Si 4 + NIL 6
Neighbours	Consecutive digits	Two-digit runs count as a Cascade	Vesper 7 + Ba 8 · Ba 8 + Hina 9 · Prima 2 + Maeve 3 · Maeve 3 + Bai Si 4
Twins	Same digit (NIL chose a companion's digit)	That digit's Jackpot ×2, plus a shared ultimate gauge	NIL 7 + Vesper
Prime Party	All three digits prime	Every strike pierces; Prime Trio pattern unlocked (×5)	Prima 2 + Maeve 3 + Vesper 7
Powers of Two	Digits are 1/2/4/8	Each pattern doubles next round's mult	Prima 2 + Bai Si 4 + Ba 8
Śūnya	Shunya in the party	0 is wild for every bond above	Shunya + anyone
Old World / New World	One hero from an "old ledger" land + one from Meridian/Crossroads	Charms from both lands' sets count for each other's set bonuses	Hina + Vesper, Prima + Fibi

Why it's deep: the player picks NIL's number in the opening minute, before knowing any of this, and that quietly decides who completes NIL all game. Pick 3 and Vesper is your Complement; pick 1 and it's Hina; pick 6 and only Bai Si completes you.

2.4 Link Arts (duo anime cinematics)

Every Complement and Neighbour pair has a 3–5 s combination attack that blends both heroes' **art styles**: - **Vesper + Maeve, *Lucky Strike*:** Maeve hurls a burning horseshoe; Vesper shoots it in mid-air into seven ricochets. - **Ba + Prima, *Golden Sieve*:** coins pour through Prima's shield as golden spears; gongbi gold meets Greek marble. - **Hina + Vesper, *Midnight Parade*:** ukiyo-e yokai stampede through a Western saloon door. - **Bai Si + Shunya, *Zero Count*:** lanterns go out one by one and the observatory's sundial shadow cuts the enemy.

These are the trailer moments and the reward for team-building.

Every lucky number has partners. With the secret recruits, each digit NIL can choose has a Twin (same digit) and a Complement (sums to 10) among the 11 companions: 0↔Shunya, 1↔Varro/Hina, 2↔Prima/Ba, 3↔Maeve/Vesper, 4↔Bai Si/Liu Shun, 5↔Kyūsuke, 6↔Liu Shun/Bai Si, 7↔Vesper/Maeve, 8↔Ba/Prima, 9↔Hina/Varro.

Full kits, all ultimates, every Awakening branch and the Link Art for every pair are in `SYSTEMS.md` §4–§6.

3. Build systems (the Balatro layer), and the land each comes from

SYSTEM	BALATRO ANALOGUE	WHAT IT IS	LAND THAT TEACHES IT
Charms of the World	Jokers	Lucky charms from every culture; slots 3 → 6; ordered left to right	Hyakki (shrine charms, *omamori*)
Wheel crafting	Deck building + enhancements	Remove, duplicate or enhance faces on each wheel	River Empire (the Luck Exchange)
Oracle sticks (求签)	Tarot cards	Shake the cylinder and a fortune stick falls: a one-shot effect with a fortune poem	River Empire
Ink (the Judge's Brush)	Spectral cards	Rewrite one reel digit after it lands	River Empire (Ten Courts)
Karma	—	A resource banked by parries and Zero-Sums, spent to re-spin or to block a lethal hit	Śūnyapura
Navaratna gems (the nine planetary gems)	Planet cards	Permanently level up a pattern (Ruby → Jackpot, Pearl → Pair, Emerald → Cascade…)	Śūnyapura
Awakenings (the Moirai's thread)	—	Each hero picks one of two class branches	Kroton Nova
Crossroads Deals	Risky vouchers	A powerful boon with a price ("+2 charm slots, but your Omen max is 77")	Crossroads
Wagers	Stakes	Bet fen or Balance before a fight for bigger rewards	Crossroads
Omen Levels	Stakes	Difficulty tiers 1–8	Post-game
Hap's Favour, Hap's Whispers, Open Hand shrines	—	Luck turns after bad luck; coincidence events on the map; rest nodes where you can re-deal an offer	Everywhere (the god of luck)
Land Laws	Boss blinds	Each land's superstition is a rule on every fight there (curses on 4/9, interest, karma, the sieve, the loop)	Each land
Tsukumogami faces	Card editions	A face stopped 100 times wakes up with a small soul (an extra effect)	Hyakki

3.1 Charms of the World (jam needs 12, full game ~90)

Organised in **sets by land**. Holding several from one set grants a set bonus, so collecting is a build axis.

CHARM	SET	RULE
Beckoning Cat (*maneki-neko*)	Hyakki	+1 fen per strike ending in your digit
Daruma	Hyakki	When you land your digit three times in a fight, paint the second eye: ×2 Mult for the rest of the fight
Seven Lucky Gods (7 charms: Ebisu, Daikokuten, Bishamonten, Benzaiten, Fukurokuju, Jurōjin, Hotei)	Hyakki	Each has a small effect; holding all 7 makes every 7 a Jackpot by itself
Ofuda ward	Hyakki	The first enemy strike each fight is parried
Red Envelope (红包)	River Empire	+8 fen on a Jackpot
Door God (门神)	River Empire	Parries also deal the enemy's strike back
Peach-Wood Sword (桃木剑)	River Empire	×2 Mult vs ghosts and Noiselings
Lemon and Seven Chillies (*nimbu-mirchi*)	Śūnyapura	Cancels one enemy curse per fight; +7 tens per 7 on the reel
108 Beads	Śūnyapura	Every 108th point of damage in a round counts double (rewards huge numbers)
Evil-Eye Bead (*mati*)	Kroton Nova	The enemy's wheel shows its next digit faintly
Fortuna's Wheel	Kroton Nova	Once per fight, spin the reel itself: all three digits re-roll
Owl Coin	Kroton Nova	+1 Mult per prime strike
Horseshoe	Crossroads	Your digit's cells are 50% wider
Rabbit's Foot	Crossroads	One free re-spin per fight
Lucky Penny	Crossroads	+1 fen when you land a 1

CHARM	SET	RULE
Seven-Come-Eleven Dice	Crossroads	If a round's strikes total 7 or 11 (or 70/110/700…), ×7 Mult
Four-Leaf Clover	Meridian (Celtic)	1-in-4 chance per strike to duplicate its digit on the reel
Thirteenth Floor	Meridian	×13 Mult if the round total ends in 13
Black Cat	Meridian	The first Bust each fight becomes a Pair

3.2 Wheel crafting (faces)

Remove · Duplicate · Enhance: **Gold** (+3 fen) · **Jade** (+4 Mult) · **Glass** (×2 strike; 1-in-4 chance it shatters) · **Iron** (+20 tens while *not* stopped) · **Lantern** (the wheel slows passing it) · **Ink** (wild) · **Horseshoe** (Maeve: lands as your digit 1 time in 4) · **Cursed** (+50 tens, curses the striker). Every enhancement is visible on the wheel itself.

3.3 Economy

Fen (the Tally's small coins; Lore Lexicon §1) come from fights (with interest: 1 per 5 held, capped), charms and gold faces. They buy charms, sticks, gems, crafting and rerolls at Markets.

4. Structure: acts, levels and bosses

4.1 An act is a map of ~8 nodes ending in a boss

The map is an in-world object in the land's style: a hand-scroll (Hyakki, River Empire), a palm-leaf manuscript (Śūnyapura), a mosaic floor (Kroton Nova), a frontier road map (Crossroads).

NODE	PURPOSE
Small Audit / Big Audit	Fights
Elite / **Keeper's Trial**	Hard fight with a special rule; Keepers are revered figures who test you (Enma, Judge Cui, Chitragupta, the Moirai)
Market	Charms, sticks, gems, crafting
Shrine / Forge	Free charm or face enhancement (Maeve's forge upgrades charms)
Teahouse / Campfire	Swap party, bond scenes, rest
Story	Cutscene + choice
Boss Audit	The land's Auditor and its unique rule

Each act ≈ 25–35 minutes. The full campaign ≈ 6–8 hours.

4.2 Difficulty curve (exponential, like Balatro's blinds)

ACT	SMALL / BIG / BOSS ENEMY BALANCE
Prologue (Meridian)	300 / 500 / 1,200
I (Hyakki)	1,500 / 2,500 / 6,000
II (River Empire)	8,000 / 14,000 / 35,000
III (Śūnyapura)	50k / 90k / 200k
IV (Kroton Nova)	300k / 600k / 1.5M
V (Crossroads)	2.5M / 5M / 12M
Finale	99,999,999 ("the whole Ledger")

These are Balances, not Omens (§1.6). Linear growth (tens) can't keep up past Act II. Players **must** build multiplier engines. Losing a fight returns you to the act start with the build you had there.

4.3 Bosses: each changes the rules

BOSS	RULE	COUNTER-PLAY IT TEACHES
Varro, the Ledgerhound	Steals your last reel digit into a debt jar; 3 parries shatter it	Defence matters

BOSS	RULE	COUNTER-PLAY IT TEACHES
Tamamo, the Nine-Tailed	9 tails, each with its own small Balance; foxfire **illusion faces** on your wheels (show one digit, land as another); 9s curse you unless Hina is in the party	Read carefully; the Hina synergy
Jin Chan, the Golden Toad	His Balance gains interest each round; taxes your Jackpots; phase 2 swallows a hero (land a Pair to free them)	Burst damage; economy
Rahu & Ketu, the Eclipse	Two bodies: Rahu darkens one reel tube each round; Ketu strikes with a blind random wheel; if their Balances drift >10% apart they rejoin and heal	**Balance damage between two targets**: Zero-Sum hits both equally
Archon Seventeen	Strikes digits off *your* wheels each round (Sieve of Eratosthenes); composite strikes halved	Wheel crafting; Prima's primes
The House	Rewinds time every 3 rounds (your reel and charms persist); copies your last pattern; offers a mid-fight deal	Finish within one loop; read the deal
Comptroller Thale	P1 Regression to the Mean · P2 Fixed Fate (wheels don't spin: a puzzle) · P3 Zero (NIL alone; companions return as Link Arts)	Everything

4.4 Variability

Enemy pools per land with distinct wheels (kasa-obake wheels hop; Golden Koi multiply when hit; Vetālas ask a riddle and the answer is a digit; Harpies steal a reel tube; the Headless Horseman's wheel has no top half). Random charm, gem and stick offers; branching maps; Elite modifiers ("Noise storm: all wheels reversed"). Party composition: 28 companion pairs × 10 NIL digits. Post-game **Endless Ledger**: every land shuffled into a true roguelike, Omen Levels 1–8, and a Daily Omen seed.

5. The complexity ladder (each land teaches what its culture is known for)

WHEN	NEW	TAUGHT BY
Prologue, fight 1	Stop the wheel; reel; Jackpot/Pair/Bust; Parry	NIL alone vs Debt Hounds
Prologue, fight 2	Choose who strikes; hero wheels differ; Cascade, Zero-Sum	Vesper joins
Prologue, boss	Ultimates	Vesper's Requiem saves NIL
Act I, Hyakki	**Charms** (3 slots) and charm sets; curses (4 *shi*, 9 *ku*); Reaper's Toll; **party of 3, bonds, Link Arts**	The Seven Lucky Gods' shrine; Hina joins
Act II, River Empire	**Fen economy, Markets, wheel crafting, oracle sticks, Ink**	The Luck Exchange; Judge Cui's Brush
Act III, Śūnyapura	**Karma; Navaratna gems (pattern levels)**; NIL's **Ledger Tree**	Chitragupta's Hall; Shunya
Act IV, Kroton Nova	**Awakenings**; number properties (prime, perfect 6 and 28, golden ratio)	The Moirai's three trials
Act V, Crossroads	**Crossroads Deals; Wagers; loops**	The midnight crossroads
Finale	Everything, then taken away	Thale freezes your wheels
Post-game	Endless Ledger, Omen Levels, Daily Omen	—

6. NIL's growth (the most important progression)

ACT	VERB	WHAT IT DOES	WHY THIS LAND
Prologue	**Splice**	Two wheels (tens and ones) for high-control strikes	NIL's own nature
I	**Carry**	A strike over 50 carries +1 to the next hero's digit	Learned from Hina's century of counting
II	**Rewrite**	Free Ink each fight (rewrite one reel digit)	The Judge's Brush
III	**Place Value**	Move one stopped digit into the tens place of the next strike	India's place-value system
IV	**Factor**	Split one strike into two reel digits (56 → 7 and 8)	Greek number theory
V	**Deal**	Once per fight, take a Crossroads Deal mid-battle	The black book
Finale	**Divide by Zero**	Ultimate: NIL's digit is wild on every wheel for a round	Everything

The **Ledger Tree** (from Act III) has one branch per Seal (7), with 3–4 nodes each, fed by recovered memory digits. Examples: *Remainder +1* (permanent 4th reel tube, late), *Audit Eye* (see the enemy's next wheel), *Compound Remainder* (+Mult per Settled enemy this act).

7. Hero progression

- **Levels** (shared XP): +10 tens per level (digits never change). Cap 30.
- **Bond arcs:** 4 scenes per hero → second passive, Link Art upgrade, true-ending flag.

- **Awakening** (Act IV): one of two class branches per hero (e.g. Vesper: *Dead Man's Hand* crits vs *Blackjack* bonus at 21; Hina: *Hundred Years* becomes real vs *Kasa-obake* stays a trickster).
- **Gear:** one personal charm slot per hero.

8. Minimal UI, still

SCREEN	IN-WORLD OBJECT
Combat	Diorama + land-styled reel: neon sign (Meridian) · torii with three hanging lanterns (Hyakki) · golden abacus (River Empire) · observatory sundial dials (Śūnyapura) · three marble columns with mosaic numerals (Kroton Nova) · slot machine (Crossroads) · Basilica clock (Meridian)
Map	Scroll / palm-leaf / mosaic / road map, by land
Market	A stall with wares on the table
Teahouse / Campfire	Heroes sit around; drag two beside NIL; constellation lines show bonds
Charms	Pinned to the reel frame, left to right
Lucky-number choice	The Naming terminal: ten Nixie tubes

9. Jam slice scope (ships Nov 13)

"Prologue & Act I: The Remainder and the Night Parade": the intro cinematic (a five-culture montage), the Prologue (Meridian Undercity, 3 nodes + Varro) and Act I (Hyakki, ~6 nodes + Tamamo), then an ending cinematic teasing the River Empire and Thale.

SYSTEM	IN THE JAM SLICE
Core verb, reel, all 7 patterns, parry	✅
Heroes	NIL + Vesper (7) + Maeve (3) + Hina (9): choose 2 of 3 in Act I
Bonds / Link Arts	Complements, Neighbours, Twins; Link Art *Lucky Strike* (Vesper + Maeve) and *Midnight Parade* (Hina + Vesper)
Ultimates	✅ every hero
Charms	12: Hyakki set (beckoning cat, daruma, ofuda, 3 of the Seven Lucky Gods) + Meridian/Crossroads (horseshoe, rabbit's foot, black cat, four-leaf clover, thirteenth floor, seven-come-eleven); 3 slots
Wheel crafting	Remove + Gold/Jade/Glass at one Hyakki shrine; Maeve's forge upgrades a charm
NIL verbs	Splice + Carry
Bosses	Varro, Tamamo (nine tails, foxfire illusions)
Keeper's Trial	Enma (optional elite)
Not in jam	Economy beyond fen, oracle sticks, Ink, Karma, gems, Awakenings, Acts II–V

10. Build & tech summary

Unity 6 LTS · URP · WebGL 2 (WebGPU optional) · Unity 2D Animation (PSB rigs) + Cinemachine 3 + Timeline + PrimeTween · VideoPlayer for cinematic clips from StreamingAssets · pure C# rules core with EditMode tests · ScriptableObject data for heroes, wheels, faces, charms (with set tags), enemies, boss rules and maps. Details in `PRODUCTION_PLAN.md` §5.

The full rules behind `GDD.md`. Numbers live in `TUNING.md` (jam) and `ECONOMY_PROGRESSION.md` (full game). Everything here obeys GDD §0: **one verb** (stop the number), everything else changes what that verb can do.

1. The luck model in one picture

```
 Hap (the god)            → Hap's Favour, Whispers, Open Hand shrines      (luck turns; coincidences)
 The land                 → Land Law (superstition as a rule)              (curses, blessings,
interest, karma, sieve, loop)
 Your party               → lucky digits → bonds, Link Arts                (who you bring)
 Your wheels              → faces: width = odds, enhancements              (what can land)
 Your tap                 → the stopped digit                              (skill)
 The reel                 → pattern × Mult                                 (the payoff)
 Your charms, gems, sticks → modify every layer above                      (the build)
```

Every layer is visible on screen: the sigil on the reel frame (Hap), the Land Law icon, the constellation lines (bonds), face widths, the reel, the charms pinned to the frame.

2. Luck mechanics

2.1 Fate Wheels and faces

- A wheel is a vertical strip of faces; **a face's height is its odds**. Starter wheels have **8 faces**: each hero omits two digits (below).
- The ones digit spins; the tens are fixed (hero level + gear). *Tens are safe, ones are fate.*
- Spin speed per hero (Slow 380, Medium 470, Fast 620 px/s at 1080p); assist mode −30%.

HERO	STARTER FACES (OMITTED)	WIDTHS
NIL	all but the two digits farthest from your number (e.g. digit 7 omits 2 and 1)	your digit ×2
Vesper	omits 4, 9	7 ×2
Maeve	omits 4, 8	3 ×2; 1 Horseshoe face
Hina	omits 4, 0	9 ×2; bounces
Ba Jinling	omits 4, 1	8 ×2; 2 Gold faces
Bai Si	omits 9, 0	4 ×3
Shunya	omits 4, 9	0 ×3
Prima	only 2, 3, 5, 7, 0 (5 faces, by design)	equal
Fibi	omits 4, 6	her current digit ×2 (moves every round)
Varro	omits 4, 9	1 ×3
Kyūsuke	omits 4, 9	5 ×2
Liu Shun	omits 4, 7	6 ×2; Jade faces

2.2 Reel patterns (GDD §1.3, priority §1.5)

Overwrite (×21 for every digit) → Reaper's Toll (×4, 44% of every Balance with a 4 in its Omen) → Jackpot (×digit, min ×2) → Cascade (×3, extra round) → Zero-Sum (×2, cancel enemy action) → Pair (×2) → Bust (×1). **Unlockable patterns:** Prime Trio (all three prime, ×5; Prime Party bond), Fibonacci Run (1-1-2, 2-3-5, 3-5-8: counts as Cascade with Fibi or the Golden Ratio charm), Four-tube patterns (NIL's *Remainder Release*, the Ledger Tree's *Remainder +1*): Two Pair ×3, Four of a Kind ×digit^2, Long Cascade ×5; Five-tube (Fibi's *Spiral Deal*): Full House ×6, Five of a Kind ×digit^3.

2.3 Pattern Sight

When the third striker's wheel appears, faces that would complete a pattern shimmer in that pattern's colour (Pair silver, Zero-Sum cyan, Cascade blue, Jackpot gold, Overwrite white-gold). It teaches patterns without text and makes the last tap a decision. Accessibility: each shimmer also carries a shape (dot, ring, chevron, star).

2.4 Hap's Favour (luck turns)

- **Warm** (after 1 Bust; one die on the Open Hand sigil lights): next round all wheels –20% speed; faces matching a digit already on the reel are ×1.5 wide.
- **Hap's Hand** (after 2 Busts in a row; both dice light): the next round **cannot Bust**. If it would, Hap's hand flicks a die across the reel and it becomes a Pair (a small cinematic, never hidden).
- **Cools** after any pattern. Some charms and Whispers warm it on purpose.
- **Why:** the fantasy "my luck turned", and no player ever suffers three Busts in a row. Measured with `python3 tools/sim_patterns.py --favour` (jam party, 8-face wheels, 8-round fights):

PLAYER	FIGHTS WITH 3+ BUSTS IN A ROW	BUST RATE
Random tapper	52% → **0%**	56% → 46%
Average (25% aim)	25% → **0%**	40% → 35%
Skilled (50% aim)	6% → **0%**	23% → 21%

2.5 Land Laws (each land's superstition as a rule)

LAND	LAW	EFFECT	COUNTER IN THE SAME LAND
Meridian	Debt	Every 3rd round, –5% party Balance unless the round scored a pattern	Vesper's Lucky Seven; any pattern
Hyakki	Shi and Ku	4 curses the striker (–10% next strike); 9 curses your next parry; 7 and 8 bless (+1 Mult)	Hina (9s safe, +9 Mult); Ofuda ward
River Empire	Interest	10% of each round's Mult carries to the next; resets on Bust; enemies too	Jinling's Compound; streak play
Śūnyapura	Karma	Parries and Zero-Sums bank Karma; Busts vs cursed enemies cost it	Shunya; 108 Beads
Kroton Nova	Indivisibility	Prime strikes pierce; composite strikes deal half; every 3 rounds the Sieve removes one composite face from each wheel	Prima; Factor
Crossroads	The House	Wagers before fights; every 3 rounds the fight loops (reel and enemy Balance reset; your build persists)	Fibi; Deals
Basilica	Fixed Fate	Wheels –5% speed and lose their least-used face each round	Outrun it

2.6 Curses and blessings

- **Curses** (from Land Laws, enemies, Deals): shown as a small black ribbon on the hero's wheel. Lemon and Seven Chillies cancels one per fight; Ofuda cancels the first; Karma can buy one off.
- **Blessings** (from Shrines, Whispers, keepers): a gold ribbon. Lasts one fight unless stated.
- **Unlucky numbers by land** (13 in Meridian and the Crossroads, 17 in Kroton Nova, 4 and 9 in Hyakki, 4 in the River Empire) appear as enemy weak points: hitting an enemy whose Omen shows its land's unlucky number deals +13%.

2.7 Hap's Whispers (coincidence events)

Map nodes with a single choice, always about luck, always in the land's idiom (lists per land in `ACT_DESIGN.md`). Rules: one choice, two lines of text at most, no traps without a visible price, at least one Whisper per act is pure good luck (a gift from Hap).

2.8 Open Hand shrines

Rest nodes with a stone hand and a die. Choose one: **re-deal** one Market or reward offer; **pray** (start the next fight with Favour warm); **offer** fen (every 99 fen offered across the game unlocks one of Hap's faces in the gallery and a small permanent blessing).

2.9 Omen and Balance

Every combatant has a **Balance** (luck at stake); the Omen over its head shows the share left, 99 → 00 (GDD §1.6). The party shares one Balance. Rules that "read an Omen" read the two-digit display.

3. Fantasy mechanics (one per land, all built on the same verb)

LAND	MECHANIC	WHAT IT DOES
Hyakki	**Tsukumogami faces**	A face stopped 100 times wakes up with a small soul: a random extra effect (e.g. +1 fen, a free re-spin once per fight). Faces can only wake once. Hina's kit interacts with this (her 99 → 100 arc)

LAND	MECHANIC	WHAT IT DOES
River Empire	**Ink** (Judge Cui's Brush)	Rewrite one reel digit after it lands. NIL gets one free per fight from Act II (Rewrite); more from charms and sticks
River Empire	**Oracle sticks** (求签)	Shake the cylinder; a stick falls with a one-shot effect and a two-line fortune poem. 24 sticks; rarity by fortune (great luck → great misfortune, and misfortune sticks are powerful with a price)
Śūnyapura	**Karma**	Banked from parries and Zero-Sums; spend 1 to re-spin a wheel, 3 to block a lethal hit; carries between acts
Śūnyapura	**Navaratna gems**	Nine planetary gems, each permanently levels one pattern (below)
Śūnyapura	**Place Value** (NIL)	Move a stopped digit into the tens of the next strike (×10 its weight)
Kroton Nova	**Awakenings** (the Moirai's thread)	Each hero chooses one of two branches (§5)
Kroton Nova	**Factor** (NIL)	Split one strike into two reel digits (56 → 7 and 8)
Crossroads	**Deals**	A powerful boon with a lasting price (list below)
Crossroads	**Wagers**	Bet fen or Balance before a fight; win ×2 fen or a rare reward; lose the stake
Crossroads	**Loops**	The fight rewinds; your build persists
Basilica	**Divide by Zero** (NIL's final verb)	NIL's digit is wild on every wheel for one round

Navaratna gems

GEM	PLANET	LEVELS
Ruby (*Manik*)	Sun	Jackpot +1 level (+×1 base)
Pearl (*Moti*)	Moon	Pair (+×0.5)
Red coral (*Moonga*)	Mars	Reaper's Toll (spares one hero per level)
Emerald (*Panna*)	Mercury	Cascade (+×0.5, extra round chance)
Yellow sapphire (*Pukhraj*)	Jupiter	Omen Overwrite (+×1)
Diamond (*Heera*)	Venus	Link Arts (+15% each)
Blue sapphire (*Neelam*)	Saturn	Zero-Sum (cancels two enemy actions at level 3)
Hessonite (*Gomed*)	Rahu	Bust (+tens on a Bust)
Cat's eye (*Lehsunia*)	Ketu	Hap's Favour (warms after 1 Bust at level 2)

Crossroads Deals (sample of 12; each is "boon / price")

+2 charm slots / Omen display capped at 77 · Every Jackpot ×2 / Pairs pay ×1 · Free re-spin every round / enemies' wheels +20% speed · Start every fight with Favour hot / Shrines closed for the act · All Gold faces ×2 fen / Markets cost +50% · +1 reel tube / lose one hero's ultimate · Karma never decays / Karma can't block lethal hits · Overwrite charges ultimates twice / Busts curse the striker · See enemy wheels' next digit / your Pattern Sight is off · One free Wager win per act / Wagers can't be refused · Your digit ×3 wide / your digit's Jackpot pays ×1 · Revive once per act / the House takes 10% of all fen.

4. Hero kits, ultimates and secret recruits

HERO	DIGIT	PASSIVE	ULTIMATE (3–5 S CINEMATIC)	CHARGE
NIL	yours	Grows by act: Splice · Carry · Rewrite · Place Value · Factor · Deal · Divide by Zero	***Remainder Release***: the reel gains a 4th tube this round, filled by whatever NIL taps on a free wheel	Your digit lands 3 times, or an Overwrite
Vesper Sette	7	*Lucky Seven*: each 7 on the reel adds +7 to every strike this round	***Seven-Chamber Requiem***: 7 auto-stopping mini-wheels; each 7 doubles the shot	Her 7 lands 3×
Maeve Triskel	3	*Third Time Lucky*: the round's 3rd strike counts double if Maeve made it; upgrades charms at forges	***Triskele***: three spinning horseshoes each land a free 3 (a Jackpot of 3s, or Twins with NIL 3)	3 lands 3×
Hina	9	*Ninety-Nine*: 9s don't curse her party; each 9 gives +9 Mult while her Omen is 99 or below	***Hundred-Year Night***: her umbrella opens its eye; for one round every wheel shows only the party's lucky digits	9 lands 3×
Ba Jinling	8	*Compound*: +0.8 Mult per consecutive patterned round	***Eight Bells***: ×1.1 Mult per bell (×2.14) and +8 fen each	8 lands 3×
Bai Si	4	*Fourfold*: every 4th strike of a fight marks the enemy (×1.44 damage taken)	***Four Lanterns Out***: a Reaper's Toll that spares the party	4 lands 3×
Shunya	0	*Pūrṇa*: 0s on the reel are wild for every pattern	***Place Value***: move any reel digit into the tens of every strike next round	0 lands 3×

HERO	DIGIT	PASSIVE	ULTIMATE (3–5 S CINEMATIC)	CHARGE
Prima	2	*Indivisible*: prime strikes pierce; +1 Mult per prime on the reel	***Sieve of Light***: shatter every composite number on screen (shields, armour)	a prime lands 3×
Fibi Leone	Fib.	*Golden Ratio*: Cascades ×1.618; Fibonacci triples count as Cascade; her digit advances each round	***Spiral Deal***: the reel extends to 5 tubes this round	her current digit lands 3×
Varro Kade	1	*First and Only*: strikes first; on a Bust his digit fills the empty slots	***Debt Collector***: steal the enemy's last reel digit into a jar, then smash it for ×6	1 lands 3×
Kyūsuke	5	*Bow*: once per fight, a 5 right after his bow makes the enemy skip a strike	***Cucumber Tide***: every 5 on the reel heals the party 5% of Balance	5 lands 3×
Liu Shun	6	*Six-Six Smooth*: two 6s make the rest of the round land in order (a free Cascade)	***Thirteen Orphans***: the reel becomes 13 mahjong tiles for one round; any pair pays ×13	6 lands 3×

5. Awakenings (Act IV: two branches per hero)

HERO	BRANCH A	BRANCH B
NIL	***The Remainder***: Divide by Zero charges 2× faster	***The Open Hand***: Hap's Favour warms after 1 Bust for the whole party
Vesper	***Dead Man's Hand***: crits (×3) on aces-and-eights (1 then 8 in a round)	***Blackjack***: if the round's strikes total 21 (or 210…), ×21
Maeve	***Forge-Mother***: every charm gets +1 level at forges	***Triskelion***: every third round is a guaranteed pattern
Hina	***Hundred Years***: she turns 100: +100 tens, loses the 9 immunity	***Kasa-obake***: stays a trickster: her wheel bounces into your best face once per round
Ba Jinling	***Bank of Bells***: interest on fen (1 per 4 held, uncapped)	***Golden Fan***: abacus beads hit every enemy for 8% of the round
Bai Si	***True Count***: the 4th-strike mark is permanent	***Mourning White***: Reaper's Toll heals the party instead of hurting it
Shunya	***Śūnya***: 0s count as any digit for bonds too	***Astrolabe***: she sees and shows every enemy wheel's next digit
Prima	***Even Prime***: 2 counts as prime *and* composite (best of both)	***Aegis***: the shield blocks the first two strikes each fight
Fibi	***1,597***: her digit can be frozen for one round	***Golden Spiral***: Cascades extend to 4 tubes
Varro / Kyūsuke / Liu Shun	***Bailiff*** / ***Full Dish*** / ***Heavenly Hand***	***Free Man*** / ***River God's Friend*** / ***Mahjong Queen***

6. Bonds and Link Arts

Bonds (GDD §2.3): Complements (sum to 10), Neighbours (consecutive), Twins (same digit), Prime Party, Powers of Two, Śūnya, Old World / New World.

LINK ART	PAIR	BOND	EFFECT	LOOK
Lucky Strike	Vesper + Maeve	Complement 7+3	Ricochet ×7 hits	Burning horseshoe shot in mid-air
Golden Sieve	Ba + Prima	Complement 8+2	Pierces all armour; +8 fen per hit	Coins pour through the sieve shield as spears
Ninety-Nine and One	Hina + Varro / Fibi on 1	Complement 9+1	The enemy's Omen display is forced to 01 for a round (×9 damage)	A hundred umbrellas open at once
Six Lanterns, Four Lanterns	Bai Si + Liu Shun	Complement 4+6	Reaper's Toll that heals the party	Mahjong tiles burn like lanterns
Lucky Eight	Vesper + Ba	Neighbours 7-8	Next 2 rounds' Pairs count as Jackpots	Gunshots ring eight bells
Lantern Fan	Ba + Hina	Neighbours 8-9	Fan blows the enemy's wheel one face	A gold fan scatters paper lanterns
Spark and Shield	Prima + Maeve	Neighbours 2-3	Shield bash + forge sparks: guaranteed Cascade	Hammer on shield
Third Lantern	Maeve + Bai Si	Neighbours 3-4	Removes the enemy's next 3 faces	A horseshoe cuts a talisman
Midnight Parade	Hina + Vesper	Old/New World	Every wheel shows the party's digits for a round	Ukiyo-e yokai stampede through a saloon door
Zero Count	Bai Si + Shunya	Śūnya	Lanterns go out; the sundial shadow cuts the enemy (×10 to the last digit)	Shadow of the observatory gnomon
Golden Ratio Deal	Fibi + Shunya	Śūnya	Spiral of cards; a 5-tube reel with 0 wild	φ drawn in light
Prime Directive	Prima + Vesper + Maeve	Prime Party (2-3-7)	Prime Trio guaranteed	Three columns, three shots

LINK ART	PAIR	BOND	EFFECT	LOOK
Powers of Two	Prima + Bai Si + Ba	Powers of Two (2-4-8)	Doubles the next round's Mult, twice	Binary lanterns
Twins	NIL + anyone with NIL's digit	Twins	Shared ultimate gauge; that Jackpot ×2	Mirror split-screen

7. Charms of the World (63: nine per set, seven sets)

Holding 3 from one set: set bonus I; 5: set bonus II. Slots grow 3 → 6.

SET	CHARMS (RULE IN SHORT)	SET BONUS I / II
Meridian (street luck)	Four-Leaf Clover (1 in 4: duplicate a digit) · Thirteenth Floor (×13 if the round total ends in 13) · Black Cat (first Bust → Pair) · Lucky 77 Jacket (+7 tens per 7) · Nixie Pendant (your digit +1 width) · Bazaar Ribbon (+1 charm slot, −10% damage) · Collector's Badge (+20% vs Collectors) · Naming Token (your digit's Pair ×3) · Gus's Receipt (Markets −15%)	+1 Mult per different culture held / charms trigger twice on Overwrite
Hyakki (shrine luck)	Beckoning Cat (+1 fen per strike ending in your digit) · Daruma (land your digit 3× → ×2 Mult for the fight) · Ofuda (first enemy strike parried) · Ebisu (+fen on Pairs) · Daikokuten (+tens on 8s) · Bishamonten (+parry window) · Benzaiten (Link Arts +20%) · Jurōjin & Fukurokuju (heal 3% per 7) · Hotei (Busts give +3 fen)	7s are blessed twice / all Seven Lucky Gods: every 7 is a Jackpot by itself
River Empire (prosperity)	Red Envelope (+8 fen on a Jackpot) · Door God (parries strike back) · Peach-Wood Sword (×2 vs ghosts and Noiselings) · Money Toad (interest +1) · Jade Bi Disc (Jade faces +2 Mult) · Firecracker String (first round ×2) · Lucky Knot (Pairs heal 2%) · Dragon Boat Oar (Cascades +1 extra round chance) · Joss Paper (on Settle, +88 fen)	Interest doubled / every 8 on the reel is gold (+3 fen)
Śūnyapura (auspicious)	Lemon and Seven Chillies (cancel one curse per fight; +7 tens per 7) · 108 Beads (every 108th point counts double) · Rangoli Chalk (first Zero-Sum each fight ×3) · Peacock Feather (Favour warms 1 Bust sooner) · Brass Diya (+1 Karma per Pair) · Kolam Thread (Neighbours bonds active for any two digits) · Shankha Shell (enemies' wheels −10% speed) · Navagraha Ring (+1 gem slot) · Palm-Leaf Page (see the boss rule early)	Karma ×2 / 0 is wild for bonds
Kroton Nova (proportion)	Evil-Eye Bead (see the enemy's next digit) · Fortuna's Wheel (once per fight, re-roll the reel) · Owl Coin (+1 Mult per prime strike) · Olive Wreath (+10% Balance) · Golden Ratio (Fibonacci triples count as Cascade) · Tetractys Token (if strikes sum to 10, ×10) · Amphora (store a digit for later) · Laurel of Seven (×7 on 7-7-x) · Thread of Lachesis (one Atropos-style cut per act: remove an enemy face)	Primes pierce / perfect numbers (6, 28) on the round total pay ×6
Crossroads (deals)	Horseshoe (your digit's cells 50% wider) · Rabbit's Foot (one free re-spin per fight) · Lucky Penny (+1 fen per 1) · Seven-Come-Eleven Dice (×7 if the round totals 7 or 11) · Pumpkin Lantern (×2 vs the headless) · Silver Dollar (Wager wins +50%) · Route 7 Sign (+7% speed on enemy wheels, +7 Mult) · Dead Man's Card (1–8 in a round: ×8) · Last Train Ticket (loop rounds keep the reel)	Deals' prices halved / one free Deal per act
The Open Hand (Hap; legendary, from shrines and Whispers)	Hap's Die (once per fight, set any reel digit) · Open Palm (Favour never cools below warm) · Shooting Star (once per act, re-deal a whole Market) · Coin on Its Edge (a Bust has 1 in 10 to become an Overwrite) · Rainbow at Night (+1 Mult per different digit on the reel) · Lucky Stranger (a random companion's passive each fight) · The Long Shot (Jackpots of 0 or 1 pay ×20) · Second Chance (revive once per run) · Hap's Laugh (on Overwrite, every charm triggers again)	Hap's Whispers appear twice as often / the finale's Fixed Fate is halved

8. Other build systems

- **Wheel crafting** (River Empire Markets): Remove · Duplicate · Enhance (Gold +3 fen · Jade +4 Mult · Glass ×2 strike, 1 in 4 shatters · Iron +20 tens while not stopped · Lantern slows the wheel passing it · Ink wild · Horseshoe lands as your digit 1 in 4 · Cursed +50 tens, curses the striker · Foxfire shows as another digit to enemies).
- **The Ledger Tree** (from Act III): 7 branches (one per Seal), 3–4 nodes each, bought with memory digits from Seals and side quests. Capstones: *Remainder +1* (permanent 4th tube), *Audit Eye*, *Compound Remainder*, *Open Hand*.
- **Levels:** shared XP; +10 tens per level; cap 30. Digits never change.
- **Gear:** one personal charm slot per hero.

9. Interaction matrix (what combos, what breaks)

	CHARMS	WHEEL CRAFTING	GEMS	KARMA	DEALS	LINK ARTS	HAP'S FAVOUR
Charms	sets	Horseshoe + wide faces ⇒ reliable Overwrites	pattern charms × gem levels	Diya feeds Karma	Crossroads set halves prices	Benzaiten +20%	Open Hand set
Wheel crafting		removing faces raises odds			+1 tube Deal		Favour widens matching faces
Gems				Cat's eye ⇒ Favour		Diamond	
Karma					Karma Deal		

	CHARMS	WHEEL CRAFTING	GEMS	KARMA	DEALS	LINK ARTS	HAP'S FAVOUR
Deals							"Favour hot" Deal

Watch list (likely to break the game, test first): removing faces to 3 per wheel (guaranteed Jackpots: cap minimum faces at 4) · Hap's Laugh + Overwrite loops (cap at one re-trigger) · Tetractys Token + Factor (allow, but cap ×10 once per round) · Liu Shun's Thirteen Orphans with Twins (cap ×13 once per fight).

Every number in the jam slice lives in `../tuning/jam_tuning.csv`; Unity imports it, so there are **no magic numbers in code**. This page explains the targets and shows the simulation that backs them. Re-run with `python3 tools/sim_patterns.py` and `python3 tools/sim_patterns.py --favour`.

1. Targets (what "fun" means in numbers)

MEASURE	TARGET	WHY
Small Audit length	3–4 rounds, 60–90 s	Long enough for a pattern, short enough to stay snappy
Big Audit	5–6 rounds	Room for a Cascade or an ultimate
Boss	9–12 rounds, 3–4 min	Two phases and at least one big moment
Pair or better	~65% of rounds (average player)	Something good happens most rounds
Jackpot or Overwrite	1 in 12 rounds (average player with own-digit Twins), about once per fight	The dream moment is common enough to chase
Three Busts in a row	**0%**	Bad luck never feels like a wall (Hap's Hand)
Damage growth	"No synergy" build stalls by Tamamo; "good synergy" build finishes Tamamo in $\leq$ 10 rounds	Balatro's rule: the build must matter
Losing	Party Settles $\approx$ 1 in 5 boss attempts for an average first-timer	Tension without frustration

2. Pattern odds (simulated, 60,000 rounds per cell)

Party: NIL (digit 7) + Vesper (7) + Maeve (3). "Skill" = chance a tap lands the face the player aims at. The **average player** is modelled at 25%.

SETUP	OVERWRITE	JACKPOT	CASCADE	ZERO-SUM	PAIR	BUST
10-face wheels, aim at own digit (original GDD)	1.0%	0.4%	1.4%	9.8%	30.1%	**57.3%**
10 faces + **Pattern Sight** (3rd striker aims to complete)	4.7%	1.3%	3.3%	14.5%	31.6%	**44.5%**
8-face starter wheels + Pattern Sight	5.8%	1.8%	2.6%	13.5%	35.3%	**41.1%**
8 faces + Pattern Sight + **Hap's Favour** (8-round fights)	—	—	—	—	—	**34.5%**

Findings that changed the design: 1. **Aiming at your own digit in a mixed party makes Busts more likely** (mixed party, 50% skill: 72% Busts). The fix is *reading the reel*: Pattern Sight shows the third striker which faces complete a pattern (GDD §1.1). 2. **10-face wheels are too wide for a starter.** Starters have 8 faces; removing faces is the core of wheel crafting later. 3. **Bad-luck streaks** happened in 25% of an average player's fights before Hap's Favour, and in 0% after (SYSTEMS §2.4). 4. **Overwrite must not depend on the digit** (×3·digit gave digit 0 a ×0 and digit 9 a ×27). It is now a flat **×21** for every digit.

Expected Mult per round (from the table, 8 faces + Pattern Sight): **$\approx$ 1.6 for a random tapper, $\approx$ 2.8 for an average player**, most of the gap coming from Overwrites. That gap is the skill expression we want.

3. Damage and Balance curve (jam slice)

Strike = tens (10 × level) + digit. Round damage = Σ strikes × Mult.

POINT IN THE SLICE	PARTY LEVEL	TYPICAL ROUND (AVG PLAYER, NO CHARMS)	WITH A GOOD BUILD	ENEMY BALANCE	ROUNDS TO WIN
Fight 1 (NIL alone)	1	(10+5)×3 × 2.8 $\approx$ 125	—	Debt Hound 300	3
Fight 2 (NIL + Vesper)	1	$\approx$ 125	—	3 × 300 (one at a time)	3 each
Collectors (Big)	2	(20+5)×3 × 2.8 × 0.75 (armour) $\approx$ 160	$\approx$ 220	500	3–4
Varro	2	$\approx$ 210 (jars cost ~1 tube in 3 rounds)	$\approx$ 300	1,200	6–9
Act I Small	3	(30+5)×3 × 2.8 $\approx$ 295	$\approx$ 450	1,500	4–5
Act I Big	3	$\approx$ 295	$\approx$ 500	2,500	5–8
Tamamo	4	(40+5)×3 × 2.8 $\approx$ 380	$\approx$ 700 (Hina + charms + Link Art)	6,000	9–16

The plain build needs ~16 rounds against Tamamo (and will likely Settle), the good build ~9: the build matters, as intended. If playtests show the plain build winning easily, raise Tamamo to 7,500 before touching anything else.

4. Enemy damage

Enemy strike = enemy tens + stopped digit; the player stops enemy wheels too, so parries (0) and Fortune (your digit) depend on skill. Starter enemy wheels have 10 faces (no omissions) so parry odds are 10–12%.

ENEMY	TENS	AVG HIT	HITS TO SETTLE A 300-BALANCE PARTY
Debt Hound	20	~24	12
Collector	30	~34	9
Varro P1 / P2	40 / 80	~44 / ~88	7 / 3–4
Tamamo	60	~64 (+curses)	7 at Balance 420

5. Economy (jam)

Fen: Small 8 · Big 15 · Elite 25 · Boss 40; interest 1 per 5 held (cap 5). Prices: common charm 6, rare 12, reroll 3 (+1 each time), face removal 5, enhancement 8. Expected fen at Ofuku's shrine ≈ 45: enough for one rare charm *or* two enhancements, never both (a real choice).

6. How to tune during the jam

1. Change the CSV, never the code.
2. Re-run the simulation if you touch wheels, faces or patterns.
3. Change one number per playtest round, and write down why in the CSV note column.
4. Priority when a fight feels wrong: enemy Balance → enemy tens → wheel speed → pattern mults (last; they're the game's language).

Done when: Unity loads every value above from the CSV, the simulation matches §2, and two playtest rounds have confirmed or corrected §3.

The full-game model behind `TUNING.md` (jam). Build power must grow faster than linear stats so that synergy, not grinding, wins (GDD §4.2).

1. Progression layers

LAYER	GROWS BY	RESETS?	WHERE
Party level	XP from fights (cap 30)	No	Tens +10 per level
NIL's verbs	Story (one per act)	No	GDD §6
Companions	Story + secret quests	No	8 core + 3 secret
Bonds	Teahouse scenes (4 per hero)	No	Second passive, Link Art upgrade, true-ending flag
Awakenings	Act IV choice per hero	No (respec at Act IV shrines)	SYSTEMS §5
Ledger Tree	Memory digits from Seals and side quests	No	7 branches
Charms, faces, gems	Fen, rewards, shrines	**Per act** (see §4)	The build
Fen	Fights, interest, gold faces	Per act (half carries)	Markets
Karma	Parries, Zero-Sums	Decays 25% per act	Śūnyapura+

2. Enemy Balance curve (GDD §4.2, exponential)

ACT	SMALL	BIG	BOSS	PARTY LEVEL (EXPECTED)	EXPECTED MULT NEEDED
Prologue	300	500	1,200	1–2	~2.5
I	1,500	2,500	6,000	3–4	~4
II	8,000	14,000	35,000	6–8	~10
III	50k	90k	200k	10–12	~35
IV	300k	600k	1.5M	14–17	~150
V	2.5M	5M	12M	19–22	~700
Finale	—	—	99,999,999	24–26	~3,000 (over Phase 3 with Link Arts)

Linear tens alone give ~5× damage from Prologue to finale; Balances grow ~80,000×. The rest must come from Mult: patterns × gems × charms × bonds × Link Arts. That is the Balatro curve and the reason builds matter.

3. Fen economy (per act)

SOURCE	FEN
Small / Big / Elite / Boss	8 / 15 / 25 / 40 (+50% per act)
Interest	1 per 5 held, cap 5 (Jinling's Bank of Bells removes the cap)
Gold faces	+3 per stop
Whispers, side quests	5–30

SINK	PRICE (ACT I → V)
Common / rare / legendary charm	6 / 12 / 25 → ×1.5 per act
Re-roll a Market	3, +1 each re-roll in the same Market
Face removal / duplicate / enhancement	5 / 8 / 8
Oracle stick	4
Gem	10
Wager	any

Rule of thumb: each Market should offer one thing the player wants and can't quite afford. Expected fen at each Market ≈ 1.2× the price of one rare item.

4. What resets between acts (the roguelite spine)

Within the **story campaign**, each act is a mini-run: your **charms and face enhancements carry over**, but you choose which **3 charms** to keep at each act's end ("pack light: the ferry only takes three"); fen halves; Karma decays 25%. This keeps each act's build fresh and each new land's charm set relevant, without throwing away the player's work. In the **Endless Ledger**, everything resets per run.

5. Difficulty and losing

- **Losing a fight** returns you to the act's start **with the build you had there** plus any memory digits earned (never a total loss).
- **Assist mode** (any time): wheels −30%, enemy Balance −25%.
- **Omen Levels 1–8** (post-game): +25% enemy Balance per level, faster enemy wheels, harsher Land Laws, fewer Shops, no Hap's Hand at level 8.

6. Unlock cadence (something new every ~15 minutes)

Prologue: patterns, parry, charms · Act I: party of 3, bonds, Link Arts, curses · Act II: fen economy, wheel crafting, sticks, Ink · Act III: Karma, gems, Ledger Tree · Act IV: Awakenings, number properties · Act V: Deals, Wagers, loops · Finale: everything, then taken away · Post-game: Endless Ledger, Omen Levels, Daily Omen, Hap's Gallery, New Game+.

7. Validation

Extend `tools/sim_patterns.py` into a run simulator (party + charms + gems) and check that a "no synergy" build fails the Act III boss while a "synergy" build passes it with ~10 rounds. Tune boss Balances, not pattern mults, first.

Stopping a number is the whole game. If that one tap doesn't feel incredible, nothing else saves us. This spec defines the tap frame by frame, the payoff ladder, and the speed rules that keep spectacle from becoming waiting. All times in milliseconds at 60 fps.

1. Principles

1. **Response before spectacle.** The wheel reacts to the tap within one frame (16 ms). Spectacle follows; it never delays the response.
2. **Readability first.** The stopped digit is legible within 150 ms, under every effect, at 720p, on a phone.
3. **Payoff scales with rarity.** A Bust is a shrug, a Jackpot is a fireworks finale. The ladder (§4) is strict so players learn value by feel.
4. **Every animation earns its time.** The first time you see a pattern, it plays in full. The third time, it's shortened. Hold to fast-forward anything.
5. **Sound is half the feel.** Every beat below has a sound (see `AUDIO.md`); the stop sound always cuts through the mix.

2. Anatomy of one strike

T (MS)	EVENT	VISUAL	CAMERA	SOUND
0	Hero selected	Hero steps forward; wheel unfurls above them in 180 ms (scale 0.6 → 1.05 → 1.0)	Push in 5% on the hero over 300 ms	Wheel "unfurl" whoosh
180	Spinning	Ones digit scrolls (speed per hero); tens fixed; faces sized by odds; Pattern Sight shimmer on the 3rd striker	Idle sway ±0.5°	Tick per face passed (pitch rises with speed)
tap	**Stop**	Wheel decelerates over 90 ms with ease-out-back (overshoot 6%) and settles on the face the tap was over	2-frame micro-shake (2 px)	"Clack" (layered: mechanical + Nixie zap)
+90	Settle	Stopped face flashes white 1 frame, then the land's accent colour; other faces dim to 30%	—	Short resonance tail
+120	Hit-stop begins	Hero snaps into attack key pose (smear frame); enemy flinch	Hit-stop 80 ms (Bust/plain) to 140 ms (your lucky digit)	Hit + voice bark (30% chance)
+260	Damage number	Strike value tears off the wheel: the tens in white, **the stopped digit in the hero's colour**, 1.4× size, then shrinks to 1.0 over 120 ms	Small zoom-out	Number "tear"
+380	Digit to reel	The ones digit flies along an arc (300 ms, ease-in-out) and **slams** into the next reel tube	Tube-impact shake 3 px	Slam (pitch by tube: low, mid, high)
+680	Ready	Next striker can be chosen; input buffer accepts a tap from +500	—	—

One strike ≈ 0.7 s after the tap. A three-strike round, excluding player thinking time, ≈ 2.1 s + the payoff.

Input

- Tap anywhere, **Space**, or the hero's number key. Mouse, touch and keyboard identical.
- Latency budget: input → wheel stop visible ≤ 50 ms end to end (measure on WebGL Chrome and Safari).
- Input buffer: a tap during the last 180 ms of the previous animation is queued, not lost.
- Double tap within 120 ms counts once. A tap during the unfurl (first 180 ms) is ignored, with a soft "too early" tick so it never feels like a dropped input.
- Assist mode: wheels −30% speed; **hold-to-stop** alternative (press and release; stop on release).

3. Enemy strike and parry

T (MS)	EVENT
0	Enemy wheel unfurls over the enemy, faces sized by its odds; its special faces show icons
0–∞	The player stops it (same verb). 0 → **Parry**: the strike bounces, counter-flash, 120 ms hit-stop, "ting". Your lucky digit → **Fortune**: reflect with a gold trail. Otherwise: damage, hero hurt pose, Balance drop, Omen ticks down with a mechanical flip
Telegraphs	Any special move is shown one round ahead: an icon on the enemy's wheel and a distinct sound (BESTIARY rules)

4. The payoff ladder

PATTERN	HOLD BEFORE REVEAL	HIT-STOP	SCREEN	CAMERA	MUSIC	CINEMATIC	REPEAT RULE
Bust	0	—	Tubes flicker out	—	—	—	Always short (0.4 s)

PATTERN	HOLD BEFORE REVEAL	HIT-STOP	SCREEN	CAMERA	MUSIC	CINEMATIC	REPEAT RULE
Pair	150	60	Silver flourish over the two tubes	Small punch-in	Stinger (2 notes)	—	0.8 s
Zero-Sum	200	100	Mirror wipe; ledger lines	—	Ledger chime	Enemy's next action icon crossed out	1.2 s
Cascade	200	100	Time-slow bands; blue	Slow 0.5× for 300 ms	Rising arpeggio	"YOU ACT AGAIN"	1.4 s
Jackpot	300	160	Squad panels slash down; screen crack	Crash zoom	Full fanfare	Counting Mult up to the huge number	2.5 s first time → 1.5 s after 3
Reaper's Toll	300	200	Red wash; four bells	Slow push	Four tolls	Every Omen with a 4 loses 44%	2.0 s
Omen Overwrite	400	220	Everything gold; your number rings bells (count = your digit)	Spin around the party	Your hero theme, full	The city rewrites itself (land-styled)	3.0 s first → 2.0 s after 3
Link Art	—	—	Split-screen duo cut-in	Authored	Duo motif	3–5 s	Skippable after first view
Ultimate	—	—	Hero's cut-in band (land-styled)	Authored	Hero theme	3–5 s	Skippable after first view

The count-up: round damage counts up over 600 ms (ease-out), then the Mult slams in and the total counts again over 900 ms, with a "ching" every power of ten. The final number stays on screen until the next tap (Balatro's best trick: let people *look* at a big number).

5. Speed and respect for time

- **Hold anywhere** to fast-forward (4×) any animation; a setting makes every repeat cinematic short.
- After a pattern has played **3 times** in a session, its reveal uses the short version.
- A full fight (60–90 s target) has at most **one** long cinematic unless the player triggers more.
- Never block input for more than 3 s without a skip prompt.

6. Juice checklist (every interaction)

Anticipation (a tell before) · Response (within one frame) · Impact (hit-stop, shake, flash, sound) · Follow-through (settle, particles) · Feedback in numbers (the digit in the hero's colour) · Recovery (back to readable calm within 1 s).

7. Reference clips to collect (in `art/reference/feel/`)

Balatro (score count-up; card shake) · Hi-Fi Rush (beat-synced hits) · Persona 5 (all-out attack cut-ins; menu motion) · Hades (hit-stop, dash feel) · Peggle (the "Ode to Joy" finale and slow-mo last peg) · Pachinko and slot machines (reel stop, overshoot, win lights) · Cyberpunk: Edgerunners (smear frames, flashes).

Done when: the browser prototype implements §2–§4 and 4 of 5 first-time testers say stopping a wheel "feels good" without being asked about it (`PLAYTEST.md`).

Jam players decide in 60 seconds; raters play 10–20 minutes. This is the game they actually judge. Second by second for minute one, then beat by beat to minute ten.

Rules 1. The **first tap** happens within **20 s** of pressing Start. 2. The **lucky number** is chosen in **minute 1**. 3. The **first Jackpot is guaranteed** in the first fight (scripted wheel), so everyone sees the dream moment early. 4. No text box over **2 lines**. Every mechanic is taught by **doing it once, safely**. 5. The **hook** (REMAINDER DETECTED) lands before **minute 3**. 6. Returning players skip straight to the Naming shrine (title → "Continue" or "New number").

Minute 1, second by second

T	PLAYER SEES	PLAYER DOES	LEARNS	FEELS
0:00	Black. A single Nixie tube flickers **NaN**. Title card slams: ZERO-SUM: OMEN	Taps "Start" (unlocks audio)	—	Curiosity
0:03	Kinetic type over rain: "MERIDIAN." / "EVERY SOUL WEARS A NUMBER." / "IT DECIDES WHO SETTLES."	Watches (tap skips)	The world rule	Intrigue
0:09	Rain Alley 9 at dawn. Ama's Omen ticks **03 → 02 → 01**. Tobi stares	Watches	Numbers are life	Unease
0:14	Collectors arrive. A hooded figure steps in; **NaN** glitches over their head	—	This is you	"Who is that?"
0:17	A wheel of digits spins around Ama's 01. Prompt: **TAP** (one word, pulsing)	**First tap**	Tapping stops the number	Power
0:18	The wheel slams on **07** (scripted: any tap lands 07). Ama's Omen rings like a bell. Collector: "That one has no number."	—	Stopping numbers changes fate	Delight
0:24	Chase cut (8 s, motion graphics over charm-hung alleys)	—	The world is multicultural	Momentum
0:32	Brother Nought's shrine; the Naming terminal: ten Nixie tubes	—	—	Ritual
0:38	"Every soul needs a number, child. **STATE YOUR LUCKY NUMBER.**" Hovering a digit shows one line of what it means across the Tally	**Chooses 0–9**	Your number matters	Ownership
0:55	The terminal overloads. City-wide screens: **REMAINDER DETECTED**. The Basilica's eye opens	—	You are hunted	Thrill

Minutes 1–10, beat by beat

TIME	BEAT	TEACHES	DESIGN NOTES
1:00	Logo slam with your digit; Debt Hounds leap from the rooftops	—	Straight into play
1:10	**Fight 1**, NIL alone. Round 1: "TAP" prompt; strike; digit flies to the reel. Round 1's reel is scripted to a **Jackpot** of your digit's neighbour (big fanfare)	Stop, strike, the reel, Jackpot	The fanfare is full length the first time
2:00	Enemy strike: its wheel shows a wide **0** face. Prompt: "STOP ON 0"	Parry	The first enemy wheel has 0 triple-wide
2:20	Round 2: a Pair or Bust happens naturally; one line: "Two match: a PAIR."	Pair; Bust is OK	Bust shows a shrug, not a failure
2:45	Win. Results flash: biggest number	Reward	5 s, skippable
3:00	**Vesper drops in**: "Need a hand? Seven bucks." (3 stills + motion graphics, 12 s)	—	Her first line is the game's tone
3:15	**Fight 2**: tap a hero to choose who strikes. Third strike: **Pattern Sight** shimmers; one line: "Shimmering faces finish a pattern."	Choose strikers; read the reel; Cascade; Zero-Sum	Vesper's wheel is fast: contrast
4:30	A Bust, then a second Bust (likely or scripted): the **Open Hand** sigil lights: "Hap's watching." Next round can't Bust	Hap's Favour	The first mention of Hap; mystery, no explanation
5:00	Map appears (in-world: a city map on a wall). Two paths: **Charm stall** or **Maeve's forge**	Choice	Both are good; one is a fight
5:20	Charm stall: pick 1 of 3; charm pins onto the reel frame	Charms	One line per charm
6:00	(or) Maeve's forge: defend it; Maeve joins	—	
7:00	Collectors (Big Audit): armour icon; "Pairs bounce off armour"	Enemy rules; telegraphs	Varro is foreshadowed in the background
8:00	Livia's window (5 s, no prompt)	—	A thread for later
8:15	**Varro** at the drain gates: debt jar rule shown with one line and a demo	Boss rules	Boss intro card: name, Omen 61, one-line rule
9:30	Varro phase 2; Vesper's gauge fills: "**ULTIMATE READY**: tap Vesper's portrait"	Ultimates	The Requiem saves NIL: the second big wow

TIME	BEAT	TEACHES	DESIGN NOTES
10:00	Varro spared; rooftop map reveal (six lands light up)	The journey ahead	Ends the first session on a promise

Teaching rules

- One new thing per fight. Never teach in a menu.
- Every first time plays in full; every repeat is shorter (GAME_FEEL §5).
- If a player hasn't tapped for 6 s, the prompt pulses; at 12 s it adds one line of help.
- The **Codex** (pause menu) explains everything taught so far, with a GIF each.

Done when: 5 first-time players reach Varro without asking a question, and at least 3 can explain Pair, Jackpot and Parry afterwards (`PLAYTEST.md`).

Minimal UI only works if every state is still readable. Every screen is an object from the world (GDD §8); every piece of information has one place. Accessibility for the full game is in `ACCESSIBILITY.md`; this is the jam-ready baseline.

1. Rules

1. **Diegetic first.** If it can be an object in the world, it is (the reel is a sign, the map is a scroll, the shop is a stall).
2. **One glance.** In combat the player can always answer at a glance: whose turn, which hero is next, what the enemy will do, how much Balance is left.
3. **Two lines max** of text at a time; no tooltips during a spin.
4. **Colour never carries meaning alone:** every colour cue has a shape or a number.
5. **Everything is reachable with one hand**: tap/click, Space/Enter, number keys 1–3, Esc.

2. Screen flow

```
Title → (New game) Intro → Naming shrine → Combat ⇄ Map → Market / Shrine / Teahouse / Story → Boss
→ Act end → Hap's Dream → next act
  |
|
  └→ Continue → Map              Pause (any time): Resume · Codex · Settings · Quit to title
Results / Game over → Retry act / New number
```

3. Screens

SCREEN	IN-WORLD OBJECT	ALWAYS VISIBLE	ON DEMAND	INPUT
Title	A rain-wet Nixie sign: ZERO-SUM: OMEN; NaN flickers	Start / Continue / Settings	Credits	Tap
Naming shrine	Ten Nixie tubes in a shrine niche	The ten digits	One line per digit on hover	Tap a tube; confirm by tapping again
Combat	The diorama stage; the land's reel object at top centre; charms pinned to its frame	Omens over every head (two digits); the reel; whose turn (the active hero steps forward and glows); enemy intent icon over the enemy (one icon); Open Hand sigil; Land Law icon	Hold a hero: their wheel, passive and ultimate gauge. Hold the enemy: its rule. Hold a charm: its rule	Tap hero / 1–3 to choose; tap / Space to stop; hold to fast-forward
Ultimate ready	The hero's portrait card flips gold at the screen edge	Gauge as a ring around the portrait	—	Tap the portrait
Map	Land-styled (scroll, palm leaf, mosaic, road map)	Nodes as icons; your path; the boss at the end	Hold a node: one line	Tap a node
Market	A stall with wares on the table	3–5 items with price tags	Hold an item: rule + flavour line	Tap to buy; tap the bell to re-roll
Shrine / Forge	Ofuku's shrine / Maeve's anvil	One choice of 2–3	—	Tap
Open Hand shrine	A stone hand with a die	Re-deal · Pray · Offer	—	Tap
Teahouse / Campfire	Heroes sit around a fire; NIL at centre	Party slots beside NIL; constellation lines between digits (bonds)	Hold a hero: kit card	Drag a hero beside NIL, or tap to cycle
Story	Full-screen stills + motion graphics, or video	Speaker name tab + 2 lines	Log (last 20 lines)	Tap to advance; hold to fast-forward
Results	A receipt printed by a Basilica terminal	Biggest number, patterns found, rounds, time	—	Tap
Game over	Your Omen flips to 00 like a departure board; "SETTLED" stamp	Retry act (keep the build you had at act start) · New number	—	Tap
Pause	The ledger book closes over the screen	Resume · Codex · Settings · Quit	—	Esc / tap the corner
Codex	NIL's notebook	Patterns, charms seen, enemies met, people met, lore, each with a GIF or sketch	—	Tabs

4. Combat layout (1920×1080 reference; scales to 16:9 and 16:10; safe area 5%)

- **Top centre:** reel object (3 tubes), Land Law icon left of it, Open Hand sigil right of it, charms along its frame.
- **Left half:** party (up to 3), standing; Omens over heads; ultimate portraits in the lower-left corner.
- **Right half:** enemy (or bosses that fill the right two-thirds); its Omen and intent icon above it.
- **Centre:** the active wheel appears over the striker.
- **Bottom:** clear (the hit numbers and Link Art bands need it).

5. Text and numbers

- Fonts: display Big Shoulders Display (numbers, banners), body Barlow Semi Condensed, code-ish JetBrains Mono (ledger/receipts); Noto for CJK and Devanagari.
- Minimum sizes at 1080p: body 28 px, numbers over heads 44 px, damage numbers 64 px+.
- Numbers: tabular figures; Omens always two digits ("07"); large damage numbers use thin spaces (12 000 000) and switch to short form above a billion (1.2 B) with the full number on hold.

6. Settings (jam)

Master/music/SFX/voice volume · **Reduced motion** (no shake, no flashes, short cinematics) · **Assist** (wheels –30%, hold-to-stop) · **Text size** (100/125/150%) · **Colour-safe digits** (shapes on pattern colours) · Subtitles (on by default) · Fast cinematics · Screen shake intensity · Language (English for the jam).

7. Accessibility baseline (jam must-haves)

- Reduced motion; flash limit (no more than 3 flashes per second; checked with a photosensitivity tool).
- Assist mode and hold-to-stop for players who can't time taps.
- Every pattern has a shape and a sound, not just a colour.
- All story text is subtitled; speaker names always shown.
- Full mouse-only, touch-only and keyboard-only play.

Done when: each screen above has a wireframe in `art/ui/wireframes/` and every setting in §6 works in the jam build.

What the player should feel, minute by minute and act by act, and the loops that make "one more" irresistible without manipulation.

1. The three loops

LOOP	LENGTH	WHAT HAPPENS	THE PULL
The tap	3 s	Choose a striker → stop the wheel → digit slams into the reel	Skill + suspense, every three seconds
The round / fight	10 s / 60–90 s	Three digits → a pattern → the count-up → the enemy's turn → parry	A small payoff every round, a big one most fights
The run / act	25–35 min	Map choices → charms, faces, gems → companions and bonds → the boss	The build: "what if I add *this*?"
The journey	6–8 h	Lands, story, bond arcs, NIL's verbs, the reveal, the endings	Mystery and people you care about
The ledger (post-game)	endless	Endless Ledger, Omen Levels, Daily Omen, Hap's Gallery	Mastery and discovery

2. Why it's moreish (and where we draw the line)

DRIVER	HOW WE USE IT	THE LINE WE DON'T CROSS
Variable reward	The reel's pattern odds; the Jackpot is rare enough to chase	No real-money randomness, ever; the casino is aesthetic only (World Bible §10)
Near miss	Two matching digits and a third that *almost* landed are shown as "so close"	We never fake a near miss; the wheel is honest and Pattern Sight shows real odds
Agency over luck	Every tap is a choice; the third striker reads the reel; builds bend the odds	Luck never decides a fight without a player decision first
Pity / luck turns	Hap's Favour: three Busts in a row can't happen	—
Build discovery	Synergies between digits, charms, faces and gems (Balatro's "oh, *that* works")	Every rule is readable; no hidden maths
Collection	63 charms, Hap's faces, Codex entries, Link Arts	No completion that requires grinding
Short sessions	Saves at every node; a fight is 60–90 s	No energy timers, no daily login pressure
Social	Daily Omen seed and leaderboard; screenshot-friendly big numbers	Opt-in only
Story pull	Every act ends on a Thale scene and a Hap's Dream question	Cliffhangers resolve; no paid endings

3. Emotional arc of the campaign

ACT	TARGET EMOTION	INTENSITY (1–10)	REST BEAT	PEAK BEAT
Prologue	Curiosity → thrill	4 → 7	Rooftop map reveal	REMAINDER DETECTED; Vesper's Requiem
I · Hyakki	Wonder, mischief	5 → 8	Teahouse with Hina	Hundred Demons' Night vs Hundred-Year Night
II · River Empire	Ambition, injustice	6 → 8	The silent Ten Courts	Jinling burns the books; Jin Chan's fall
III · Śūnyapura	Awe, identity shock	4 → 9	The quietest Teahouse (after the reveal)	"NIL is the Remainder"; the eclipse
IV · Kroton Nova	Moral anger → resolve	6 → 8	The undercity of the fallen	Prima holds the children over the drop
V · Crossroads	Fatalism → hope	5 → 9	The motel at 3 a.m., the 1,597th time	Dawn breaks
Finale	Dread → catharsis	7 → 10 → 3	The last campfire	Phase 3: every companion returns; the Open Hand

Intensity rules: never more than two acts in a row peaking at 9+; every act has one scene with no music; every boss is followed by a quiet scene.

4. Moment-to-moment feelings (one fight)

Anticipation (enemy reveal and its rule) → control (choosing strikers) → suspense (the spin) → payoff (the stop, the pattern) → threat (the enemy wheel) → relief (parry) → triumph (the count-up and Settle). If any of these is missing for most testers, the fight is redesigned.

5. Player types and what each gets

PLAYER	WANTS	GETS
The story player	Characters and the mystery	Assist mode, fast cinematics off, bond arcs, Hap's Dreams
The build player	Synergies and big numbers	63 charms, gems, faces, Deals; the Endless Ledger
The skill player	Timing and mastery	Fast heroes, Omen Levels, Daily Omen leaderboard

PLAYER	WANTS	GETS
The collector	Completion	Codex, Hap's Gallery, all Link Arts, secret recruits
The jam rater	A great 20 minutes	The First 10 Minutes plan; the full slice in 25–35 min

Done when: each act in ACT_DESIGN has a named rest beat and peak beat, and playtests confirm the peak for most testers.

PART IX

Production

From jam slice to full game.



How we get from today (Fri 6 Nov 2026, jam day 1) to a polished jam slice by **Fri 13 Nov**, then to a full game. Read with `GDD.md` and `WORLD_BIBLE.md`.

Hard jam deadline: Sat 14 Nov 2026, 12:00 UTC. A WebGL build is required. AI use must be disclosed (log it in `AI_DISCLOSURE.md` as you go).

1. Reality check

The full vision (a capital plus five lands inspired by Japanese, Chinese, Indian, Greek and American myth; 9 playable heroes; ~30 enemies; 7 bosses; 5 keeper trials; 4 endings; ~20 minutes of cinematics) is a **12–15 month** production for a small team. In 7 days the right move is a **vertical slice** that looks finished. Make a short game that feels like the opening of a AAA anime RPG, not a long game that looks like a jam game. The jam judges and players will see 20–30 minutes; make every one of them look like the trailer.

APPROACH	VERDICT
All 7 acts, rough	❌ Loses on art, polish and cohesion
Intro + Prologue + Act I, polished, ending on a cliffhanger	✅ Wins on theme (lucky number in the first minute), art, narrative and fun
Intro + Prologue only	✅ Fallback if the team is 1–2 people

2. Phases

PHASE	DATES	GOAL	EXIT CRITERIA
0 · Pre-production & look-dev	Fri 6 Nov (today)	Lock story, systems, style; prove the art pipeline	Bibles merged ✅ · style anchor image chosen · NIL key art approved · Unity WebGL "hello" build loads on a private itch page
1 · Jam vertical slice	Sat 7 – Fri 13 Nov	Ship *"Prologue & Act I: The Remainder and the Night Parade"*	Submitted, with disclosure, by Fri 13 Nov
1.5 · Jam rating	14 Nov – 10 Dec	No build changes (jam rule). Collect feedback; prep Steam page; R&D	Feedback log; Act II design locked; Steam page draft
2 · Demo: Acts 0–II	Dec 2026 – Mar 2027	The River Empire (Bāfā Harbor + the Ten Courts): economy, wheel crafting, oracle sticks, Ink; 6 heroes (+ Ba Jinling, Bai Si); NIL's *Rewrite*	Public demo for the next Steam Next Fest; trailer; wishlists
3 · Production: Acts III–V	Apr – Sep 2027	Śūnyapura (India-inspired), Kroton Nova (Greece-inspired), the Crossroads Territory (America-inspired); full roster of 8; Karma, Navaratna gems, Awakenings, Deals, Wagers, Ledger Tree; VO; localisation (Japanese, Chinese, Hindi); cultural reviewers per land	Content complete through Act V
4 · Finale, endgame, launch	Oct 2027 – Jan 2028	Meridian, 4 endings, Endless Ledger, Omen Levels, Daily Omen, QA	Launch on PC; mobile (the one-tap verb suits touch) after
5 · The Far Lands (post-launch)	2028	Expansion acts: Norse North (the Norns), Kemet (Ma'at's scales), the Long Count (Maya zero), Anansi's Coast (Akan folklore), the House of Wisdom (al-Khwārizmī), each with cultural consultants	One act per update

3. The jam slice (Phase 1) in detail

3.1 Content list

CATEGORY	ITEMS
Cinematics	Intro (≈2:30, World Bible §8, including the five-culture ledger montage) · Prologue end: world-map reveal with six lands in six art styles (≈30 s) · Act I end: Tamamo's question + Thale teaser (≈45 s) · 2 Link Arts (*Lucky Strike*, *Midnight Parade*, 4 s each) · ultimates for NIL, Vesper, Maeve, Hina (3 s each)
Playable heroes	**NIL** (Splice, Carry), **Vesper** (7, Prologue), **Maeve** (3, Prologue, optional), **Hina** (9, Act I): choose 2 of 3 in Act I
Bosses	**Varro the Ledgerhound** (debt jar) · **Tamamo, the Nine-Tailed** (nine tails, foxfire illusion faces, the curse of 9) · Keeper's Trial: **Enma** (optional elite)
Enemies	Meridian: Debt Hound, Collector, Noiseling · Hyakki: Kasa-obake (umbrella spirit), Chōchin-obake (lantern ghost), Kappa, Oni enforcer
Regions	Meridian Undercity (battle stage + map) · Hyakki (2 battle stages: the red bridge of the Night Parade, Tamamo's foxfire palace; hand-scroll map) · Naming shrine (intro)
Systems	Everything in GDD §9: verb, reel, 7 patterns, parry, bonds (Complements/Neighbours/Twins), 12 Charms of the World incl. Hyakki set bonus (3 slots), shrine face crafting (Remove, Gold, Jade, Glass), Maeve's forge, ultimates

CATEGORY	ITEMS
Audio	Meridian theme (synth noir), Hyakki theme (shamisen, shakuhachi and taiko over synth pads), Tamamo boss theme; every land later gets its own instrumentation (erhu/guzheng, sitar/tabla, lyre/bouzouki, slide guitar/harmonica); each in 4 stems that fill with the reel; SFX vocabulary; optional AI voice for the Basilica and Thale (disclose it)
Length	25–35 minutes

3.2 Team shape

SIZE	ROLES	SCOPE
5 (ideal)	Lead dev (combat/sequencing) · Dev 2 (UI, map, data, build) · Character artist/animator · Environment + VFX artist · Cinematics/audio/writing	Full slice
3	Dev · Art/animation · Cinematics/audio/design	Full slice, with 1 Link Art and enemies as 3-pose sprites
1–2	Everything	Intro + Prologue (NIL + Vesper + Maeve, Varro), then a "to be continued" Hyakki teaser

3.3 Day-by-day (all dates UTC)

DAY	DATE	CODE	CHARACTER ART & ANIMATION	ENVIRONMENTS & VFX	CINEMATICS, AUDIO, WRITING	EXIT TEST
1	Fri 6	Unity 6 project, URP, WebGL; **empty build on a private itch page**; port the prototype's rules to the C# `Core` with tests	Higgsfield look-dev: style anchor, **NIL key art**, Vesper, Maeve, Hina, Tamamo concepts (§4)	Meridian Undercity + Hyakki establishing concepts	Intro script and shot list final; temp music	Build loads on itch; style anchor chosen
2	Sat 7	Combat greybox with wheels, reel, patterns, parry, hero picking, scoring formula	NIL / Vesper / Maeve / Hina: turnarounds, expressions, **layered PSDs**	Meridian battle stage split into parallax layers; in-world neon reel	Intro storyboard frames (stills); generate intro video shots batch 1	A fight is fun with greybox art
3	Sun 8	Map scroll + nodes; charms (ordered effect pipeline); bonds	**Rig** NIL / Vesper / Maeve / Hina (2D Animation); attack key poses ×3 each	Hyakki stages ×2 (layers); ukiyo-e woodblock VFX (flat wave and cloud shapes); torii-and-lantern reel	Intro shots batch 2; edit intro v1	Prologue playable start-to-finish
4	Mon 9	Ultimates and Link Arts (video + in-engine); NIL's Splice and Carry; enemy wheel profiles	Varro + Tamamo boss rigs (Tamamo's nine tails on spring bones); 7 enemies (3-pose sprite animation)	Land-styled cut-in bands (neon glitch vs woodblock print); hit sparks; foxfire	Ultimate clips; Link Art clips; music stems v1	Act I playable with placeholder bosses
5	Tue 10	Boss rules (debt jar; nine tails; foxfire illusion faces); shrine face crafting; Maeve's forge; Teahouse party select. **FEATURE FREEZE**	Secondary motion (hair/cloth springs), blinks, reactions to the reel	Lighting pass, post-processing, weather	Prologue-end (world map) + Act I-end cinematics; SFX pass	Full slice playable
6	Wed 11	Balance (wheel speeds, targets) with 5 playtests; WebGL perf + size. **CONTENT LOCK**	Polish poses, victory animations	Polish, optimisation (atlases, compression)	Final mix; subtitles; credits	60 fps; < 250 MB; no blockers
7	Thu 12	Bugs only; reduced-motion + assist modes; release candidate	Capture GIFs and screenshots	itch page art	AI disclosure finished; page copy	RC on the private page
Buffer	Fri 13	**Submit.** Rate your 3-game queue.				Submitted

4. Art pipeline with Higgsfield

4.1 What's available

NEED	HIGGSFIELD TOOL	NOTES
Character and environment stills	**3D Jutsu** → `generate_angles` (Seedream V5 Pro) from a blockout render; or the Higgsfield web app's image models (Nano Banana Pro, GPT Image 2.6, Seedream, Cinema Studio)	The blockout fixes pose and framing, which keeps the roster consistent
Consistent characters across images and videos	**Elements** (reusable reference characters/environments)	Create one per hero from the approved key art, then use it in every later prompt
Video cutscenes	**3D Jutsu** → `generate_video` (Seedance 2.5: text-to-video or start image), or the web app (Seedance 2.0, Kling 3.0 with a start frame, Cinema Studio Video 3.0)	720p, 4–30 s per clip; use start frames or Elements for consistency
Animated idle loops (optional)	Genjutsu motion presets on a character sheet	For dialogue portraits; check loop seams

Composition guides already rendered (free) in this repo: `art/lookdev/pose_blockout.png` (full-body hero pose, 2:3) and `art/lookdev/street_blockout.png` (deep street with a tower, 16:9). The 3D Jutsu project is **"ZERO-SUM OMEN - Look Dev"**.

4.2 Character pipeline (per hero, ~1 day for a 2D artist plus AI)

1. **Key art.** Use the 3D Jutsu pose blockout and the prompt from `art/HIGGSFIELD_PROMPTS.md`. Generate 4 variations and pick 1.
2. **Element.** Create a Higgsfield Element from the approved image (name it after the hero).
3. **Sheets.** Using the Element: front/side/back turnaround, 6 expressions, 3 attack key poses, hurt, victory, ultimate splash.
4. **Paint-over.** Fix hands, the costume's signature details (NIL's decoder is on the *right* wrist; Vesper's cylinder has 7 chambers) and the palette. This is where AI art becomes *our* art, and where coherence is won.
5. **Layer split.** Separate hair front/back, face, eyes, mouth, torso, arms, coat tails and weapon into a PSB (manual + segmentation tools; ~2–3 h).
6. **Rig.** Unity 2D Animation bones + mesh deformation (a Live2D-style idle: breathing, hair and coat on spring bones, blinks).
7. **Action.** Anime *limited animation*: snap between the 3 drawn key poses with smear frames, holds on 2s/3s, impact frames, VFX and camera. This is a genuine anime technique, not a shortcut, and it reads as high-end.
8. **Ultimate.** A 3–4 s video from the splash illustration (image-to-video), played full-screen.

Making characters feel alive: they **react to numbers**. Vesper grins and spins her cylinder when a 7 lands; Maeve kisses her clover on a 3; Hina's umbrella sticks out its tongue on a 9; Tamamo's tails flare when a 4 appears; everyone flinches on a Bust; idle barks reference the reel. Reaction poses are cheap, and they make the theme personal.

4.3 Environment pipeline (per stage, ~half a day)

1. Blockout in 3D Jutsu with the battle camera (heroes left, enemy right, reel object top-centre).
2. Generate the painting at 2k using the region's style prompt.
3. Split into 3–5 depth layers (depth estimation + inpaint the hidden areas).
4. In Unity: layered planes at depth, Cinemachine moves give real parallax; add particles (rain, mist, lanterns, joss paper, petals) and shader motion (water, flags, flicker).

4.4 Cinematic pipeline

1. Storyboard each shot as a still (image generation, same Elements).
2. Animate with Seedance 2.5 or Kling 3.0 from that still as the **start frame**, 5–10 s per shot, 720p.
3. Edit, then grade to the region palette, add sound design, music and subtitles.
4. Export H.264 720p; play via Unity `VideoPlayer` from `StreamingAssets` (WebGL plays video from a URL; audio is unlocked by the title-screen tap).
5. Mix video with in-engine shots: the Naming terminal choice and the logo slam must be in-engine so they use the player's digit.

4.5 Consistency rules (pin these up)

- Always reference the hero's **Element**, and append the **style suffix** from the prompt pack, unchanged.
- One palette per land; every hero keeps their signature colour (NIL orange-teal, Vesper crimson, Maeve copper-green, Hina red-white, Tamamo gold-violet foxfire, Varro rust).
- QC every image: hair streak, eye colour, signature prop, handedness, digit glyphs readable.
- Keep a log: every generation that ships goes in `AI_DISCLOSURE.md` (tool, model, what, how it was edited).

4.6 Credit budget

Paid generations spend Higgsfield credits. Rough counts for the slice: **~150–250 images** (including iterations) and **~35–45 video clips** (intro ~10, other cinematics ~10, ultimates/Link Arts ~6, iterations). Check current pricing in your Higgsfield plan before starting, and confirm commercial-use rights for your plan before any post-jam release.

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5. Technical plan

AREA	DECISION
Engine	Unity 6 LTS (6000.3); WebGL 2 baseline (WebGPU optional on 6.6+ only if tested in every major browser)
Rendering	URP; 2D characters on lit quads (Shader Graph, alpha clip + rim light); baked or unlit painted backgrounds; Bloom/LUT/vignette/grain
Characters	Unity 2D Animation (PSB importer, bones, Sprite Skin), spring bones for hair and cloth

AREA	DECISION
Camera & cutscenes	Cinemachine 3, Timeline, VideoPlayer (StreamingAssets URLs)
Tweening	PrimeTween
Code layout	`Core/` (pure C#: wheels, reel, patterns, scoring, bonds, charm pipeline, deterministic RNG; EditMode tests) · `Sequencing/` (async beat scripts: HeroStrike, EnemyStrike, PatternPayoff, LinkArt, Ultimate) · `Data/` (ScriptableObjects: HeroDef, WheelDef, FaceDef, CharmDef, EnemyDef, BossRuleDef, ActMapDef) · `Presentation/` · `Audio/` (stem controller, beat clock)
Save	PlayerPrefs (IndexedDB on web): act checkpoints, bonds, codex
WebGL budget	< 250 MB total; character atlases $\leq 2048^2$; Crunch compression; music streamed Vorbis; videos 720p H.264 at ~2.5 Mbps; Gzip with decompression fallback; first fight playable within 20 s of load
Accessibility	Reduced motion, assist (slower wheels), subtitles, colour-independent signals

The **rules core comes straight from the prototype** (`prototype/index.html` : wheel odds, pattern evaluation, round flow). Port it on Day 1 so art can land on working gameplay from Day 2.

6. Risks

RISK	LIKELIHOOD	MITIGATION
AI art inconsistency across poses/videos	High	Elements + paint-over + QC checklist; start frames for every video shot
Animation scope (rigs take longer than planned)	High	Limited-animation key poses; enemies as 3-pose sprites; cut Link Arts to 1
WebGL size and performance with high-res art and video	Medium	Budgets in §5; test the uploaded build daily from Day 1
Some community raters dislike visible AI art	Medium	Heavy paint-over on heroes; disclose honestly; let the hand-built motion, systems and writing carry the identity
Cultural inaccuracy across five cultures	Medium–High	Reviewers from each culture (Japanese, Chinese, Indian, Greek, American) for names, scripts, clothing and terms; revered deities are keepers, never enemies; no Indigenous sacred material (World Bible §10)
Theme too buried under RPG systems	Low	The lucky-number choice comes in minute one, the reel is on screen every second, and bonds are digit relationships
Credits run out mid-week	Medium	Day 1 look-dev decides the style before mass generation; batch only approved prompts

7. Step-by-step: the next 24 hours

Status at the end of Day 1 (Fri 6 Nov): story, systems, art direction v2, the World & Art Bible PDF and the art book draft are done; the prototype matches the canon; 3D bases and colour keys exist for the Remainder Wards and the Hyakki Moon Pier; the vector identity kit (emblems, Omen displays, charm icons, reel objects, cast figures) is in `art/graphics/` .

1. **Unblock image generation.** Higgsfield has 6 paint passes stuck in "submission unknown" (check credits and job history on higgsfield.ai; don't resubmit blindly, it may double-charge). Alternatives: enable Hugging Face Spaces invoke for this connector, or connect Meshy.
2. **Paint the style frames** over the existing 3D bases: the Wards street, the Moon Pier, NIL in action, and one battle frame (Art Book Production Guide §7, images 1, 4 and 9 first).
3. **Lock NIL**: key art → Higgsfield Element → turnaround, checked against the canon locks and the figure-kit silhouette in the art book.
4. **Unity:** project, URP, WebGL settings; upload an empty build to a private itch page; confirm it loads.
5. **Port** `Core` from the prototype with tests (wheel, reel, pattern priority, Balance/Omen readout, scoring formula from GDD §1.2 and §1.6).
6. **Lock the intro script** (World Bible §8) and generate its storyboard stills, including one still per culture for the ledger montage.
7. End-of-day review: one screenshot-quality mock of a Hyakki battle on the Moon Pier made from the approved art. If it doesn't look like a trailer frame, fix the style before anything is mass-produced.

Art status (end of Nov 6)

ASSET	STATUS	NEXT STEP
Art Direction v2 style frames (NIL action, Meridian dawn, Hyakki pier, battle frame)	6 paint passes stuck in Higgsfield "submission unknown"	Check credits; reconcile; then paint over the 3D bases

ASSET	STATUS	NEXT STEP
3D bases and colour keys	Done: Remainder Wards, Moon Pier, line-up base	Paint pass (register A)
Design figures for all 16 characters + 4 Meridian eras	Done (vector stand-ins, art book and model pages)	Replace with key art and turnarounds
Graphic identity: emblems, Omen displays, charm icons, reel objects	Done (art/graphics/)	Refine by hand; reuse as UI icons
Character key art, v2 style (16 characters)	Pending	Generate after the style frames are approved
Environment establishing + battle stage per land (7 lands)	Meridian and Hyakki explored in pass 1; colour scripts for all	Redo in v2; then the five remaining lands
Enemy sheets (Meridian, Hyakki)	Pending	3-pose sprites per enemy
Intro cinematic shots	2 test shots done	Storyboard stills, then video
Turnarounds and attack key poses for jam heroes	Pending	Needed for rigs

What has to be written **before** we build, so that what we build is fun, polished and loved, and what we can leave until later. Read with `PRODUCTION_PLAN.md` and `GDD.md`.

0. The honest rule first

Documents don't make a game fun. **Playtests make a game fun.** Balatro, Hades and Into the Breach were found by playing builds every day, not by writing. So pre-production docs have one job: **make sure every hour of building goes into the right thing, with a clear bar for "good enough"**, and leave room to discover the fun in the prototype.

That means: 1. **Lean.** Each doc is the shortest thing that answers its questions. A table beats a page; a GIF beats a paragraph. 2. **Testable.** Every doc ends with a "done when" check someone can verify. 3. **Gated.** Only the **Gate A** docs block the jam build. Everything in Gate B can wait until after the jam, when we know what players loved. 4. **Living.** The Markdown files in `docs/` stay the source of truth; the bible and art book rebuild from them.

The jam deadline is **Sat 14 Nov 2026, 12:00 UTC**. Gate A has to be finished by the end of **Sat 7 Nov** (Day 2) at the latest, or it costs build time we don't have.

1. What we already have

DOC	STATUS	GAP THAT STILL MATTERS
GDD	✅ Strong	Needs numbers (Gate A #2) and a moment-to-moment feel spec (Gate A #1)
World Bible, Lore Lexicon, Locations	✅ Strong	Full act scripts are Gate B
Character Bible	✅ Strong	Bond-arc scripts are Gate B
Narrative Style	✅ Good	Jam script itself is missing (Gate A #8)
Art Direction, Art Book Production, art book PDF	✅ Strong	Asset list with specs (Gate A #6); painted style frames still blocked
Production Plan	✅ Good	Needs a cut list per feature (Gate A #3) and a definition of done (Gate A #10)
AI Disclosure	✅ Running log	Final summary at submission
Prototype	✅ Playable core verb	Must be playtested against Gate A #9 before Unity work locks the rules

The **missing** layer is everything between "great idea" and "great to play": feel, numbers, the first ten minutes, sound, and the quality bar. That's what Gate A covers.

2. Gate A: write before jam development (11 docs, about 1 day of work in total)

Ordered by how much they decide whether people love the game.

#	DOC (FILE)	OWNER	TIME	WHY IT MATTERS FOR FUN
1	**Game Feel Spec** (`GAME_FEEL.md`)	Lead dev + design	2 h	The tap-to-stop is the whole game. If stopping a wheel doesn't feel incredible, nothing else saves it
2	**Tuning Sheet** (`tuning/jam_tuning.csv` + `TUNING.md`)	Design	2 h	Fun lives in the numbers: fight length, how often patterns land, how fast Mult grows
3	**Jam Slice Spec + Cut List** (`JAM_SLICE.md`)	Producer	1 h	A finished small game beats an unfinished big one; cuts are decided calmly now, not at 3 am on Day 6
4	**First 10 Minutes (FTUE) Beat Sheet** (`FIRST_10_MINUTES.md`)	Design + writing	1.5 h	Jam players decide in 60 seconds. Raters play 10–20 minutes. This is the game they actually judge
5	**UX, Diegetic UI and Accessibility Spec** (`UX_SPEC.md`)	Design + UI	1.5 h	Minimal UI only works if every state is still readable; a timing game must be playable by everyone
6	**Asset List and Specs** (`ASSET_LIST.md`)	Art lead	1.5 h	Art is the bottleneck. Every asset needs a size, format, layer split and owner before anyone generates or rigs
7	**Audio Direction and Sound List** (`AUDIO.md`)	Audio	1 h	Sound is half the game feel; a reel that clicks on the beat feels twice as good
8	**Jam Script and Storyboards** (`JAM_SCRIPT.md` + `storyboards/`)	Writing + cinematics	2 h	Cutscenes are our biggest wow and our biggest risk; boards first, generation second

#	DOC (FILE)	OWNER	TIME	WHY IT MATTERS FOR FUN
9	**Playtest Plan and Fun Metrics** (`PLAYTEST.md`)	Design	45 min	Tells us what "fun" means in numbers, and when to change course
10	**Quality Bar and Definition of Done** (`QUALITY_BAR.md`)	Whole team	45 min	"GOTY quality" has to mean a checklist, or it means nothing
11	**Technical Design Doc** (`TECH_DESIGN.md`)	Lead dev	1.5 h	WebGL limits, deterministic rules, data-driven content: the things that are expensive to change on Day 5

A1. Game Feel Spec (`GAME_FEEL.md`)

Must answer - The exact anatomy of one strike, frame by frame: wheel appears → spin → tap → stop (ease, overshoot, settle) → hit-stop → camera → number tear-off → digit slams into the reel. Timings in milliseconds. - Input: latency target (< 50 ms tap-to-visual), input buffering, what happens on a mistimed or double tap, keyboard/mouse/touch parity. - Payoff ladder: what Bust, Pair, Zero-Sum, Cascade, Jackpot, Overwrite and Reaper's Toll each look and sound like, scaled by importance (hit-stop length, shake strength, flash, slow-mo, music sting). - Speed: how a round speeds up after the player has seen a pattern 3 times (shortened cinematics, hold to fast-forward), so spectacle never becomes waiting. - "Readability first": the stopped digit must be legible within 150 ms, under every effect. - Reference clips (GIFs) from games that nail it: Balatro (score counting), Hi-Fi Rush (beat sync), Persona 5 (all-out attack), Hades (hit-stop), Peggle (Ode to Joy finale).

Done when the prototype implements it and 4 of 5 playtesters say stopping a wheel "feels good" without being asked about it.

A2. Tuning Sheet (`tuning/jam_tuning.csv` + `TUNING.md`)

Must answer - Every number in the jam slice in one sheet: wheel speeds and face sizes per hero, base tens, enemy Balances and wheels, pattern mults, charm values, fen rewards, shop prices. - Targets: an ordinary fight lasts **60–90 s**, a boss **3–4 min**; a Pair or better lands in about **50%** of rounds, a Jackpot in **5–10%**, an Overwrite about once per fight with a good build. - The growth curve: damage per round at each node for a "no-synergy" build vs a "good synergy" build. The good build has to clearly win by the boss, as in Balatro; the plain build should struggle by Tamamo. - A simple simulation (a script running 10,000 random rounds) that checks the pattern odds against the targets.

Done when the Unity build loads its values from this sheet (no magic numbers in code) and the simulation output matches the targets.

A3. Jam Slice Spec + Cut List (`JAM_SLICE.md`)

Must answer - The exact content of the slice: every node, fight, enemy, charm, hero, cutscene and line count. (Production Plan §3 has the outline; this is the locked list.) - Every item tagged **Must / Should / Could**, with the day by which a Should or Could is cut if it isn't working. - The **"minimum lovable slice"**: what ships if everything goes wrong (e.g. intro + Prologue + one Hyakki fight + Tamamo).

Done when the team signs it and nothing new enters without something else leaving.

A4. First 10 Minutes Beat Sheet (`FIRST_10_MINUTES.md`)

Must answer - Second-by-second for minute 1, then beat-by-beat to minute 10: what the player sees, does, learns and feels. - Rules: the first tap happens within **20 s** of pressing Start; choosing the lucky number happens in **minute 1**; the first Jackpot is guaranteed in the first fight (scripted wheel); no text box over 2 lines; every new mechanic is taught by doing it once, safely. - The "hook moment": the REMAINDER DETECTED reveal must land before minute 3. - What the skip button skips, and what returning players never see again.

Done when 5 first-time players reach the first boss without asking a question, and at least 3 can explain Pair, Jackpot and Parry afterwards.

A5. UX, Diegetic UI and Accessibility Spec (`UX_SPEC.md`)

Must answer - Every screen and state (combat, choose striker, enemy turn, reel payoff, map, market, teahouse, pause, settings, game over, victory), each as the in-world object from GDD §8, with a wireframe. - What is always visible vs shown on demand; how the player knows whose turn it is, which hero is next and what the enemy will do, without HUD panels. - Accessibility: reduced motion (no shake or flashes), slower-wheel assist, hold-to-stop alternative, colour-independent digits (shape and number, never colour alone), subtitles and text size, photosensitivity check for flashes, one-hand play, rebinding. - Mouse, touch and keyboard parity; safe areas for itch's embedded player.

Done when each screen has a wireframe and every accessibility option is in the settings menu of the jam build.

A6. Asset List and Specs (`ASSET_LIST.md`)

Must answer - One row per asset in the slice: id, type, owner, source (3D base / AI paint / hand), resolution, layer split for parallax or rigs, animation states, status, and the day it's due. - Specs per type: character PSB layer naming for Unity 2D Animation, background parallax layer count and sizes, VFX sheet sizes, video codec and length, UI atlas. - File naming and folders, so assets drop into Unity without renaming.

Done when every Must item from A3 has a row, an owner and a due day.

A7. Audio Direction and Sound List (`AUDIO.md`)

Must answer - Music: tempo per land (the reel and stops sync to the beat clock), stems that build as the reel fills, the boss themes, cutscene cues. - The SFX vocabulary: wheel tick, stop, hit, each pattern sting, parry, Settle, UI. Each pattern needs a sound you'd recognise with your eyes closed. - Mix rules (the stop sound always cuts through), loudness target, and sources and licences (plus AI-use disclosure for any generated audio).

Done when every Must interaction in A1 has a named sound in the list, and the temp music runs on the beat clock in the prototype.

A8. Jam Script and Storyboards (`JAM_SCRIPT.md` + `storyboards/`)

Must answer - Every line of dialogue in the slice, in the Narrative Style format, with speaker, emotion and trigger. - Storyboards (thumbnails are fine) for the intro (World Bible §8), Vesper's rescue, the world-map reveal, Hina's entrance, Tamamo's defeat and the cliffhanger, each with timing and "skippable / not" marked. - Which shots are video (Higgsfield/Seedance), which are in-engine, which are still frames with motion graphics.

Done when an animatic (boards with temp audio) of the intro plays end to end and the team agrees it hooks.

A9. Playtest Plan and Fun Metrics (`PLAYTEST.md`)

Must answer - When: a playtest every day from Day 2 (the prototype counts), with 3–5 new players per round, plus one cultural reviewer before release. - What we watch: time to first tap; time to first smile or "oh!"; deaths and quits by node; "one more run?" said or done; which pattern players cheer. - What we ask afterwards (5 questions max): what was the best moment, when were you confused, when were you bored, which hero do you want next, would you play again. - Thresholds that change the plan, e.g. if fewer than half want one more run, stop adding content and fix the core.

Done when the first session has happened on the browser prototype and its findings are logged.

A10. Quality Bar and Definition of Done (`QUALITY_BAR.md`)

Must answer - One checklist per feature type (fight, cutscene, character, background, UI screen, sound), drawn from Art Direction §5, Art Book Production §8 and A1/A5. A feature is done only when it passes its checklist **and** has been playtested. - The "trailer test": would a 3-second clip of this feature make someone click? If not, it isn't done. - What we never ship: placeholder text, stray AI artefacts (wrong digits, broken hands), untranslated or unreviewed cultural details, flashes that fail the photosensitivity check.

Done when the team uses it in the Day 5 feature freeze review.

A11. Technical Design Doc (`TECH_DESIGN.md`)

Must answer - Architecture: the pure C# `Core` (rules, patterns, Balance/Omen, scoring, RNG with seeds), `Sequencing`, `Data`, `Presentation` (Production Plan §5), and the boundaries between them. - Data: ScriptableObject schemas for heroes, wheels, faces, charms, enemies, boss rules and maps; how the tuning sheet (A2) imports. - Determinism and testing: seeded RNG; EditMode tests for every pattern and the priority order (GDD §1.5); a replay log for bug reports. - WebGL: memory and size budgets, video playback, audio unlock, load time (first fight within 20 s), browser test matrix (Chrome, Firefox, Safari, mobile Chrome). - Build and release: daily build to a private itch page, versioning, save data (localStorage via PlayerPrefs) and what happens when it's cleared.

Done when an empty WebGL build is live on itch and the `Core` tests pass for all seven patterns.

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3. Gate B: write after the jam, before full production

These matter for the full game, but they need what the jam teaches us. Writing them now would be guessing.

#	DOC	WHAT IT ANSWERS	START AFTER
B1	**Player Experience and Emotional Arc**	Target emotion per act; an intensity curve for the whole campaign (spikes, rest beats, the midpoint reveal)	Jam feedback
B2	**Systems Design, one doc per system**	Charms (all ~90, with set bonuses), wheel crafting, oracle sticks, Ink, Karma, Navaratna gems, Awakenings, Deals, Wagers, Ledger Tree; an **interaction matrix** of which systems combo, and which break the game	B3
B3	**Economy and Progression Model**	Fen income and costs per act; difficulty curve vs expected build power (a simulation like A2, for 6–8 hours); run length; how losing feels fair	Jam tuning data
B4	**Act and Encounter Design, one doc per land**	Node maps, encounter tables, Elites, the Keeper trial, the boss script with phases and telegraphs, set pieces	B1, B3
B5	**Enemy and Boss Sheets**	Per enemy: wheel profile, behaviour, telegraph, counter, art and animation needs	B4
B6	**Full Script, Bond Arcs and Dialogue Database**	Every scene and line; 4 bond scenes per hero; the branching and ending flags for the four endings	B1
B7	**Cultural Consultation Plan and Review Log**	Reviewers per culture (Japanese, Chinese, Indian, Greek, American), what each signs off, issues found and fixed (World Bible §10)	Before any public Act II–V material
B8	**Localisation Guide**	String rules, font coverage (kanji, hanzi, Devanagari, Greek), text expansion, glossary for translators, the Japanese, Chinese and Hindi plan	B6
B9	**Accessibility Plan (full game)**	Extends A5 against the Game Accessibility Guidelines and Xbox Accessibility Guidelines; difficulty and assist modes	Jam feedback
B10	**Telemetry and Balancing Plan**	Events to log (opt-in), dashboards for win rates, pick rates and quit points, the balance-patch process	B3
B11	**QA and Test Plan**	Test cases per system, regression suite, platform matrix, bug triage rules	B2
B12	**Market, Audience and Positioning**	Target players, comparable games (Balatro, Persona, Slay the Spire, Hades), the hook in one sentence, price, platforms, Steam tags	Jam ratings
B13	**Marketing Beat Plan and Trailer Script**	Steam page, wishlists, festivals (Next Fest), trailer shot list built from Link Arts and Overwrites, community plan	B12
B14	**Legal, IP and AI Policy**	Originality review (inspired by Balatro and anime, never copying), asset licences, music licences, AI-generated asset policy and disclosure, trademark check on the name	Before the Steam page
B15	**Team, Pipeline and Tools Handbook**	Who owns what, review cadence, the art pipeline end to end (3D base → paint → overpaint), version control rules for large assets	When the team grows

4. Order of work for the next 36 hours

1. **Today (Fri 6 Nov), 2 h:** A3 Jam Slice + Cut List, then A1 Game Feel. Everyone needs to know what we're making and how it should feel.
2. **Today, 2 h:** A2 Tuning Sheet; wire the browser prototype to it and run the first playtest (A9) tonight.
3. **Sat 7 Nov, morning:** A4 First 10 Minutes, A5 UX, A11 Tech Design (lead dev starts Unity in parallel).
4. **Sat 7 Nov, afternoon:** A6 Asset List, A7 Audio, A8 Script + boards, A10 Quality Bar.
5. **From Sun 8 Nov:** no new docs. Update Gate A docs only when a playtest changes something.

5. One-page checklist

Written means the document exists and answers its questions. Each doc's **"done when"** check (a playtest, a build, a sign-off) still has to happen in production; those are tracked in the right-hand column.

Gate A (blocks the jam build) | | Doc | Written | "Done when" still needs | |---|---|---|---| | A1 | Game Feel Spec | ✅ | Prototype implements §2–§4; 4 of 5 testers say it feels good | | A2 | Tuning + `jam_tuning.csv` + `tools/sim_patterns.py` | ✅ (simulated) | Unity loads the CSV; two playtest rounds confirm §3 | | A3 | Jam Slice + Cut List | ✅ | Team signatures | | A4 | First 10 Minutes | ✅ | 5 testers reach Varro without help | | A5 | UX, Menus, Diegetic UI, Accessibility | ✅ | Wireframes in `art/ui/wireframes/`; settings in build | | A6 | Asset List and Specs | ✅ | Owners assigned; status kept daily | | A7 | Audio Direction + Sound List | ✅ | Temp music on the beat clock | | A8 | Jam Script + Storyboards | ✅ script | Thumbnails in `art/storyboards/`; intro animatic | | A9 | Playtest Plan | ✅ | First session logged | | A10 | Quality Bar | ✅ | Used at the Tue 10 freeze review | | A11 | Technical Design | ✅ | Empty WebGL build live; Core tests pass |

Gate B (after the jam; written now so production starts from a full plan) | | Doc | Written | |---|---|---| | B1 | Player Experience, Loops, Emotional Arc | ✅ | | B2 | Systems Design (luck and fantasy mechanics, kits, ultimates, Awakenings, Link Arts, 63 charms, interaction matrix) | ✅ | | B3 | Economy and Progression | ✅ (run simulator still to build) | | B4 | Act and Encounter Design (all lands) | ✅ | | B5 | Bestiary: Enemies, Elites, Keepers, Bosses | ✅ | | B6 | Full Story + Bond Arcs and Dialogue Database | ✅ (outline + key lines; full line-by-line script per act still to write) | |

B7 | Cultural Consultation Plan | ✅ (reviewers to hire) | | B8 | Localisation Guide | ✅ | | B9 | Accessibility Plan | ✅ | | B10 | Telemetry and Balancing | ✅ | | B11 | QA and Test Plan | ✅ | | B12 | Market and Positioning | ✅ (update with jam data) | | B13 | Marketing Plan + Trailer Script | ✅ | | B14 | Legal, IP and AI Policy | ✅ (lawyer review before Steam) | | B15 | Team Handbook | ✅ |

Rescoped Nov 7 (after the refinement review): "Prologue: NaN". Ship one complete, polished **10–15 minute Meridian Prologue** with **NIL, Vesper and Varro**. Act I (Hyakki) is a **stretch goal** that starts only after the Prologue passes the gate in §1A. The first engineering milestone is **one correct, readable Audit running in the browser in Unity WebGL**. Deadline: **Sat 14 Nov 2026, 12:00 UTC (17:30 IST)**; we submit **Fri 13 Nov**. AI use disclosed; WebGL build required.

Tags: **M** = must (no ship without it) · **S** = should (cut on the date shown if not working) · **C** = could (only if everything else is done).

1A. The gate before any Hyakki work

All of these must be true in the WebGL build on the private itch page: - [ ] One full Audit is correct (scoring matches GDD §1.3 and `node tools/test_scoring.js`) and readable at a glance - [ ] Beats 1–12 play start to finish in 10–15 minutes with no blocker - [ ] GAME_FEEL §2 timings in place; 4 of 5 testers say stopping a wheel feels good - [ ] Varro's Versus card, phases and Vesper's ultimate work (Art Direction v3 §0.4–§0.5) - [ ] Load to first tap ≤ 20 s; 60 fps on the reference laptop

1. Player journey (Prologue 10–15 min, then Act I as stretch)

#	BEAT	CONTENT	TAG	CUT DATE
1	Title	Logo, tap to start, audio unlock, settings	M	—
2	Intro cinematic	Shots 1–4 of World Bible §8 (myth montage + title), ~45 s	M (shortened)	—
3	Rain Alley 9	Ama's Omen drops to 01; NIL stops it on 07 (scripted interactive stop)	M	—
4	Naming shrine	Brother Nought; **choose your lucky number**; REMAINDER DETECTED	M	—
5	Fight 1	Debt Hounds ×2; NIL alone; scripted Jackpot	M	—
6	Vesper joins	Rescue scene (stills + motion graphics)	M	—
7	Fight 2	Debt Hounds ×3; choose strikers; Cascade/Zero-Sum	M	—
8	Maeve's forge	Defend the forge (Big Audit: Collectors) → Maeve joins	C	Tue 10
9	Charm stall	Gus Mendoza: pick 1 of 3 charms	M	—
10	Boss: Varro	Versus card (Art Direction v3 §0.5), Debt Jar rule, 2 phases, Vesper's Requiem	M	—
11	World-map reveal	Six lands light up; Seal 1	M	—
12	Hap's Dream 0	15 s dream: "Pick a number…"	S	Wed 11
13	Ferry → Moon Pier	Fight 3: Chōchin-obake on the pier	Stretch	after §1A gate
14	Night Parade	Hina joins (party of 3, bonds)	Stretch	after §1A gate
15	Map choice	Ofuku's shrine (charm + sets) **or** fox-wedding Whisper	Stretch	after §1A gate
16	Fight 4	Kasa-obake pack	Stretch	after §1A gate
17	Teahouse	Party swap; one bond scene (Vesper + Hina)	Stretch	after §1A gate
18	Market / Elite	Kappa market (Kyūsuke cameo) **or** Kitsune twins elite	Stretch	after §1A gate
19	Enma's trial	Keeper trial (no parry, 4 rounds); NIL learns Carry	Stretch	after §1A gate
20	Boss: Tamamo	Nine tails, illusion faces, Shi and Ku; *Hundred Demons' Night* vs *Hundred-Year Night*	Stretch	after §1A gate
21	Ending	Seal 9; Thale's line "The Remainder is learning to count"; "To be continued: the River Empire"	Stretch	after §1A gate
22	Results	Biggest number, patterns found, time; "Play again with a new number"	M	—

Minimum lovable slice (if everything goes wrong by Wed 11): beats 1–7, 9–11, 22 (the Prologue alone, ~10 minutes). **Full jam target:** beats 1–12 + 22. **Stretch:** beats 13–21 (Act I) only after the §1A gate. If Act I is not built, the Prologue ends after beat 11 (or Hap's Dream 0) on a manga splash card: **"TO BE CONTINUED: THE NIGHT PARADE"**, then Results.

2. Systems in the slice

SYSTEM	TAG	NOTES
Fate Wheels (8-face starters), reel, all 7 patterns, priority	M	GDD §1, SYSTEMS §2.1–2.2
Pattern Sight	M	Biggest single fix for Busts (TUNING §2)
Hap's Favour (warm + Hap's Hand)	M	Removes 3-Bust streaks

SYSTEM	TAG	NOTES
Parry / Fortune on enemy strikes	M	
Omen over a Balance, party Balance	M	GDD §1.6
Heroes: NIL, Vesper, Hina (M); Maeve (S)	M/S	Party of 3 max
Ultimates for every hero in the slice	M	Video or in-engine; 3–5 s
Bonds: Complements, Neighbours, Twins	M	Constellation lines in the Teahouse
Link Arts: *Midnight Parade* (M), *Lucky Strike* (S)	M/S	
Charms: 12 (Hyakki 6 + Meridian/Crossroads 6), 3 slots, sets	M	SYSTEMS §7
Land Laws: Debt (Meridian), Shi and Ku (Hyakki)	M	
NIL verbs: Splice (M), Carry (S)	M/S	
Wheel crafting: Remove + Gold/Jade/Glass at Ofuku's shrine	S	Tue 10
Hap's Whispers (1 per act)	S	Wed 11
Tsukumogami faces	C	
Economy beyond fen rewards, oracle sticks, Ink, Karma, gems, Awakenings, Deals	✗	Not in jam

3. Assets in the slice (owners and specs in `ASSET_LIST.md`)

- **Characters (rigged, 2D):** NIL, Vesper, Hina (M); Maeve (S). Bosses: Varro, Tamamo (M). Enemies (3-pose sprites): Debt Hound, Collector, Chōchin-obake, Kasa-obake (M); Kitsune twins (S).
- **Stages (parallax):** Rain Alley 9, drain gates (Meridian); Moon Pier, foxfire palace (Hyakki). 4 stages (M).
- **Reel objects:** Nixie sign, torii with three lanterns (M).
- **Cinematics:** intro (M, ~45 s), Vesper rescue (M, stills), world-map reveal (M), Hina's entrance (M), Tamamo defeat (M), ending (M); Hap's Dream 0 (S).
- **Music:** Meridian theme, Hyakki theme, Tamamo boss theme (M); stems (S).

4. Rules for change

1. New content needs a matching cut of equal size, agreed in the daily stand-up.
2. A **Should** that isn't working on its cut date is cut, not "nearly done".
3. Feature freeze **Tue 10 Nov**; content lock **Wed 11 Nov**; bugs only from **Thu 12 Nov**.
4. Every day ends with a build on the private itch page.

Done when: the team has signed this list (names below) and the Day 2 build matches beats 1–7.

Signed: _____ (design) · _____ (code) · _____ (art) · _____ (audio/cinematics)

Art is the bottleneck. Every asset in the jam slice has an id, an owner, a spec and a due day before anyone generates, paints or rigs. Status: ☐ not started · ◐ in progress · ☑ done. Due days follow `PRODUCTION_PLAN.md` §3.3.

1. Specs by type

TYPE	FORMAT	SIZE	LAYERS / STATES	NOTES
Character (rigged)	PSB for Unity 2D Animation	2048 px tall master; atlas ≤ 2048^2	head, hair_front, hair_back, torso, arm_L_upper/lower/hand, arm_R_…, leg_…, coat_tail_L/R, props; 3 expressions on the head	Flat cel, 2 tones + 1 rim; coloured lines (Art Direction §2.2)
Character poses	PSB frames	as above	idle (rig), attack key poses ×3, hurt, victory, ultimate start/end	Smear frames drawn
Enemy (sprite)	PNG sheet	1024 px tall	idle, attack, hurt (3 poses)	
Boss (rigged)	PSB	2560 px tall	per boss sheet	Tamamo's tails on spring bones
Battle stage	PSD → PNG layers	3840×2160	sky, far, mid, near, floor, foreground (6 layers)	Empty plates; deck clear for actors
Reel object	PSD → PNG	1200×500	frame, 3 tube backs, glass, glow	Digits rendered in engine (TextMeshPro)
Cut-in band	PNG + shader	3840×900	2–3 layers	Land-styled
VFX	Flipbook PNG	1024^2 per sheet, 4×4	—	Hand-drawn, flat shapes
Story still	PNG	3840×2160	2–4 layers for motion graphics	
Video cinematic	MP4 H.264	1280×720, ~2.5 Mbps	—	Streamed from StreamingAssets
UI	PNG atlas	2048^2	—	9-slice where possible
Icons (charms, emblems)	SVG → PNG	256^2	—	From `art/graphics/`

Naming: `{type}_{id}_{variant}_{version}.{ext}`, e.g. `chr_nil_attack01_v02.psb`, `stg_moonpier_mid_v01.png`. **Folders:** `Assets/Art/{Characters,Enemies,Bosses,Stages,Reels,VFX,UI,Story,Video}`. **Pipeline per asset** (Art Book Production §4): 3D base → paint pass → overpaint → layer split → engine. AI use is logged in `AI_DISCLOSURE.md`.

2. Jam slice list

Characters and enemies

ID	ASSET	TAG	OWNER	DUE	STATUS
chr_nil	NIL: rig, 3 attacks, hurt, victory, ultimate	M	Char artist	Sun 8	☐
chr_vesper	Vesper: as above	M	Char artist	Sun 8	☐
chr_hina	Hina + umbrella (separate rig with its own expressions)	M	Char artist	Mon 9	☐
chr_maeve	Maeve: as above	S	Char artist	Tue 10	☐
boss_varro	Varro: rig, jars as separate parts, 2 phases	M	Char artist	Mon 9	☐
boss_tamamo	Tamamo: rig, 9 tails (spring bones), giant-fox phase still	M	Char artist	Tue 10	☐
enm_debthound / enm_collector / enm_chochin / enm_kasa	3-pose sprites	M	Char artist	Mon 9	☐
enm_kitsune_twins	3-pose sprites ×2	S	Char artist	Tue 10	☐
npc_*	Ama, Tobi, Brother Nought, Gus, Ofuku, Tetsu: portrait busts for story	M	Char artist	Tue 10	☐

Stages and reels

ID	ASSET	TAG	DUE	STATUS
stg_rainalley	Rain Alley 9 (from the Wards 3D base + colour key)	M	Sun 8	◐ base + key done
stg_draingates	Drain gates (Varro)	M	Mon 9	☐
stg_moonpier	Moon Pier (3D base + colour key exist)	M	Sun 8	◐ base + key done
stg_foxpalace	Tamamo's foxfire palace	M	Tue 10	☐
reel_meridian / reel_hyakki	Nixie sign; torii with three lanterns	M	Sun 8	◐ vector designs in `art/graphics/`

VFX and UI

ID	ASSET	TAG	DUE
vfx_stop, vfx_hit, vfx_parry	Stop flash, hit spark, parry ring	M	Sun 8
vfx_pattern_*	Pair, Zero-Sum, Cascade, Jackpot, Reaper, Overwrite	M	Mon 9
vfx_foxfire, vfx_rain, vfx_fireflies	Land VFX	M	Mon 9
ui_*	Title, Naming tubes, map (Meridian wall map, Hyakki scroll), market stall, teahouse, results receipt, pause ledger, Codex	M	Tue 10
ico_charms ×12, ico_landlaw ×2, ico_intent ×6	Icons	M	Mon 9

Cinematics and stills

ID	ASSET	TAG	DUE
cin_intro	Intro montage (~45 s video)	M	Mon 9
cin_vesper, cin_mapreveal, cin_hina, cin_tamamo_end, cin_ending	Stills + motion graphics	M	Tue 10
cin_hapdream0	Hap's Dream 0 (15 s)	S	Wed 11
ult_*	Ultimates: NIL, Vesper, Hina (M), Maeve (S); Link Arts: Midnight Parade (M), Lucky Strike (S)	M/S	Mon 9–Tue 10

Audio (see `AUDIO.md`)

Meridian theme, Hyakki theme, Tamamo theme (M); SFX vocabulary (M); stems (S).

Done when: every Must item from `JAM_SLICE.md` has a row, an owner and a due day, and the status column is updated at each stand-up.

Sound is half the game feel. A reel that clicks on the beat feels twice as good. Every interaction in `GAME_FEEL.md` has a named sound here.

1. Direction

- **One musical identity, many lands:** a synth-noir core (Meridian) that every land re-voices with its own instruments. The player should hear the journey.
- **The beat clock rules the reel.** Music runs on a beat clock; wheel ticks are quantised to the beat; the stop "clack" is not quantised (response comes first), but its tail lands on the next beat.
- **Stems fill with the reel:** stem 1 (pads/rhythm) always; stem 2 (bass) when the first digit lands; stem 3 (lead) on the second; stem 4 (choir/brass) on a pattern.
- **Silence is a tool:** the Ten Courts, Chitragupta's Hall and Hap's Dreams use near-silence.

2. Music

CUE	LAND / MOMENT	TEMPO	INSTRUMENTS	STEMS	TAG (JAM)
Meridian theme	Prologue	96 BPM	Analog synth bass, Nixie-hum pads, rain, muted trumpet	4	M
Varro boss	Prologue boss	120	Distorted bass, sirens, stomps	4	S (use Meridian battle remix)
Hyakki theme	Act I	90	Shamisen, shakuhachi, taiko over synth pads, wind chimes	4	M
Tamamo boss	Act I boss	132	Taiko ensemble, koto runs, choir, synth arps	4	M
Hap's Dream	Interludes	free	Music box of dice, celesta, reversed bells	1	S
River Empire	Act II	100	Erhu, guzheng, dizi, gongs over synth	4	—
Ten Courts	Act II underworld	—	Near silence, brush strokes, low drones	1	—
Śūnyapura	Act III	84	Sitar, tabla, tanpura drone, rain on stone	4	—
Kroton Nova	Act IV	108	Lyre, bouzouki, frame drum, male chant of primes	4	—
Crossroads	Act V	92	Slide guitar, harmonica, upright bass, slot-machine choir	4	—
Thale	Finale	72 → 140	Cathedral organ, brass, then every land's instrument in Phase 3	6	—
Title	Title	80	Solo Nixie hum → the main motif on synth	1	M

The main motif: seven notes (the "lucky seven"), rising; every hero theme quotes it in their land's mode.

3. SFX vocabulary (jam must-haves)

ID	SOUND	RECOGNISABLE AS
sfx_wheel_unfurl	Paper-and-brass whoosh	"A wheel appeared"
sfx_wheel_tick	Soft mechanical tick per face, pitch rising with speed	Spinning
sfx_stop	**Clack**: mechanical latch + Nixie zap + low thump	The stop. Must cut through everything
sfx_hit_light / heavy	Punch + digital crunch	Damage
sfx_digit_slam_1/2/3	Tube slams, low → mid → high	The reel filling up
sfx_bust	Tubes flicker out, a sigh	Nothing
sfx_pair	Two bright notes	Something good
sfx_zero_sum	Ledger chime + paper flip	Balanced
sfx_cascade	Rising three-note arpeggio	Again!
sfx_jackpot	Slot fanfare + coins + crowd "oh"	Big win
sfx_reaper	Four funeral bells	Danger
sfx_overwrite	Bells rung your digit's number of times + choir	Miracle
sfx_parry	"Ting" + whoosh	Saved
sfx_fortune	Gold shimmer + reversed hit	Reflected
sfx_favour_warm / haps_hand	Two dice rattling in a palm / dice thrown and landing	Luck turning
sfx_settle	Departure-board flip to 00 + wind	Defeat (never gory)

ID	SOUND	RECOGNISABLE AS
sfx_count_up	Ticks accelerating, "ching" per power of ten	The big number
sfx_ui_tap / back / buy	Paper, stamp, bell	Menus

4. Voice

- Jam: **barks only** (3–5 per hero: strike, hurt, pattern, ultimate) + a few story lines; subtitles always.
- Voice direction per character from `CHARACTER_BIBLE.md` (Voice rows).
- Optional AI voice for the Basilica's announcements and Thale's narration **must be disclosed**; human actors preferred for heroes.

5. Mix rules

- Priority: sfx_stop > dialogue > pattern stingers > hits > music > ambience.
- Duck music −6 dB during stingers and dialogue.
- Loudness target −16 LUFS integrated (web); true peak −1 dBTP.
- Every sound has a reduced-intensity variant for Reduced Motion mode (softer transients).

6. Sources and licences

Original compositions; SFX from licensed libraries (record which in `audio/LICENSES.md`); culturally specific instruments recorded or licensed from performers of that tradition where possible. Any AI-generated audio is logged in `AI_DISCLOSURE.md`.

Done when: every Must interaction in GAME_FEEL has a named sound here, and the temp music runs on the beat clock in the prototype.

Every line in the jam slice, in the `NARRATIVE_STYLE.md` format, plus shot-by-shot boards. Full campaign: `STORY.md`. Lines are short; **NIL speaks least**. Production key: **V** = video (Higgsfield/Seedance) · **E** = in-engine · **S** = stills + motion graphics. ⏩ = skippable after first view.

1. Intro (V, ~45 s, ⏩)

SHOT	IMAGE	LINE (THALE, VOICE-OVER)	LENGTH
1	Abstract ink and static; dice fall like hail	"Before numbers, there was the Noise."	6 s
2	Montage: the Moirai's thread (red-figure) → palm leaves (miniature) → the Book (ink-wash) → Enma's ledger (ukiyo-e) → a black book at a crossroads (woodcut)	"Every people kept its own accounts with fate."	15 s
3	Brass chains pull every ledger into the Basilica; a beam sweeps the world; a number appears over every head	"Then we built one machine for all of them. Every soul a number. Every number a share."	12 s
4	Title card: ZERO-SUM: OMEN, Nixie	—	4 s

2. Prologue

Rain Alley 9 (S → E)

> *Dawn. The Tally beam sweeps the alley. AMA's Omen ticks 03… 02… 01.* **TOBI:** Mum? **COLLECTOR:** Ama Okoro. Your account's closed. *A hooded figure steps between them. Above their head: NaN.* **COLLECTOR:** Move. *(Interactive: digits spin around Ama's 01. TAP. The wheel slams on 07.)* **COLLECTOR:** …That one has no number. **NIL:** Neither do you, now. *(Beat.)* Run.

Naming shrine (S → E)

> **BROTHER NOUGHT:** In here. Quick. *(He bolts the door.)* Every soul needs a number, child. **NIL:** I don't have one. **BROTHER NOUGHT:** Then pick. Out loud. The machine is listening. *(STATE YOUR LUCKY NUMBER.) Every screen in the city: REMAINDER DETECTED. The Basilica's eye opens.* **BROTHER NOUGHT:** *(quietly)* Hap's watching.

Fight 1 barks: NIL (stop): "Fourteen." · NIL (Jackpot): "…Huh." · NIL (hurt): "Noted."

Vesper's rescue (S, 12 s)

> *Three Debt Hounds circle NIL. A shot. A hat lands on the cobbles, then a woman lands beside it.* **VESPER:** Need a hand? Seven bucks. **NIL:** I don't have seven bucks. **VESPER:** Then you owe me. Everybody owes somebody, kid.

Fight 2 tutorial lines (E, one line each): "Tap a hero to choose who strikes." · "Shimmering faces finish a pattern." · *(after two Busts)* **VESPER:** "Feel that? Luck's turning. Hap's watching."

Gus's stall (E)

> **GUS:** Charms from seven countries, all genuine, most of them. First one's on credit. **VESPER:** Nothing's on credit with Gus. **GUS:** Everything's on credit with Gus. That's the business.

Maeve's forge (S, optional)

> **MAEVE:** Third time lucky! You're the third stranger today. The first two were Collectors. **MAEVE:** *(after the fight)* Right. I'm coming. Somebody has to fix your charms.

Varro (E + S)

> **VARRO:** NaN. Worth more than my whole street. **VESPER:** He's got a sister. Debt eats her Omen every morning. **VARRO:** Don't talk about her. *(Boss intro card: VARRO KADE · THE LEDGERHOUND · OMEN 61 · "He steals your last digit.")* **VARRO:** *(phase 2)* Final notice. **VESPER:** *(ultimate)* Seven chambers. Pick one. **VARRO:** *(defeated)* Go on. Settle me. **NIL:** No. **VARRO:** ...You'll regret that. *(He limps away.)*

Rooftop: the map reveal (S, 20 s)

> **VESPER:** The Upper Gate wants seven Seal Digits. One from every land's Auditor. *(Six lands light up across the horizon.)* **NIL:** That's six. **VESPER:** The seventh's yours. The number you just told the whole city. *Varro's Seal shatters over the drain gates: 1.*

Thale (S, 8 s)

> *The Basilica nave. The Comptroller reads the alert and closes his eyes.* **THALE:** There you are.

Hap's Dream 0 (V, 15 s, should)

> *A void full of stars. A vast figure with dice for eyes, wearing a street fortune-teller's face, leans down.* **HAP:** Pick a number. Any number. *(It sees NIL's number.)* ...Oh, *that* one. Interesting. *Dice rattle like laughter. NIL wakes.*

3. Act I

Ferry (S)

> **TETSU:** I don't ask where you're going. I used to ask. I used to work for the people who ask.

Moon Pier barks: HINA (later): "Hina approves!" · MAEVE: "Horseshoe goes open end UP." · VESPER: "Lucky seven, unlucky you."

The Night Parade / Hina joins (V + S)

> *The party hides inside the parade, disguised as yokai. Maeve's disguise is a hat. An umbrella hops out of a shrine stand, opens one eye, and becomes a small girl.* **HINA:** You smell like a hundred wishes. **NIL:** I've never made one. **HINA:** Then they're other people's. Hina likes other people's. *(She attaches herself to NIL's sleeve.)* Hina is coming. **VESPER:** Does it bite? **HINA:** *She* bites.

Ofuku's shrine (E)

> **OFUKU:** One eye when you wish. One eye when it comes true. I paint one a day, for whoever needs it.

Teahouse bond scene: Vesper + Hina (S, should)

> **HINA:** You flip that coin and never look. Why? **VESPER:** If I look, I have to believe it. **HINA:** Hina looks at everything. That's how Hina got real. Almost.

Enma's trial (E, should)

> **ENMA:** The living don't pass this gate. *(He opens his ledger. A page is missing.)* ...You are not written here. You are not written anywhere. **ENMA:** *(after the trial)* Pass, then. Something unwritten should see what it's missing.

Tamamo (E + V)

> **TAMAMO:** Little NaN. Do you know what happens to a story nobody tells? It fades. Every fox and lantern on these islands is fading. **HINA:** So you sold us. **TAMAMO:** I *saved* us. Fixed Fate keeps everything. *(Boss intro card: TAMAMO · THE NINE-TAILED AUDITOR · OMEN 99 ×9 · "Her foxfire lies about your digits.")* **TAMAMO:** *(ultimate)* Hundred demons. One night. **HINA:** *(counter-ultimate)* Hina has lived a hundred years of nights. *(The umbrella opens its eye.)* **TAMAMO:** *(defeated)* What is a yokai worth, without belief? **NIL:** What's anyone worth without a number? *Tamamo laughs softly, touches Hina's head with one tail, and fades into the lanterns. Seal 9.*

Ending (S)

> *The Basilica. Thale watches a second Seal alarm.* **THALE:** The Remainder is learning to count. *Card: TO BE CONTINUED: THE RIVER EMPIRE.*

4. Storyboard checklist

Thumbnails (any quality) for every S and V beat above, with timing and the skip flag, in `art/storyboards/`. Then an **animatic** (boards + temp audio) of the intro.

Done when: the intro animatic plays end to end and the team agrees it hooks.

The decisions that are expensive to change on Day 5: architecture, data, determinism, WebGL limits, build and save. Engine: **Unity 6 LTS, URP 2D, WebGL 2** (WebGPU optional later).

1. Architecture

```
Core/          pure C#, no UnityEngine: wheels, faces, reel, patterns (+priority), Balance/Omen, scoring,
               bonds, charms pipeline, Land Laws, Hap's Favour, seeded RNG. EditMode tests.
Sequencing/    async beat scripts (UniTask): HeroStrike, EnemyStrike, PatternPayoff, LinkArt, Ultimate,
               StoryBeat. Reads Core results; drives Presentation; owns timing from GAME_FEEL.
Data/          ScriptableObjects: HeroDef, WheelDef, FaceDef, CharmDef(+SetTag), EnemyDef, BossRuleDef,
               LandLawDef, ActMapDef, StoryDef. Tuning CSV importer.
Presentation/  actors (2D Animation rigs), wheels, reel object, VFX, camera (Cinemachine 3), UI
(UGUI/TMP).
Audio/         beat clock, stem controller, SFX bank.
Meta/          save, settings, codex, results, telemetry (opt-in, post-jam).
```

Rules: Core never references Unity; Presentation never decides outcomes; Sequencing is the only place time passes. Every random draw goes through `Core.Rng` with a seed.

2. Core model (C# sketch)

```
public sealed record Face(int Digit, float Width, FaceEnh Enh);
public sealed class Wheel { public IReadOnlyList<Face> Faces; public float Speed; }
public enum Pattern { Bust, Pair, ZeroSum, Cascade, Jackpot, ReapersToll, Overwrite }   // priority
reversed
public static class Patterns {
  public static Pattern Evaluate(int a, int b, int c, int lucky) { /* GDD §1.5 order; mirrors prototype
evaluate() */ }
}
public sealed class Combatant { public long Balance, MaxBalance; public int Omen => Balance <= 0 ? 0 :
Math.Clamp((int)Math.Ceiling(Balance * 99.0 / MaxBalance), 1, 99); }
public sealed class Favour { public int BustStreak; public bool Warm => BustStreak >= 1; public bool
HapsHand => BustStreak >= 2; }
```

Damage uses `long` (Balances reach 99,999,999 and damage can exceed `int` with stacked Mult). Mult is `double`, rounded at the end of the round.

3. Data and tuning

- `tuning/jam_tuning.csv` → imported by an editor tool into a `TuningTable` asset on change; runtime reads only the asset.
- Content ids match `ASSET_LIST.md` (`chr_nil`, `enm_debthound`...).
- Strings live in a string table (Unity Localization) from day one, even in English only (LOCALISATION.md).

4. Determinism and testing

- Seeded RNG per run; per-fight sub-seeds. A **replay log** (seed + inputs with timestamps) is written for every fight and attached to bug reports.
- **EditMode tests:** every pattern, the priority order, Omen readout, Favour/Hap's Hand, each Land Law, each charm's rule. Run in CI on every push.
- **PlayMode smoke test:** boot → Naming → Fight 1 auto-played with a fixed seed → Results.
- The pattern simulation (`tools/sim_patterns.py`) is the spec's reference; a C# port of its numbers must match within 0.5%.

5. WebGL budgets

BUDGET	TARGET
Download size	< 250 MB total; first scene < 40 MB (Addressables for later acts and video)
Load to first fight	≤ 20 s on a mid laptop (cold cache)
Frame rate	60 fps on a 2020 laptop; 30 fps floor on mid phones

BUDGET	TARGET
Memory	< 1.5 GB WebGL heap
Textures	Character atlases $\leq 2048^2$, Crunch; stages streamed per scene
Audio	Vorbis, music streamed; SFX in a bank
Video	720p H.264 ~2.5 Mbps from StreamingAssets via URL; audio unlocked by the title tap
Compression	Brotli with Gzip fallback; decompression fallback on

Browser matrix: Chrome, Firefox, Safari (macOS), Edge; Chrome on Android; Safari on iOS (best effort).

6. Build and release

- **Daily build** to a private itch page from Day 1 (GitHub Actions + GameCI, or a local script); the build number shows on the title screen.
- Versioning `jam-0.DAY.BUILD`. Tag the submitted build.
- Branches: `main` always builds; feature branches merge daily.
- Large files: Git LFS for PSB/PSD/PNG/WAV/MP4.

7. Save data

- PlayerPrefs (IndexedDB on WebGL): lucky number, act progress, settings, codex unlocks, best scores.
- If storage is cleared or blocked, the game still runs from the title (no crash); "Continue" is hidden.
- Save at node boundaries only (never mid-fight in the jam).

8. Performance and safety

- Pool every VFX and damage number; no allocations per frame in combat.
- Cap particles; Reduced Motion disables camera shake and full-screen flashes at the shader level.
- No network calls in the jam build (telemetry is post-jam and opt-in).

Done when: an empty WebGL build is live on the private itch page and the Core tests pass for all seven patterns and the priority order.

Playtests find the fun; documents only point at it. We test **every day from Day 2**, starting with the browser prototype.

1. Schedule (jam)

DAY	BUILD	TESTERS	FOCUS
Fri 6 (tonight)	Browser prototype	3 friends	Does stopping a wheel feel good? Do they want another round?
Sat 7	Prototype + tuning CSV	3 new	Pattern Sight, Hap's Favour, Bust feel
Sun 8	Unity greybox: Prologue	3–5 new	First 10 minutes (FIRST_10_MINUTES rules)
Mon 9	Prologue + Act I start	3–5 new	Charms, party of 3, bonds
Tue 10	Full slice (freeze)	5 new + 2 returning	Difficulty curve, Tamamo, pacing
Wed 11	Content lock	5 new	Balance; bugs; one cultural reviewer for Hyakki
Thu 12	Release candidate	3 new on itch (WebGL, 3 browsers)	Load time, performance, last confusions

New testers each day (fresh eyes beat returning ones). At least one tester per round who doesn't play games much.

2. Protocol

1. Give them the link and **say nothing**. No explanation, no help.
2. Observe (screen + face if they agree); the observer only takes notes.
3. Stop at 15 minutes or when they stop on their own.
4. Five questions afterwards (no leading).

3. What we watch (log per tester)

METRIC	HOW	TARGET
Time to first tap	Timestamp	$\leq$ 20 s
First smile / "oh!"	Observer notes the moment	Before 2:00
Confusion moments	Any pause > 6 s or a question	$\leq$ 2 before Varro
Busts that look frustrating	Sighs, "ugh"	Rare; never 3 in a row (Hap's Hand)
Pattern that got the biggest reaction	Observer	Jackpot/Overwrite, then Link Art
Quits	Where and why	No quits before Varro
"One more"	Did they replay or say so?	$\geq$ 60%
Can explain Pair, Jackpot, Parry	Ask after	$\geq$ 3 of 5

4. The five questions

1. What was the best moment?
2. When were you confused?
3. When were you bored?
4. Which character do you want to know more about?
5. Would you play again? (Yes / maybe / no, and why.)

5. Thresholds that change the plan

- If fewer than half want "one more": **stop adding content** and fix the core loop (feel, tuning, pacing) the next day.
- If anyone can't find the tap or the third-striker choice: fix the prompt, not the tutorial text.
- If a fight takes more than 2× its target rounds for most testers: lower enemy Balance first (TUNING §6).
- If a cultural reviewer flags something: fix it before release, no exceptions.

6. Log

Keep findings in `playtests/YYYY-MM-DD.md`: tester id (no names), metrics table, quotes, top 3 issues, what we changed.

Done when: the first session on the browser prototype has happened and its findings are logged.

"GOTY quality" has to mean a checklist, or it means nothing. A feature is **done** only when it passes its checklist **and** has been playtested.

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0. The trailer test (applies to everything)

Would a 3-second clip of this make a stranger click? If not, it isn't done.

1. A fight

- [ ] Teaches or tests exactly one idea (its enemy rule or Land Law)
- [ ] Hits its round target (TUNING §1) for an average tester
- [ ] Every special move is telegraphed one round ahead
- [ ] The stop, hit, pattern and Settle feel match GAME_FEEL §2–§4
- [ ] At least one moment worth a GIF
- [ ] No input blocked for more than 3 s without a skip prompt

2. A boss

- [ ] Its rule is explained in one line on its intro card
- [ ] Two or more phases, each visibly different
- [ ] A cinematic ultimate, and a counter the player can find in the same land
- [ ] Its defeat changes the world (story beat) and the music resolves

3. A cutscene

- [ ] Storyboarded and timed before production
- [ ] Skippable after first view; subtitled; speaker names shown
- [ ] No line longer than 15 words (NIL: shorter)
- [ ] One image in it is good enough for the store page

4. A character (art)

- [ ] Reads as a black silhouette at 64 px (the art book's readability page)
- [ ] Signature colour is the most saturated thing on them
- [ ] Two tones + one rim; coloured variable-weight lines; no airbrush
- [ ] Every canon lock respected (Character Bible)
- [ ] Hands, faces and prop handedness correct after overpaint
- [ ] Cultural details checked against the World Bible §10 and the consultation log

5. A background

- [ ] One light source and a time of day; atmospheric depth
- [ ] 2–3 deliberate "little things" (Locations Guide)
- [ ] Neon only where the fiction allows
- [ ] The battle floor is clear and readable behind actors
- [ ] Split into parallax layers per ASSET_LIST

6. A UI screen

- [ ] It's an object from the world (UX_SPEC §3)
- [ ] Answers its questions at a glance; two lines of text max
- [ ] Works with mouse, touch and keyboard; meets the text-size minimums
- [ ] Every colour cue has a shape or number

7. A sound

- [ ] Recognisable with eyes closed
- [ ] Sits right in the mix (AUDIO §5); has a reduced-intensity variant

- [ ] Licensed and logged

8. Never ship

Placeholder text · stray AI artefacts (wrong digits, "Nan" for "NaN", broken hands) · unreviewed cultural details · flashes that fail the photosensitivity check · a crash or soft-lock in the critical path · an undisclosed AI asset.

9. Reviews

- **Daily:** each owner checks their finished items against this list at stand-up.
- **Tue 10 (feature freeze):** whole-team review of every Must feature; anything failing is fixed or cut.
- **Thu 12 (release candidate):** "never ship" sweep.

Done when: the team uses this list in the Tue 10 feature-freeze review.

1. Automated

LAYER	TESTS	WHEN
Core (EditMode)	Every pattern; priority order; Omen readout; Favour and Hap's Hand; each Land Law; each charm; each boss rule; Balance overflow (`long`)	Every push (CI)
Simulation parity	C# pattern odds vs `tools/sim_patterns.py` within 0.5%	Nightly
PlayMode smoke	Boot → Naming → Fight 1 (fixed seed) → Results	Every build
Replay	Re-run saved replay logs; outputs must match	Nightly
WebGL	Build size, load time, memory ceiling	Every build

2. Manual test passes

PASS	SCOPE	WHEN
Critical path	New game → Tamamo → results, with each of the 10 lucky digits	Daily from Day 4 (jam)
Fight matrix	Every enemy × every hero pair present in the slice	Feature freeze
Input	Mouse, touch, keyboard; hold-to-stop; assist	Feature freeze
Browsers	Chrome, Firefox, Safari, Edge; Android Chrome; iOS Safari	RC
Accessibility	Every option on; reduced motion; colour-blind presets; text 150%	RC
Save	Quit at every node type and resume; storage blocked	RC
Localisation (post-jam)	Text overflow, fonts, line breaks per language	Per language
Cultural	Reviewer sign-off per land	Before public release

3. Bug triage

SEVERITY	DEFINITION	RULE
S1 Blocker	Crash, soft-lock, data loss, critical path broken	Fix now; no build ships with one
S2 Major	Wrong rule, broken feature, bad cultural error	Fix before RC
S3 Minor	Visual glitch, typo, small balance issue	Fix if time
S4 Polish	Nice-to-have	Backlog

Every bug report includes: build number, browser/device, steps, expected vs actual, and the **replay log** if it happened in a fight.

4. Exit criteria (jam RC, Thu 12 Nov)

Zero S1 · zero S2 on the critical path · all ten lucky digits complete the slice · load $\leq$ 20 s · 60 fps on the reference laptop · QUALITY_BAR "never ship" sweep passed · AI disclosure complete.

> **Superseded style rules:** the art direction changed to *painted world, inked people* (Ghibli-inspired painted backgrounds + Edgerunners-inspired cel characters and action). Use the prompt grammar in `docs/ART_DIRECTION.md` §6 instead of the style suffix in §1.2 below. Subject descriptions below are still valid.

Paste-ready prompts for character sketches, key art, environments and video cutscenes across every land of the Tally: Meridian (modern capital), Hyakki (Japanese folklore), the River Empire (Chinese folk religion), Śūnyapura (Indian myth and mathematics), Kroton Nova (Greek myth) and the Crossroads Territory (American folklore). Priorities: **[JAM]** = needed for the Nov 13 slice; **[LATER]** = demo and beyond.

1. How to use

1.1 Two routes

- **3D Jutsu** (Claude can drive this from the session): project **"ZERO-SUM OMEN - Look Dev"** has two layouts.
- `CAM_pose` renders `pose_blockout.png` (2:3, full-body three-quarter pose) for every character key art, so the roster shares framing and scale.
- `CAM_street` renders `street_blockout.png` (16:9, deep street with a tower) for city shots.
- Call `generate_angles` with that render plus a prompt below at 2k, and start every character prompt with: *"Use the grey mannequin only as a pose and framing guide; replace it completely."*
- Video: `generate_video` (Seedance 2.5) with an approved still as `start_image`.
- **Higgsfield web app:** image models (Nano Banana Pro, GPT Image 2.6, Seedream, Cinema Studio Image) and video models (Seedance 2.0, Kling 3.0 with a start frame, Cinema Studio Video 3.0). After a hero's key art is approved, **create an Element** and put it in every later prompt for that hero.

1.2 Style suffix (append to every image prompt, unchanged)

> premium anime game key art, 2.5D anime RPG character art, crisp clean lineart, cel shading with painterly soft lighting, high-detail costume design, cinematic rim light, expressive face, consistent anime proportions (about 7 heads tall), rich but controlled palette, ultra high resolution, clean silhouette that reads at small size, no text, no watermark, no logo, no extra limbs, correct hands

1.3 Land style lines (add to environments, VFX and cut-in textures; characters stay anime)

LAND	STYLE LINE
Meridian Undercity	*anime cyber-noir: heavy rain, neon in amber and teal, nixie-tube signage, wet reflections, stacked tenements*
Hyakki (Japan-inspired)	*ukiyo-e woodblock print influence: flat graphic colour fields, bold outlines, stylised Hokusai-like waves and clouds, washi paper texture, indigo, vermilion and gold*
River Empire (China-inspired)	harbor: *gongbi fine-line painting with gold leaf, red lanterns, lacquer and jade, festival neon* · underworld: *ink-wash shan shui, misty monochrome with vermilion seals, rice-paper texture*
Śūnyapura (India-inspired)	*Indian miniature painting and Pattachitra influence: ornate borders, rich mineral pigments (lapis, saffron, peacock green, gold), monsoon light, intricate geometric stepwells and stone observatory instruments*
Kroton Nova (Greece-inspired)	*Aegean light, white marble and deep blue sea, black- and red-figure pottery motifs, mosaic patterns, Greek key borders, bronze*
Crossroads Territory (America-inspired)	*Americana: desert sunset gradients, Route-66 motel neon, pulp western poster lettering style, art-deco casino lights, dust and starry night skies*
The Basilica	*colossal brass clockwork cathedral, monochrome with nixie amber glow, walls of scrolling numbers*

1.4 Rules

- Generate 4 variations, pick 1, then iterate with the Element. Approve the style anchor before mass production.
- Log everything that ships in `docs/AI_DISCLOSURE.md`.
- Cultural review before release (World Bible §10). Deities are never shown as monsters.

2. Style anchor [JAM] (generate first, pick one)

Render: `pose_blockout.png` (2:3).

> Use the grey mannequin only as a pose and framing guide; replace it completely. Style exploration for an anime RPG called ZERO-SUM: OMEN, a world where every person has a glowing two-digit number floating above their head. A young protagonist in a long black coat standing in a rain-soaked neon alley at night, a flickering holographic "NaN" glitch above their head, glowing orange nixie-tube digits drifting in the air; around them, lucky charms from many cultures hang from wires: a horseshoe, a beckoning cat, a string of lemon and green chillies, a blue evil-eye bead, a red paper talisman. [style suffix]

Variants: add one of *(a) "bold graphic shapes, high contrast"* · *(b) "soft painterly storybook light"* · *(c) "ink outlines with brush texture"* · *(d) "glossy splash-art finish with strong bloom"*. The pick becomes `art/lookdev/style_anchor.png`.

---

3. Characters

Templates (after the key art is approved; use the Element): - **Turnaround:** "Character turnaround sheet of {ELEMENT}: front, three-quarter, side and back views, neutral standing pose, flat light grey background, identical outfit and proportions. [style suffix]" - **Expressions:** "Expression sheet of {ELEMENT}: neutral, smirk, laughing, angry, shocked, sad; head and shoulders; flat background. [style suffix]" - **Battle key poses:** "{ELEMENT} in three sequential attack key poses for 2D game animation: anticipation, strike, follow-through; full body; consistent scale; flat background; motion lines. [style suffix]" - **Ultimate splash:** "Full-screen ultimate cut-in illustration of {ELEMENT}: {ultimate}, extreme dynamic perspective, speed lines, glowing digits bursting outward. 16:9. [style suffix] [land style line]" - **Reactions:** "{ELEMENT} reacting with {emotion}: three small poses, flat background. [style suffix]"

3.1 NIL, the Remainder [JAM]

> Use the grey mannequin only as a pose and framing guide; replace it completely. Full-body character concept of NIL, protagonist of an anime RPG. Androgynous young adult about 19, sharp amber eyes, messy silver-white hair with one glowing nixie-orange streak, a broken holographic "NaN" glyph flickering above the head like a cracked halo. Long asymmetric black actuary's coat with brass ledger clasps and a high collar, fingerless gloves, tall boots, a cracked brass decoder device fused into the RIGHT wrist leaking orange light and floating digits. Right arm extended, fingers spread, a spinning ring of glowing digits 0–9 orbiting the open hand. Rainy neon night, orange and teal rim light, plain dark atmospheric background. [style suffix]

Ultimate: *a fourth nixie tube tears open in the air and every digit pours out of it like a storm.*

3.2 Vesper Sette (7), gunslinger [JAM]

> Use the grey mannequin only as a pose and framing guide; replace it completely. Full-body character concept of Vesper Sette, a gunslinger and former casino croupier. Tall woman in her late 20s, crooked grin, small scar across her lip, black hair with a crimson underlayer in a long braid, red eyes. Crimson long coat over a black croupier's waistcoat and white shirt with rolled sleeves, wide-brim black hat with playing cards in the band, a large revolver with a seven-chamber cylinder engraved 1–7 held forward, faint red glowing "7" motifs. Mix of neon noir and western frontier. [style suffix]

Ultimate: *seven muzzle flashes in a ring around her, each bullet trailing a glowing red 7.*

3.3 Maeve Triskel (3), charm-smith [JAM]

> Use the grey mannequin only as a pose and framing guide; replace it completely. Full-body character concept of Maeve Triskel, a cheerful blacksmith's daughter from a Celtic immigrant quarter of a neon city. Freckled young woman around 20, bright red hair in three braids, green eyes, leather forge apron over rolled-sleeve work clothes, heavy boots, horseshoe-shaped knuckle gauntlets glowing copper-hot, a four-leaf-clover pin on her collar, a forge hammer with a triskele (triple spiral) burned into its head, sparks around her. Warm copper and green palette against a dark forge. [style suffix]

Ultimate: *three spinning red-hot horseshoes orbit her and slam down, each stamping a glowing 3.*

3.4 Hina of the Ninety-Nine (9), tsukumogami [JAM]

> Use the grey mannequin only as a pose and framing guide; replace it completely. Full-body character concept of Hina, a playful tsukumogami spirit born from a 99-year-old Japanese paper umbrella. Small girl-shaped spirit with a short black bob and mischievous golden eyes, a furisode kimono patterned with tiny numbers 1 to 99 in indigo and vermilion, geta sandals, holding a large red-and-white oiled-paper umbrella that has one big eye and a long tongue like a kasa-obake, umbrella ribs glowing faintly, a floating "99" above her head. Paper lanterns and fireflies, night shrine backdrop. [style suffix] Ukiyo-e flat colour accents on the kimono pattern.

Ultimate: *her umbrella spins open into a giant one-eyed moon and a parade of tiny yokai pours out beneath it.*

3.5 Varro Kade, the Ledgerhound [JAM] (Prologue boss)

> Use the grey mannequin only as a pose and framing guide; replace it completely. Full-body boss concept of Varro Kade, a debt collector and bounty hunter in a neon slum. Hulking man, heavy riot-armour trench coat in rust and gunmetal, a hound-skull mask with glowing red eyes, chains of glass debt jars on his belt each holding a floating glowing digit, a massive hooked collection baton, his Omen "61" floating above his head in red. Rain, neon, steam, low angle. [style suffix]

3.6 Tamamo, the Nine-Tailed Auditor [JAM] (Act I boss)

> Use the grey mannequin only as a pose and framing guide; replace it completely. Full-body boss concept of Tamamo, an elegant and dangerous nine-tailed fox spirit from Japanese folklore. Tall, beautiful woman with fox ears, long gold-white hair, sly violet eyes with red eye-liner, layered Heian-style courtly robes in gold, violet and white, nine enormous glowing fox tails fanned behind her, each tail tip holding a floating glowing digit, blue-white foxfire flames drifting around her, a folding fan in one hand. Moonlit palace, ukiyo-e cloud shapes in the background. [style suffix] [Hyakki style line]

3.7 Comptroller Lucan Thale [JAM] (intro narrator; teaser)

> Full-body character concept of Comptroller Lucan Thale, the calm and sympathetic villain. Tall ageless man, swept-back silver hair, immaculate white and brass vestments shaped like a clockwork cathedral, a monocle with a slowly turning gear, gloved hands folded, a steel (not glowing) "50" floating above his head. Before enormous brass gears and a wall of scrolling numbers in cold gold light. [style suffix]

3.8 Later roster and bosses [LATER]

- **Ba Jinling (8), River Empire:** elegant woman in a modern jade-and-gold qipao, eight tiny golden bells in her hair ornaments and anklets, a folding fan whose ribs are abacus rods, smoky violet eyes, red lips; red lanterns and gold-leaf harbor behind. [gongbi line]
- **Bai Si (4), River Empire underworld:** very tall pale man, long white hair bound with black funeral ribbons, white mourning robes with black ink 囚 sigils, a huge guandao-style blade hung with four paper talismans, joss paper swirling, four lanterns behind him. [ink-wash line]
- **Shunya (0), Śūnyapura:** young woman astronomer-mathematician, long dark braid with brass astrolabe hairpins, indigo-and-saffron layered clothing with mirror-work embroidery, a large brass astrolabe spinning in her hand like a weapon, palm-leaf manuscripts tied at her hip, ink-stained fingers, a glowing "0" that looks like a full circle of light; stone observatory instruments behind. [Śūnyapura line]
- **Prima of Kroton (2), Kroton Nova:** earnest young paladin, short blonde hair, blue eyes, white-and-gold armour with Greek-key borders, a giant round shield shaped like a sieve set with blue glass, a halo of floating glyphs 2-3-5-7; white marble and Aegean sea behind. [Kroton line]
- **Fibi Leone (Fibonacci), Crossroads:** 18-year-old casino dealer, golden spiral-curled hair, spiral-patterned irises, tailored croupier tailcoat with headset, glowing cards flowing around her in a golden spiral; desert casino neon behind. [Crossroads line]
- **Chancellor Jin Chan, the Golden Toad:** enormous jovial lord robed in gold coins on a three-legged toad throne, coins spilling from his mouth. [gongbi line]
- **Rahu & Ketu, the Eclipse** (cosmic, tragic, awe-inspiring, *not* demonic or grotesque): a vast crowned celestial head devouring a sun's light, and a separate headless serpentine body with a comet's tail, in a star-filled monsoon sky. [Śūnyapura line] *Have a cultural reviewer check before use.*

- **Archon Seventeen:** gaunt, beautiful, severe hierarch in white and bronze, laurel crown of bronze, a staff topped with a sieve-shaped disc, a cold "17" overhead. [Kroton line]
- **The House (Mr. Eighty-Nine):** elegant old croupier in a white suit at a desert crossroads under a single streetlamp at midnight, ouroboros tattoo around his throat, a pocket watch running backwards. [Crossroads line]
- **Keepers (shown with reverence):** Enma as a towering, solemn judge in red robes with a great ledger (Hyakki) · Judge Cui with a giant brush and the Book of Life and Death (River Empire) · Chitragupta as a serene scribe writing on palm leaves by lamplight (Śūnyapura) · the Moirai spinning, measuring and cutting a glowing thread in a cave (Kroton Nova).

4. Environments

Battle-stage rule: **16:9, empty centre-left floor for 3 heroes, empty right side for the enemy, clear space top-centre for the land's reel object, horizon around 60%.**

4.1 Meridian Undercity (Prologue) [JAM]

- **Establishing** (render `street_blockout.png`): "Anime cyber-noir slum beneath a giant brass cathedral tower, stacked tenements, neon signs showing two-digit numbers, charms from many cultures hanging over doorways (horseshoes, beckoning cats, lemon-and-chilli strings, evil-eye beads, red paper talismans), heavy rain, crowds with small glowing numbers above their heads, a searchlight beam sweeping the alley at dawn. [Meridian line] [style suffix]"
- **Battle stage:** "Wide side-on battle background: a rain-soaked neon alley with a broken drain gate, puddles, a dead neon sign shaped like three nixie tubes at top centre, empty floor space, foreground pipes. [Meridian line] [style suffix]"
- **Naming shrine:** "A cramped black-market shrine, an old jury-rigged terminal of brass dials and ten glowing nixie tubes on an altar, incense smoke, charms from many cultures, a single amber lamp. [style suffix]"

4.2 Hyakki (Act I) [JAM]

- **Establishing:** "A misty Japanese-inspired island town at night under a giant full moon, a huge stylised wave curling around the harbor, a long red bridge with a Night Parade of one hundred yokai crossing it (umbrella spirits, lantern ghosts, kappa, foxes), torii gates climbing a hill, paper lanterns everywhere. [Hyakki line] [style suffix]"
- **Battle stage A (the Night Parade bridge):** "Wide side-on battle background on a red arched bridge at night, rows of paper lanterns, a torii gate at top centre with three large hanging lanterns (the reel), fog over dark water below, distant yokai silhouettes, empty floor space. [Hyakki line] [style suffix]"
- **Battle stage B (Tamamo's foxfire palace):** "Wide side-on battle background inside a moonlit palace hall with sliding shoji doors painted with nine-tailed foxes, blue-white foxfire floating, gold-leaf folding screens, empty floor space. [Hyakki line] [style suffix]"
- **Shrine of the Seven Lucky Gods:** "A small hillside shrine with seven statues of the Seven Lucky Gods, a treasure ship carving, daruma dolls and beckoning cats on the steps, morning mist. [Hyakki line] [style suffix]"

4.3 Later lands [LATER]

- **River Empire:** Bāfā Harbor Lantern Festival over the Luck Exchange (golden abacus reel), and the Ten Courts with Naihe Bridge and endless shelves of ledgers.
- **Śūnyapura:** "a monsoon city built around a colossal stone open-air observatory with giant sundials and astrolabes, a geometric stepwell descending in perfect symmetry, lamps reflecting in rain pools" (reel = three sundial dials).
- **Kroton Nova:** "a white marble island city shaped like a gigantic stone sieve above a deep blue Aegean sea, mosaics of numbers, bronze automata on patrol" (reel = three columns with mosaic numerals).
- **Crossroads Territory:** "a desert crossroads at midnight under one streetlamp, a lone signpost, starry sky; beyond it the casino boomtown of Lucky Strike glowing with art-deco neon, a ghost train crossing the horizon" (reel = a giant slot machine).
- **The Basilica:** clockwork cathedral interior whose walls show every land's original ledger.

5. Enemies [JAM]

- **Debt Hound:** "Lean cyber-dog with a debt-collector's collar, red digit display on its flank, sparking jaw, side view, flat background. [style suffix]"
- **Collector:** "Armoured slum bailiff with a riot shield stamped with a ledger seal and a glowing number overhead, side view. [style suffix]"
- **Noiseling:** "Small glitch creature made of scrambled floating digits and static, cute but unsettling, side view. [style suffix]"
- **Kasa-obake:** "Hopping one-eyed paper umbrella spirit with a long tongue and a single wooden leg, mischievous, side view. [Hyakki line] [style suffix]"

- **Chōchin-obake:** "Floating paper lantern ghost with a torn mouth and glowing eyes, flames inside, side view. [Hyakki line] [style suffix]"
- **Kappa:** "Turtle-shelled river imp with a water-filled dish on its head, webbed hands, cheeky expression, side view. [Hyakki line] [style suffix]"
- **Oni enforcer:** "Red-skinned horned ogre in a debt-collector's haori jacket with an iron club studded with numbers, side view. [Hyakki line] [style suffix]"

6. Intro cinematic (video) [JAM]

Make a still for each shot (same Elements), then animate it as the **start image**. 720p, 16:9.

#	LENGTH	START STILL PROMPT	MOTION PROMPT
1	8 s	Abstract chaos: floods, toppling towers, hail shaped like dice, rendered in ink splashes and digital static	Ink and static churn violently; dice-hail falls; slow push-in
2a	4 s	Three women of Greek myth spinning, measuring and cutting a glowing thread, in red-figure pottery style on black	The thread glows and runs off-frame to the right like a fuse
2b	4 s	A serene scribe writing on palm-leaf pages by lamplight, Indian miniature painting style with ornate border	Script glows on the palm leaves and lifts into the air as numbers
2c	4 s	A judge with a giant brush writing in an enormous Book of Life and Death, ink-wash style	Characters lift off the page and become glowing numbers
2d	4 s	A towering solemn judge in red robes opening a great ledger as a lantern-lit parade of yokai passes, ukiyo-e woodblock style	Lanterns sway; the ledger's pages flip in the wind
2e	4 s	A hand signing a black book under a lone streetlamp at a desert crossroads at midnight, Americana woodcut style	The ink signature glows red; the streetlamp flickers
3	12 s	A colossal brass clockwork cathedral rising over a modern city; brass chains reaching out to five glowing books and scrolls from different art styles	The chains drag the five ledgers into the cathedral; gears swallow them; its eye opens and a sunrise beam sweeps the world as a number appears above every head
5	15 s	Dawn in a neon slum: crowds with glowing numbers overhead; a tired mother with "01", her small son staring at it	Camera cranes down through rain; numbers tick; the mother's number flickers to 01; the boy grabs her sleeve
6	10 s	Armoured debt collectors with hound masks pushing through the crowd; one hooded figure with a broken "NaN" glitch stays still	Collectors advance in slow motion; the hood lifts to reveal NIL's amber eyes; NaN crackles
7	8 s	NIL's hand reaching toward the mother; a ring of glowing digits spinning around her "01"	Digits spin like a slot reel and lock on "07" with a flash; collectors turn toward NIL
8	12 s	NIL running through neon alleys hung with charms from many cultures, collectors and cyber-dogs behind	Fast anime chase: low tracking, splashing puddles, sliding under pipes, ducking into an amber-lit shrine doorway
10	10 s	Every screen and sign in the city glitching to "REMAINDER DETECTED"; the cathedral's eye glowing red	The glitch spreads screen to screen; numbers above every head scramble; bounty posters with NIL's face flutter down in the rain

Shots 4 (title), 9 (lucky-number choice) and 11 (logo slam) are built in-engine.

7. Ultimates and Link Arts [JAM]

- **Ultimate clips (3–4 s):** start image = the hero's ultimate splash. Motion: *"Explosive anime ultimate, fast camera push, speed lines, glowing digits bursting outward, impact flash, hold."*
- **Lucky Strike (Vesper + Maeve):** still: *"Maeve hurling a red-hot horseshoe through the air while Vesper aims her revolver at it, neon alley"*; motion: *"Vesper shoots the horseshoe mid-air; it shatters into seven glowing ricochets that converge on the target."*
- **Midnight Parade (Hina + Vesper):** still: *"Hina riding her one-eyed umbrella over swinging saloon doors as a parade of ukiyo-e yokai pours through behind her, Vesper tipping her hat"*; motion: *"the yokai stampede across the screen in woodblock style while Vesper fires through the gaps."*
- **Land cut-in bands:** neon-glitch bands (Meridian) and **woodblock-print wave bands** (Hyakki), built in-engine from generated textures.

8. Land-styled reel objects [JAM]

- Meridian: "A dead neon sign shaped like three nixie tubes in a rusted brass frame, front view, flat background."
- Hyakki: "A vermilion torii gate with three large paper lanterns hanging from its beam, each lantern able to show one painted numeral, front view, flat background."

PART X

Care, launch and policy

Cultures, players, data, the market and the rules.



Five cultures inspire this game. Getting them right is part of the quality bar, not a legal checkbox. Principles are in `WORLD_BIBLE.md` §10.

1. Principles

1. **Inspired by, never depicting.** Lands are fictional; names are invented or from widely told public folklore.
2. **Revered figures are keepers, never enemies.** Enma, Judge Cui, Chitragupta, the Moirai, the Courts' wardens Ox-Head and Horse-Face.
3. **Hap is invented** and never claimed to be any real religion's deity; real gods appear only as shrines, customs and references, handled with care.
4. **No Indigenous sacred traditions** in the Crossroads.
5. **Lived, not explained.** Characters live their customs; the Codex explains (Narrative Style §2).
6. **Paid reviewers**, credited by name if they wish.

2. Reviewers and scope

CULTURE	LANDS / CHARACTERS	REVIEWER PROFILE	SCOPE	WHEN
Japanese	Hyakki; Hina, Tamamo, Enma, Ofuku, Kyūsuke, Tetsu, the Kitsune twins	Native speaker with folklore knowledge (yokai, shrines)	Names and kanji; shrine depiction; daruma and omamori use; yokai tone; kimono details	Before the jam release (Act I is in the slice)
Chinese	River Empire; Ba Jinling, Bai Si, Liu Shun, Judge Cui, Jin Chan, Little Lantern	Native speakers (Mandarin; ideally one familiar with folk religion and Cantonese festival culture)	Names and hanzi (Simplified + Traditional); the Ten Courts and Meng Po; mourning customs; red envelopes and joss paper	Before any Act II material is public
Indian	Śūnyapura; Shunya, Vidya Rao, Meera, Baital, Chitragupta, Rahu & Ketu	Reviewers with Hindu and Jain traditions, and Indian history of mathematics; a Hindi speaker	Names and Devanagari; Chitragupta and Rahu/Ketu depiction (venerated celestial bodies: cosmic and tragic, never cheap demons); clothing; nimbu-mirchi, rangoli, navaratna	Before any Act III material is public
Greek	Kroton Nova; Prima, Archon Seventeen, the Moirai, Kallisto, Niko	Greek speaker with classics background	Names and Greek script; the Moirai; mati and Fortuna; tone of the Synod	Before Act IV is public
American	Crossroads; Vesper, Fibi, the House, the Stranger, Penny, Sheriff Vance, Gus	Reviewer familiar with the blues crossroads legend and its Black American roots; Italian- and Irish-American heritage	The crossroads legend's origins and respectful framing; immigrant charms; no Indigenous sacred material	Before Act V is public
Irish/Celtic	Maeve	Irish reviewer	Triskele use, name, accent in text (none faked)	Before the jam release

3. Review process

1. Owner prepares a **review pack**: character sheets, lines, screenshots, names with scripts, and specific questions.
2. Reviewer returns notes within a week (paid by the pack).
3. Every note gets an entry in the log below with a decision and a fix.
4. Nothing flagged as harmful ships unfixed.

4. Known questions to ask

- Is it acceptable to show Enma as a solemn keeper who tests the living? How should his ledger look?
- Is Ox-Head and Horse-Face's portrayal as wardens (a trial, not villains) respectful?
- How should Chitragupta's Hall and his records be shown? Any imagery to avoid?
- Rahu & Ketu as a boss: what framing is acceptable? Should their defeat be "returned to the sky" (current) or something else?
- Is "Shunya" appropriate as a personal name for a character?
- Are the Seven Lucky Gods appropriate as shrine imagery and charm names (Ebisu, Daikokuten…)?
- Is Lady Luck / crossroads imagery free of appropriation of the blues legend's spiritual roots?

5. Review log

DATE	CULTURE	ITEM	NOTE	DECISION	FIXED IN
–	–	–	–	–	–

English at the jam; Japanese, Simplified and Traditional Chinese and Hindi next (World Bible §10); then French, German, Spanish (LatAm), Brazilian Portuguese, Korean.

1. Engineering rules (from day one)

- All text in Unity Localization string tables; ids from the dialogue database (`BOND_ARCS.md` §3).
- No text baked into images (signs in art use digits or are layered for replacement).
- **Digits are universal**: Omens and reels always use Western Arabic numerals (0–9) in every language, because the mechanics are about those glyphs. Each language's own numerals (kanji, hanzi, Devanagari) appear only as decoration.
- Plural and gender support via Smart Strings; NIL is always they/them in English; translators choose the natural equivalent (e.g. no pronoun) per language.
- Text expansion: UI boxes fit +40% (German) and line breaks work for CJK (no spaces).

2. Fonts

SCRIPT	FONT	NOTES
Latin	Big Shoulders Display, Barlow Semi Condensed, JetBrains Mono	
Japanese	Noto Sans JP	Vertical text only for decorative signs
Chinese (SC/TC)	Noto Sans SC / TC	
Devanagari	Noto Sans Devanagari	Check conjuncts render in TMP
Greek	Noto Sans (Greek)	
Fallback chains per language in TextMeshPro; subset fonts per build to save WebGL size.		

3. Glossary for translators

Keep untranslated: **NaN**, **Omen** (or the agreed local term, decided once), the names of heroes and bosses. Translate consistently (lock once): Balance, Remainder, the Basilica, the Tally, Settled, Audit, Fate Wheel, the reel, Jackpot, Pair, Bust, Cascade, Zero-Sum, Omen Overwrite, Reaper's Toll, Hap, the Open Hand, Hap's Favour, fen. Puns to adapt, not translate: Vesper's "Seven bucks", Maeve's "third time lucky", the catechism "Every soul a number".

4. Culture-specific care

- Where a land's own language appears (Hyakki in Japanese, the River Empire in Chinese, Śūnyapura in Hindi), the native-language version should feel *more* authentic, not less: localisers may restore terms the English simplifies.
- Lucky and unlucky numbers are part of the mechanics; translators must not "localise" them away (4 stays unlucky in Hyakki in every language).

5. Process

Freeze English per act → export → translation (with the glossary, screenshots and character sheets) → in-game LQA pass → cultural reviewer pass for that land (`CULTURAL_CONSULTATION.md`) → fix → ship.

A timing game must be playable by everyone. The jam baseline is in `UX_SPEC.md` §6–§7; this is the full-game plan, checked against the Game Accessibility Guidelines and the Xbox Accessibility Guidelines.

1. Motor

OPTION	WHAT IT DOES
Assist speed	Wheels –30% / –50% / –70%
Hold-to-stop	Press and release instead of a tap; stop on release
Auto-stop on your digit	Wheels stop themselves on the hero's lucky digit (pure strategy mode); patterns still matter
Input remapping	Every action rebindable; one-switch mode (a single button cycles heroes and stops)
No simultaneous inputs	Nothing in the game needs two buttons at once
Timing window display	Optional: a soft ring shows when the aimed face is under the stop line

2. Vision

- **Colour-safe digits**: every pattern colour also has a shape (dot, ring, chevron, star) and a word.
- Colour-blind presets (protanopia, deuteranopia, tritanopia) for land palettes' key cues; tested with simulators.
- Text size 100–200%; high-contrast UI mode; damage-number size option.
- Screen-reader support for menus (Unity accessibility plugin / OS text-to-speech) post-launch.

3. Hearing

- Subtitles on by default, with speaker names and [sound captions] for meaningful sounds ("[four bells toll]").
- Every audio cue has a visual cue (pattern banners, the Open Hand sigil lights, telegraph icons).
- Separate volume sliders; mono audio option.

4. Cognitive

- **Codex** with a GIF for every mechanic; **recap** card when resuming a save.
- Hold on any object for its rule (two lines); never timed text.
- **Story mode**: assist speed + enemy Balance –50% + no game over (Settle → retry the round).
- Pause anywhere, including during cinematics.

5. Photosensitivity and motion

- No more than 3 flashes per second; full-screen flashes below the luminance threshold; validated with a PEAT-style tool on every cinematic.
- **Reduced Motion** turns off camera shake, slow-mo, full-screen flashes and parallax wobble; shortens cinematics.
- Screen-shake intensity slider.

6. Testing

Include disabled players in playtests from the full slice onward (motor, low vision, deaf/hard of hearing). Log issues in the QA tracker with the tag `a11y`.

Opt-in only, anonymous, post-jam. The jam build makes no network calls (`TECH_DESIGN.md` §8).

1. Consent and privacy

- First launch asks: "Share anonymous play data to help balance the game?" Default **off**.
- No personal data; a random install id; no IP storage; data deleted after 12 months; a "delete my data" button.
- Compliant with GDPR/CCPA; privacy policy linked from settings and the store page.

2. Events

EVENT	FIELDS	ANSWERS
`run_start`	lucky_digit, mode, omen_level	Which digits players choose
`fight_end`	fight_id, result, rounds, party, charms, damage_max, busts, patterns{…}	Fight length vs targets; pattern odds in the wild
`pattern`	type, digit, round	Real pattern distribution vs the simulation
`favour`	warm/haps_hand count per fight	How often luck turns
`boss_attempt`	boss_id, phase_reached, result	Difficulty spikes
`node_choice`	act, node_type, options, chosen	Map choice balance
`charm_pick`	offered[ ], picked	Pick rates; dead charms
`quit`	location, time	Where people stop
`setting_changed`	setting, value	Accessibility usage

3. Dashboards

Fight length by node · win rate by boss and attempt · pattern frequencies vs `tools/sim_patterns.py` · charm pick and win rates · hero pick rates and pair frequencies · quit heat-map by node · lucky-digit distribution.

4. Balancing process

1. Weekly review of dashboards.
2. Flag: any boss with first-attempt win rate < 40% or > 85%; any charm picked < 5% or with win rate > 15 points above average; any node where > 10% quit.
3. One change per item per patch; note it in the patch notes with the reason.
4. Re-run the simulation for any wheel/pattern change.

Draft for after the jam; update with jam ratings and wishlist data.

1. The hook

One sentence: *An anime roguelite RPG where every number spins and you tap to stop it, across lands built from the world's lucky-number superstitions.* **One image:** a giant slot-reel Jackpot exploding over a ukiyo-e night parade, with a hooded hero whose number reads NaN. **Elevator pitch:** Balatro's build-crafting meets Persona's style and a party you love, in a world where luck is visible and someone wants to freeze it forever.

2. Audience

SEGMENT	WHY THEY'LL LIKE IT	COMPARABLE GAMES
Deckbuilder / roguelite fans	Synergy discovery, big numbers, short runs	Balatro, Slay the Spire, Luck be a Landlord, Dicey Dungeons
Anime RPG fans	Style, party, story, cinematic ultimates	Persona 5, Metaphor: ReFantazio, Sea of Stars
Story-forward indie players	A strange, warm world and characters	Hades, Disco Elysium (tone of moral argument), Citizen Sleeper
Mobile-curious players	One verb, short fights	Balatro mobile, Vampire Survivors mobile

3. Differentiators

1. **One verb with real skill**: timing + reading the reel, not just picking cards.
2. **Theme in the maths**: lucky numbers from five cultures *are* the rules.
3. **Cinematic payoffs**: every pattern is an anime moment.
4. **A party with digits**: team synergy as number theory, readable at a glance.
5. **A god of luck you free, not fight.**

4. Platforms and price (proposal)

PC (Steam) first, Steam Deck verified; Switch and mobile after. Premium, no microtransactions, no randomness for money. Indicative price USD 19.99 (full game); the jam slice stays free on itch as a demo.

5. Store tags (Steam)

Roguelite · Deckbuilding (adjacent) · Anime · RPG · Turn-Based Combat · Rhythm (light) · Story Rich · Multiple Endings · Colorful · Mythology.

6. Risks

"Gambling" perception (mitigate: no real-money mechanics, messaging about luck and community) · AI-art scepticism (mitigate: human overpaint, honest disclosure, strong hand-made motion) · cultural missteps (mitigate: CULTURAL_CONSULTATION) · scope (mitigate: Gate B planning before production).

1. Beats

WHEN	BEAT	GOAL
Jam week	Dev GIFs on social (the stop, a Jackpot, Hina's umbrella)	First followers
Jam release (Nov 13)	itch page with the slice, disclosure, press kit	Ratings, feedback
Dec 2026	Steam page live (capsule, 5 GIFs, trailer v1)	Wishlists
Q1 2027	Devlog series "Lucky numbers of the world" (one land per post)	Theme-led reach
Q2 2027	Steam Next Fest demo (Prologue + Act I + Act II start)	Wishlist spike
Q3 2027	Festival submissions (indie showcases); creator previews	Coverage
Launch	Launch trailer; Daily Omen seed shared publicly	Sales, community

2. Assets

Capsule art (NIL + party under the Basilica's eye) · 5 GIFs: the stop · a Jackpot count-up · Midnight Parade Link Art · Tamamo's nine tails · an Overwrite · logo animation · press kit (fact sheet, screenshots, team, AI disclosure statement).

3. Trailer v1 script (60 s)

TIME	IMAGE	TEXT / AUDIO
0–4	Rain; NaN flickers	Thale: "Every soul a number."
4–8	Ama's Omen ticks to 01; the tap; 07	(The stop clack, loud)
8–12	REMAINDER DETECTED on every screen	Kick drum enters
12–24	Fast cuts: stops on beat, digits slam into reels in six land styles	Music builds; each cut on a beat
24–34	Jackpot count-up to a huge number; Link Art; ultimate	"Build your luck."
34–44	Lands montage: Hyakki, River Empire, Śūnyapura, Kroton Nova, Crossroads	"Five lands. Five ideas of luck."
44–52	Tamamo, Rahu & Ketu, the House; Thale's eye	"One man wants to freeze fate."
52–58	NIL looks up at the Basilica; Hap's dice eyes open in the dark	"Tap to stop it."
58–60	Logo; wishlist call	—

4. Community

Discord with a #daily-omen channel; share-your-biggest-number screenshots; lucky-number "teams" (players rep their digit); fan art of Hap's faces.

Not legal advice: a checklist to review with a lawyer before the Steam page and any commercial release.

1. Originality

- **Inspired by, not copying.** Balatro (build-crafting structure), anime and Studio Trigger/Ghibli-era looks (art direction), the Cyberpunk 2077 art book (method of presentation) are influences only. No names, characters, art, music, UI layouts or text are reused.
- The Cyberpunk 2077 art book PDF used for study **is not committed** to the repo and is never redistributed.
- Folklore figures used (yokai, the Ten Courts, Chitragupta, the Moirai, the Headless Horseman) are public-domain traditions; our designs are original. Hap is our own invention.
- Run a **trademark search** on "ZERO-SUM: OMEN", "Hap" (as a product term) and character names before the Steam page.

2. Assets and licences

- Fonts: Google Fonts under the SIL Open Font License (Big Shoulders Display, Barlow Semi Condensed, JetBrains Mono, Permanent Marker, Noto family). Keep licence files in `licenses/`.
- Audio: track every SFX library and its licence in `audio/LICENSES.md`; original music contracts give us full rights.
- Contractors and collaborators sign a work-for-hire or licence agreement before their work enters the repo.

3. AI policy

- **Disclose everything AI-assisted** in `AI_DISCLOSURE.md` and on store pages (jam rules require it; Steam requires a content survey disclosure).
- AI images and video are **drafts**: shipped art is overpainted by people. Prompts never name living artists or copyrighted characters.
- Check each AI tool's terms for commercial use and output ownership (Higgsfield, any Hugging Face model's licence, Meshy) and record them here before shipping AI-derived assets.
- No AI voice clones of real people. Any AI voice is disclosed and credited as such.

4. Gambling and ratings

- **No real-money gambling, no paid randomness, no loot boxes.** Casino imagery is aesthetic.
- Target ratings: ESRB T / PEGI 12 (fantasy violence, gambling themes). Answer rating questionnaires honestly about "simulated gambling" imagery.

5. Privacy

Telemetry opt-in only (`TELEMETRY.md`); privacy policy before any data collection; no data from minors knowingly collected.

6. Jam rules

Follow the Official Unity Game Jam 2026 rules: work made during the jam window, WebGL build, AI disclosed, theme "Lucky Number". Keep commit history as evidence of the timeline.

How we work. Short on purpose.

1. Ownership

AREA	OWNER (ROLE)	SOURCE OF TRUTH
Vision, pillars, scope	Director / design lead	GDD, JAM_SLICE
Combat, sequencing	Lead dev	TECH_DESIGN, GAME_FEEL
UI, map, data, build	Dev 2	UX_SPEC, TECH_DESIGN
Characters, animation	Character artist/animator	CHARACTER_BIBLE, ART_DIRECTION, ASSET_LIST
Environments, VFX	Environment/VFX artist	LOCATIONS, ART_DIRECTION
Story, cinematics, audio	Writer/cinematics/audio	STORY, JAM_SCRIPT, AUDIO
Culture	Design lead + reviewers	CULTURAL_CONSULTATION

Order of authority when docs disagree: Art Direction (visuals) → Character Bible (characters) → World Bible + Story (story) → GDD + Systems (mechanics) → Tuning (numbers).

2. Rhythm

- **Daily stand-up (10 min):** yesterday, today, blockers; update ASSET_LIST status; agree any scope swap (JAM_SLICE §4).
- **Daily build** to the private itch page; **daily playtest** (PLAYTEST).
- **Weekly (post-jam):** design review with a playtest video; docs updated in the same PR as the change they describe.

3. Repo and tools

- Docs in `docs/` (Markdown is the source; the bible and art book PDFs rebuild with `python3 tools/build_bible.py` and `python3 tools/build_artbook.py`).
- Numbers in `tuning/`; simulation in `tools/sim_patterns.py`.
- Unity project in `game/` (to be created); Git LFS for binary art and audio.
- Branches: `main` always builds; short feature branches; PRs need one review and a passing CI.
- Commit messages say what changed and why; reference the doc section.

4. Art pipeline (end to end)

Brief (Art Book Production §5) → 3D base (Blender / 3D Jutsu / Meshy) → colour key → AI paint pass (Higgsfield/FLUX, logged) → **human overpaint** → QC (QUALITY_BAR) → layer split (ASSET_LIST) → engine → playtest.

5. Communication

One channel per area plus #builds and #playtests; decisions are written down in the relevant doc within a day, or they didn't happen.

PART XI

Appendix

Writing rules and disclosures.



How the game sounds. Use it for dialogue, item text, UI copy and marketing.

1. Tone

- **Warm noir.** The world is unfair and the jokes are dry, but people are kind to each other. Think rainy city detective story told by someone who still believes in strangers.
- **Wonder before cynicism.** Every land should feel like a place you'd want to visit, even when it's in danger.
- **Stakes are personal.** Big ideas (fate, luck, conservation) always land through one person's number.
- **Age rating target:** Teen. Peril, death off-screen ("settling"), no gore, no sexual content, mild language.

2. Voice rules

1. **Short lines.** Most dialogue lines are under 15 words. Cut adjectives first.
2. **One number habit per character** (Character Bible §12). Don't overuse it: once per scene at most.
3. **Numbers are spoken, not displayed.** Avoid "+15% luck". Say "that one's lucky", "your Omen's rising".
4. **NIL speaks least.** Their silences are written in, as beats.
5. **No exposition dumps.** Lore arrives through objects, jokes and choices, three sentences at most at a time.
6. **Villains get the best arguments.** Thale's lines should make players hesitate.
7. **Cultures are lived, not explained.** Characters mention their customs casually and never stop to define them for the player; the Codex does that.

3. Formatting

- Speaker names in caps in scripts: `VESPER:`
- Omens written as two digits: "Omen 07".
- Lore terms capitalised per the Lexicon (the Basilica, the Ledger, the Rounding).
- Non-English terms in italics on first use in the Codex only, never in dialogue.

4. Sample scene (intro, after the Naming Rite)

> *The terminal sparks. Every tube on the wall flips to orange.* **PRIEST:** That's… not a number, child. **NIL:** It's mine. *(beat)* Is that a problem? *Outside, every screen in the city reads REMAINDER DETECTED.* **PRIEST:** Run.

4A. Humour (v3: witty, never winking)

The world is unfair and the stakes are real; the jokes are how the characters survive it. Inspired by the deadpan deflation of *One Punch Man* and the banter of heist films, never by meta-jokes.

Joke engines (use these, in this order of frequency): 1. **NIL's deadpan precision** against everyone else's drama. *VARRO: "Final notice!" / NIL: "You've said that three times. It's not final."* 2. **Number habits colliding** (Character Bible §12): Jinling prices a rescue at 8% compounding; Prima counts primes mid-panic; Maeve insists the third attempt will work (it's the fourth). 3. **Bureaucracy of fate:** the Basilica is a cosmic DMV. Forms, stamps, queue numbers, "please hold while your luck is processed." 4. **Running gags** that pay off: Gus's "credit" (finally called in, in Act V); Niko's missing goat (found in the Moirai's cave); Maeve's one-hat disguise (works every time, to everyone's horror); Kyūsuke's cucumber stock (sold out at the finale). 5. **Understatement after spectacle:** the bigger the Jackpot, the smaller the line after it. *(777 detonates; the screen settles.) NIL: "…Huh."*

Rules: - **Big beat, then the joke.** Never undercut a reveal, a death, a keeper, or Hap's dreams. - **Characters are funny to each other**, not to the audience. Nobody breaks the fourth wall. - **Punch up.** Jokes target power (the Basilica, the House, the rich 99s), never the poor, the Settled, or a culture. - **One gag per fight round, max**; barks stay under 8 words. - **Every comic character has one moment played completely straight** (Maeve's grandmother's charms; Fibi's 1,597th dawn).

4B. Fight dialogue

- **Pre-fight (Versus card):** one line each, trash talk with character. *VESPER: "Seven chambers, Varro. You've got one face." / VARRO: "And one bounty."*

- **Mid-fight barks** react to *what just happened on the reel* (a Bust, a Pair, a Jackpot, a parry). 3–5 per hero per category; never repeat the same bark twice in a row.
- **Phase changes:** the boss gets the best line in the fight, and it reveals their motive (Villains get the best arguments, §2.6).
- **Finishers:** short and personal. Never a pun on death.
- **After the fight:** a breath. One quiet exchange, or one gag, then move on.

5. UI and item copy

- Item names are concrete and short: "Beckoning Cat", "Door God", "Rabbit's Foot".
- Item text = rule, then flavour in one line: "+1 fen per strike ending in your digit. *It waves twice for regulars.*"
- Pattern banners in caps: JACKPOT, OMEN OVERWRITE, ZERO-SUM.
- Errors and confirmations speak plainly: "Not enough fen." / "Charm equipped."

6. Cultural care in writing

- Ask a native speaker to review all dialogue and names from Hyakki, the River Empire and Śūnyapura.
- Keepers speak with dignity; no comic relief at their expense.
- No accents rendered phonetically in text.

The Official Unity Game Jam 2026 allows generative AI **only if it is disclosed on the submission page, including how it was used.** Not disclosing disqualifies the entry from prizes. Log every AI-assisted asset that ships, as you make it, and paste the summary into the itch page.

Summary for the submission page (fill in before submitting)

> Draft, update before submitting: Generative AI was used in this project as follows. Claude (Anthropic) helped draft the design documents, world bible and browser prototype under the team's direction, and wrote the code for our vector graphics and 3D layout scenes. Higgsfield (Seedream, Seedance) generated concept art and cinematic tests from our own 3D layouts; shipped art is painted over by the team. [List any AI art, audio or video that ships, from the log below.] Everything else (final art, animation rigs, motion graphics, editing, music arrangement, final writing) was made by the team.

Log

DATE	ASSET	TOOL / MODEL	HOW IT WAS USED	HUMAN EDITS AFTER GENERATION
2026-11-06	Design docs, prototype code	Claude (Anthropic) via Claude Code	Drafted design documents and the browser prototype with the team	Reviewed and directed by the team
2026-11-06	`art/lookdev/*_blockout.png`	Higgsfield 3D Jutsu (Blender scene)	Simple 3D layout renders used as composition guides	— (not shipped)
2026-11-06	`art/concepts/*.jpg` (character key art and environment concepts)	Higgsfield 3D Jutsu "angles" (Seedream V5 Pro), image generated over our own 3D layout renders, from prompts in `art/HIGGSFIELD_PROMPTS.md`	Concept and key art for NIL, Vesper, Maeve, Hina, Tamamo, Varro, Thale; Meridian and Hyakki environments	Concepts only so far; shipped versions will be painted over, layered and rigged by the team
2026-11-06	`art/motion/*.mp4` (intro shot, NIL reveal)	Higgsfield 3D Jutsu video (Seedance 2.5), animated from our generated key art as the start frame	Cinematic pipeline tests for the intro and character reveals	Tests only; final cutscenes will be edited, graded, scored and subtitled by the team
2026-11-06	`art/base3d/*.png` (Wards, Moon Pier, line-up clay bases and colour keys)	Blender scenes built by Claude (scripted bpy) in Higgsfield 3D Jutsu, Workbench renders. No image model	Stage-1 3D bases and flat colour keys for the register-A paint passes	Not shipped; reference for painting
2026-11-06	`art/graphics/*.svg` (faction emblems, Omen displays, charm icons)	Hand-written SVG code by Claude (`tools/graphics.py`). No image model	Graphic identity for the art book and UI icon drafts	Drafts; to be redrawn or refined by the team before shipping
2026-11-06	`art/graphics/figures/*.svg` (flat model-sheet figures for 16 characters and 4 eras), `art/graphics/reel_*.svg`	Hand-written SVG code by Claude (`tools/figures.py`, `tools/graphics.py`). No image model	Silhouette and colour designs for the art book, the bible and UI drafts	Drafts; to be redrawn by the team
2026-11-07	`art/cast/*` (56 character blockouts, 4 views each), `art/envs/*` (7 land blockouts, 3 views each), `.blend` files	Procedural Blender (bpy 5.2) scenes written by Claude (`tools/blender_cast.py`, `tools/build_envs.py`); Wards and Moon Pier geometry from the 3D Jutsu project. No image model	First-draft 3D reference for scale, silhouette, layout and cameras	Drafts; base models and paintings to be made by the team over these